

Event-History Monitoring of Cryptocurrency Derivatives Instability

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Abstract

Cryptocurrency derivatives can convert market pressure into forced trading when leveraged positions meet margin constraints and automated liquidation rules. We examine which dated market histories warn of this near-term instability using a dated event-history design that connects market, derivatives, liquidity, news, and attention records to crash events, liquidation-volume shocks, and their union in daily BTC, ETH, BNB, SOL, and XRP data, and we score each candidate monitor by how well it ranks upcoming instability in a held-out historical evaluation. We report three main findings. First, the order in which past stress arrived carries the warning content. The observed history outperforms every within-token reordering of the same information. Second, how the histories are represented matters more than which forecasting model is selected. Constructed event histories improve ranking in every completed specification, while model selection on top of the same histories adds no detectable value. Third, the warning is liquidation-centered. Gains concentrate where instability includes forced deleveraging, reflect continuation and recurrence rather than earlier detection of episode starts, and transfer only partly across the five tokens. Credible warning content therefore lies in the dated record and its representation.

Keywords: Cryptocurrency derivatives; liquidation shocks; derivatives instability; event-history forecasting; forecast evaluation; machine learning

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1 Introduction

Cryptocurrency derivatives create a monitoring problem because market pressure can become forced trading through leverage, margin constraints, and automated liquidation rules (Cheng et al., 2021; He et al., 2022). Perpetual futures allow traders to maintain leveraged exposure without contract maturity, so funding pressure and collateral constraints can turn a price movement into a margin event. When liquidation engines close undercollateralized positions, the resulting order flow must be absorbed by trading venues whose prices and liquidity are fragmented across exchanges (Makarov and Schoar, 2020). This mechanism makes short-horizon instability different from ordinary return volatility. A credible monitor must identify whether risk is elevated and which dated market histories make the warning credible. Leverage, liquidations, liquidity, news, and attention matter because they describe different points in the path from market pressure to forced trading. The liquidity side is central because funding constraints and market liquidity can reinforce each other during stress, making forced selling harder to absorb (Adrian and Shin, 2010; Brunnermeier and Pedersen, 2009).

Existing risk and forecasting views do not fully answer this monitoring problem. Cryptocurrency studies show that returns, volatility, jumps, and market segmentation contain predictable structure (Catania et al., 2019; Liu and Tsyvinski, 2021), and option-implied information forecasts bitcoin volatility out of sample (Hoang and Baur, 2020). Jump and crash evidence further shows that stress can cluster rather than arrive as isolated shocks (Donier and Bouchaud, 2015; Scaillet et al., 2020). Information studies show that media tone and investor attention can affect asset-market behavior, giving news and search activity a plausible role in stress monitoring (Da et al., 2011; Tetlock, 2007). These studies establish that cryptocurrency markets are predictable, segmented, and information-sensitive. However, these studies do not identify which dated histories remain informative when derivatives pressure, liquidity, news, and attention are evaluated within the same monitoring design, and they do not separate the value of representing that dated state from the value of the forecasting model fitted to it. Our research question therefore has two parts. Which dated market-fragility and information histories contain warning information for near-term cryptocurrency derivatives instability? Does the warning content come from the representation of those histories or from the model selection applied on top of them?

We construct a daily pooled token panel for BTC, ETH, BNB, SOL, and XRP because the monitoring problem is both temporal and token-specific. Each observation records one token on one calendar date. This structure preserves the daily order of information while allowing us to evaluate whether warning content transfers across major tokens rather than appearing only in one series. The data design joins market, derivatives, liquidity, news, attention, and sentiment records because forced trading can arise through several observable channels. Price and liquidity records describe market conditions, derivatives records capture leverage and liquidation pressure, and news and attention records capture infor-

mation conditions around stress episodes. We measure instability through crash events, liquidation-volume shocks (hereafter liquidation shocks), and composite instability. This distinction is necessary because a price collapse and a forced-deleveraging episode can overlap, but they are not identical economic objects. Of the one-day composite events, 74.6% (795/1,065) involve liquidation, which motivates a liquidation-centered primary comparison. Event histories then provide the bridge from raw records to forecasting. They retain the dated pattern of prior stress, rather than treating the current raw value alone as the warning signal. Because those histories are dated, the design also fixes what the monitor is allowed to see. Predictor histories use only information observed before the forecast window, while forward targets record subsequent instability.

We evaluate these histories through a conditional event-probability monitoring framework. The empirical object is not a continuous return forecast, but the probability that a defined instability event occurs after a dated token observation. Discrete-time event-history logic connects repeated token-date observations to such probabilities (Duffie et al., 2007; Shumway, 2001). Within this framework, the design separates three questions that are often treated as one. The first asks whether chronology matters, by comparing the observed ordered histories with within-token order permutations under unchanged construction and evaluation. The second asks whether constructed histories add ranking information to persistence-and-market controls while the estimator is held fixed. The third asks whether validation-based model selection adds value beyond a fixed-memory event-history logit that uses the same dated state. Treating model form as part of the empirical question is consistent with finance-ML evidence that predictive structure varies across assets and outcomes (Gu et al., 2020). Ranking and probability monitoring are evaluated separately, because a monitor may order risky days well while its probabilities are poorly scaled for monitoring use (Elliott and Lieli, 2013; Gneiting and Raftery, 2007). Pooled evaluation measures average monitoring value, token-level evaluation checks whether gains are concentrated in one token, and paired block-bootstrap inference asks whether the gains are credible rather than only numerically positive.

We report a target-specific monitoring pattern. Warning content is credible for liquidation shocks and for composite instability, and weak for crash-only targets. Chronology comes first. None of the within-token order permutations reaches the observed ranking performance of either the selected monitor or its comparator, so the dated order of stress, not the unordered set of indicators, carries the information. The representation of that order does the next part of the work. Adding event histories raises liquidation-shock area under the receiver operating characteristic curve (ROC-AUC) by 0.039 and average precision (AP) by 0.038 relative to the benchmark, and the increment is positive in every completed specification, although the five-token sample cannot resolve its size. Model selection matters the least. It adds no detectable ranking value beyond a fixed-memory event-history logit, so the value sits in the constructed histories rather than in the model fitted to them. The channel evidence sharpens the same message. Positioning histories

add ranking content beyond own-liquidation recurrence, although the increment varies across design variants, and news and search attention operate through the represented market state rather than as a separable warning channel.

A second set of results asks whether the fitted probabilities and the implied review policy support monitoring decisions, beyond ordering risky days. They do not do so automatically. Throughout, the selected monitor, once fixed for the historical evaluation, is reported as the active monitor, and we use the two names interchangeably. The active monitor’s short-horizon probabilities are less accurate than a constant-prevalence reference, so informative ranking does not deliver a decision-ready probability. The decision evidence is therefore reported as relative decision value and alert workload under an explicit cost-loss model. The monitor detects more realized episodes than the incumbent policy, the additional detection costs additional review, and neither policy dominates across the cost-loss curve. The evidence also has boundaries. The gains reflect continuation and recurrence rather than earlier onset detection, and the primary pooled gain transfers only partly across the five tokens.

The paper’s central contribution is to show that short-horizon warning information about cryptocurrency derivatives instability resides in the dated development and persistence of stress. We make this object observable by placing market, derivatives, liquidity, news, and attention records on one information clock and converting them into event histories available at each forecast date. Ranking performance deteriorates when temporal order is removed, while adding event histories improves liquidation-shock ranking in every completed specification. Once those histories are available, validation-based model selection adds no detectable ranking value. The warning content is also target-specific, with credible gains for liquidation shocks and composite instability but weak evidence for crash-only targets. This result provides a common basis for the paper’s contributions to research on cryptocurrency instability, machine learning in empirical finance, and rare-event monitoring.

The first strand is the literature on cryptocurrency instability and crash risk ([Cheng et al., 2021](#); [Donier and Bouchaud, 2015](#); [Scaillet et al., 2020](#)). The central limitation in this literature is that stress signals are fragmented across market, derivatives, liquidity, news, and attention records, so individual channels are typically studied in isolation or through contemporaneous association ([Makarov and Schoar, 2020](#)). The event-history layer places those channels on a common dated footing, and their joint and sequential warning content becomes an empirical question rather than a modeling assumption. For crash-risk research, the concentration of warning content in liquidation-driven instability points toward the forced-trading mechanism rather than the price collapse alone.

The second strand is machine learning in empirical finance ([Chen et al., 2021](#); [Gu et al., 2020](#); [Timmermann, 2018](#)). The components of standard rare-event panel forecasting are all present. They include lagged indicators, rolling thresholds, tree models, and validation selection ([Bharath and Shumway, 2008](#); [Lessmann et al., 2015](#)). We depart from this

standard design in three aspects. First, model form is treated as part of the empirical question. A validation gate selects the monitor for each target and horizon instead of fixing one model in advance, and the identity of the selected model is read as an operating choice rather than as evidence of added value. Second, ranking and probability quality are evaluated as separate monitoring objects, and the probability side is given its own instrument. HC-TEM forms monitoring probabilities in a single in-context pass over the dated event-history table, without per-cell estimation, so probability quality can be assessed alongside the ranking monitors rather than inferred from them (Hollmann et al., 2025). Third, benchmark families are defined by information sets, so the value of the measurement layer is separated from the flexibility of the estimators. This separation is what turns the model-selection result into a finding rather than a horse race.

The third strand is rare-event forecast evaluation and stability monitoring. Distress- and crisis-prediction studies evaluate warning systems on their out-of-sample ability to order risk and to time interventions (Drehmann and Juselius, 2014; Duffie et al., 2007; Schularick and Taylor, 2012), and rare outcomes additionally require metrics read against prevalence and probability scores that reward honest scaling (Davis and Goadrich, 2006; Gneiting and Raftery, 2007). Recent studies of Bitcoin crash classification and decentralized-finance liquidation prediction show the practical demand for target-specific monitoring (Palaokrassas et al., 2024; Park and Yang, 2024). This paper treats cryptocurrency derivatives instability as such a monitoring problem and shows that credible warnings are target-specific. Which instability definition is monitored, how its history is represented, and which evaluation object the warning improves must be established jointly. A warning system certified on one target or one metric does not automatically transfer to the others.

Section 2 develops the hypotheses. Section 3 describes the pooled token panel, raw variables, and event-variable construction. Section 4 presents the monitoring framework, model pool, evaluation design, and inference methods. Section 5 reports the empirical findings. Section 6 concludes.

2 Literature Review and Hypothesis Development

2.1 Monitoring Problem

Severe stress in cryptocurrency derivatives is a short-horizon state transition rather than simply an unusually large return. Realized jumps cluster in time, and forced liquidations concentrate around margin pressure (Cheng et al., 2021; Scaillet et al., 2020). Perpetual futures have no maturity, but margin is assessed continuously and positions can be liquidated when collateral becomes insufficient. A daily monitor therefore asks whether information available before the next trading day reveals a state in which another liquidation shock is more likely (Drehmann and Juselius, 2014). Volatility persistence is relevant but incomplete (Katsiampa, 2017). The problem concerns dated market events

and trading conditions that may change the conditional probability of a recurrent rare event.

That state is fragmented across the trading process. Prices and volatility record market outcomes, funding and open interest record aspects of positioning, and liquidation volume records forced position closure. Liquidity proxies describe absorption capacity, while news and search activity record public information and attention. None measures market fragility completely. What each record can contribute is a dated history of the conditions that precede instability. Our design therefore keeps each economic construct separate from its daily proxy and admits only information observed before the forecast window.

Table 1 positions the paper against the studies closest to its empirical object. Prior work examines crash classification, price direction, tail risk, derivatives pricing, liquidation and margin risk, futures premia, or liquidity measurement. This paper brings liquidation and composite-instability targets together with ordered dated histories, rare-event ranking evaluation, and an explicit decision role. The comparison locates the contribution in the joint definition and evaluation of the warning problem.

[Table 1 about here.]

We therefore build the hypotheses on four arguments. They concern the persistence of forced deleveraging, competing readings of positioning pressure, absorption of information into observed fragility, and the relative roles of state representation and model form. To make the mapping from arguments to evidence explicit, we summarize in Figure 1 the hypothesis each argument implies, the comparison that tests it, and the criterion on which it stands or falls.

[Figure 1 about here.]

2.2 Market Fragility

2.2.1 Persistence of liquidation stress

Automatic liquidation turns a price decline into forced trading. When a margined position crosses its maintenance constraint, the exchange reduces or closes it without waiting for the trader to revise beliefs, so liquidation records are a direct trace of forced deleveraging (Alexander et al., 2023; Cheng et al., 2021), and contract design shapes these liquidation mechanics across venues (De Blasis and Webb, 2022). A recorded liquidation, however, does not imply that the market clears when the transaction ends. Funding conditions and market liquidity can deteriorate together during stress, leaving surviving positions closer to their constraints and liquidity provision thinner exactly when risk-bearing capacity is most valuable (Adrian and Shin, 2010; Brunnermeier and Pedersen, 2009). A new adverse movement can then activate another round of forced sales, the self-exciting pattern formalized in event-intensity models (Bacry et al., 2015).

Limits to arbitrage explain why this pressure can survive beyond the initial day. When arbitrage capital is constrained, dislocations and order imbalances need not be absorbed immediately (Gromb and Vayanos, 2002). Funding and market liquidity can also reinforce one another during stress. Reduced liquidity raises the price effect of a forced trade, while losses tighten financing capacity (Brunnermeier and Pedersen, 2009). Yesterday’s liquidation can therefore alter today’s distribution of collateral headroom and risk-bearing capacity. This is an economic persistence mechanism, not merely serial correlation in returns.

The daily data do not observe each margin account or the capital available to arbitrageurs. Dated crash-event and liquidation-shock histories proxy prior forced-trading pressure, not a cascade or constrained capital itself. If pressure remains unresolved, those histories should contain warning information about the next liquidation shock beyond return, volatility, and market-state persistence (Catania et al., 2019). The implication concerns continuation or recurrence. It does not require superiority for episode-start events because the mechanism begins with a market already affected by recent stress. The first hypothesis states this persistence expectation:

H1: Recent instability in cryptocurrency derivatives is associated with higher near-term instability risk.

2.2.2 Positioning pressure and its competing readings

Derivatives positioning can raise later liquidation risk through directional leverage (Adrian and Shin, 2010; Geanakoplos, 2010; Schularick and Taylor, 2012). A move against a leveraged position consumes margin, and simultaneous pressure on similarly oriented positions can make forced closure more likely. Liquidation probabilities therefore depend on leverage, direction, margin, and extreme price moves, while hedge effectiveness also depends on collateral management (Alexander et al., 2023; Cheng et al., 2021). Under this reading, rapid increases in open interest or extreme funding pressure can mark a market in which a common adverse move has unusually strong balance-sheet consequences.

Funding has a second role. A perpetual contract has no maturity price at which futures and spot must converge, so periodic funding transfers help anchor the contract price to spot. Funding premia and funding intensity are part of the no-arbitrage pricing relation, not simply a count of leveraged directional bets (Akerer et al., 2026). Extreme funding can coincide with fragile positioning, but its sign and magnitude do not identify trader leverage or remaining collateral.

Open interest shares this identification problem. Its level does not show whether the exposure behind it is directional or hedged. Basis and carry positions can increase gross futures exposure while offsetting directional risk with another leg. Cryptocurrency-futures premia vary with basis, momentum, and basis-momentum factors, showing that positioning contains carry and return-premium components as well as directional demand (Chi

et al., 2023). Positioning records also carry forward-looking information for subsequent futures price changes (Baur and Smales, 2022; Shi, 2022). Higher open interest can therefore indicate crowded exposure, intermediation, or hedged carry. Liquidation itself can also lower subsequent risk by removing weak positions. The same record may mark the start of another stress round or the completion of deleveraging.

The panel observes upper-tail funding-rate histories, open-interest changes, and liquidation histories. It cannot separate long from short leverage throughout the sample, reconstruct carry legs, observe account-level collateral, or determine whether higher open interest is hedged. H2 does not identify which force dominates. It asks whether the combined histories contain incremental information after recent liquidation recurrence is accounted for. A uniformly positive increment would favor the fragile-positioning reading. Mixed signs would be consistent with the competing anchoring, carry, and deleveraging forces. We put the combined positioning record to exactly that test:

H2: Derivatives-positioning histories add monitoring information beyond recent liquidation recurrence.

2.2.3 Information absorption

Public information can affect liquidation risk before it appears as a crash (Auer and Claessens, 2018; Morris and Shin, 2002; Tetlock, 2007). Negative news can give traders a common reason to revise downside beliefs, while abnormal search activity indicates that more investors are attending. Attention can amplify a negative signal because it reaches more decision makers and becomes more salient (Barber and Odean, 2008; Da et al., 2011; Engelberg and Parsons, 2011; Smales, 2022). Extreme sentiment in traditional markets is also reflected in bitcoin futures pricing through the basis (Conlon et al., 2024). These channels give public information an economic route into prices, positions, and collateral demand rather than treating news and search intensity as generic sentiment.

Continuous trading also creates a strong absorption mechanism. Price discovery occurs across connected cryptocurrency venues, and arbitrage transmits information even when prices are not perfectly integrated (Brauneis and Mestel, 2018; Makarov and Schoar, 2020). By the daily prediction date, much of what news and attention conveyed may already be reflected in returns, funding, open-interest changes, liquidations, and liquidity. Absorption is therefore a substantive theoretical position. Information can matter for formation of the observed market state while adding little once that state is conditioned on.

The empirical proxy is a joint dated block of negative-news and abnormal-search-attention histories. It measures recorded public information, not belief revisions or the speed of price discovery. Negative tone supplies the direction, while attention measures how broadly the signal is processed. Because connected venues can impound public signals before the daily prediction date, the absorption mechanism predicts that the block

adds no separable content once observed market fragility is conditioned on. A separable increment would instead show information reaching the monitor ahead of the market state. The third hypothesis states the absorption claim:

H3: At the daily horizon, negative news and abnormal search attention operate through observed market fragility rather than as a separable warning channel.

Taken together, H1 to H3 define the economic content of the event histories used later in the paper. H1 concerns the persistence of forced-deleveraging pressure, H2 the incremental content of the combined positioning record, and H3 the absorption of news and search attention into observed market fragility. The common structure is that each mechanism is observable ahead of the forecast window and can be transformed into a dated history. This prepares the move from market fragility to monitoring design. Once the relevant histories are defined, the remaining questions are whether their representation carries the warning content and what, if anything, model form adds on top of it.

2.3 Representation Versus Model Form

The monitoring state is distributed across market outcomes, derivatives positioning, liquidity, forced trading, news, and attention, and each record arrives before the outcome window. For a rare recurrent event, contemporaneous levels cannot distinguish an isolated observation from an accumulating sequence, which is precisely the information that self-exciting event models exploit (Bacry et al., 2015). Event histories preserve that sequence and translate heterogeneous records into a common state representation. Conditional event-probability models then map the represented state into later event risk, following the finance tradition in which time-varying covariates update distress probabilities (Duffie et al., 2007; Shumway, 2001).

This distinction separates representation from model form. A flexible model cannot recover information omitted or mistimed by the measurement layer. Once the state is adequately represented, however, several sufficiently flexible learners may approximate a similar conditional ranking or probability. A larger function class can capture nonlinearities, but rare events also create estimation variance and selection uncertainty. Flexibility is therefore an out-of-sample comparison, not an automatic improvement (Gu et al., 2020; Timmermann, 2018). The binding constraint can be the dated state rather than the choice between a transparent event-history model, a tree ensemble, or a forecast combination.

The first design question is whether the ordering inside those histories is itself informative. If histories represent an evolving state, destroying within-token order should remove information while construction and evaluation remain unchanged. If only the unordered set of indicators matters, reordering should cost nothing. This is the sharpest available test of the representation claim, because it holds everything fixed except chronology. This

yields the ordering hypothesis:

H4a: The temporal ordering of event histories carries information for daily instability monitoring.

Ordering, even if informative, says nothing yet about how much flexibility the mapping from state to risk requires. The monitored outcomes arise from different economic mechanisms. A crash is a broad repricing event that follows a sudden withdrawal of market liquidity (Donier and Bouchaud, 2015), while a liquidation shock reflects forced deleveraging when margin constraints bind (Cheng et al., 2021), so the mapping need not be the same function across outcomes. The model pool therefore spans transparent event-history probability models, which keep the economic interpretation clear when predictions feed decisions (Rudin, 2019), and tree ensembles that capture nonlinear interactions a linear specification would miss (Breiman, 2001; Geurts et al., 2006). Unlike the deep-learning direction classifiers of Kim et al. (2025), the design below asks whether model form adds value once the dated state is represented. Neither class is privileged, and forecast-comparison discipline guards against reading too much into a single winner (Giacomini and White, 2006; Hansen et al., 2011). Validation-based selection should add stable value only if the conditional mapping remains materially misspecified after the histories are supplied. We state the model-selection claim accordingly:

H4b: Forecast-model selection adds incremental ranking value beyond the event-history representation.

A representation claim that survives both tests is still a pooled claim. Transfer across tokens is plausible when the five tokens share a market process through common funding, liquidity, and forced-deleveraging conditions, and a common representation should then help each token rather than only the pooled average. Token dispersion bounds the transfer claim (Petersen, 2009). A negative token result or a token-exclusion sign reversal indicates partial transfer rather than one invariant market-wide mapping. Transfer may be complete, holding for every token, or partial, holding in the pooled comparison and for most tokens. Section 5.4 states which form the evidence delivers. The transfer expectation completes the design:

H4c: Event-history monitoring generates gains across pooled and token-level evaluations.

These are distinct comparisons. Weak added model value cannot prove that representation is sufficient, and a flexible-model gain cannot show that chronology is unnecessary.

A final design claim is evaluated as a check rather than as a numbered hypothesis.

Monitoring output should remain meaningful as a probability as well as a ranking. A well-ranked set of observations can still produce probabilities that are poorly scaled for decisions, so proper scoring rules assess probability quality directly (Gneiting and Raftery, 2007) and decision-based evaluation asks whether the implied review policy justifies its workload (Elliott and Lieli, 2013). HC-TEM is included in the model pool for this probability role. It treats the event-history table associated with a target-horizon query as the conditioning object for the event probability, following in-context tabular learning (Hollmann et al., 2025). Sections 4 and 5 report these probability and decision checks alongside the ranking evidence.

3 Data

3.1 Data and Descriptives

We construct a pooled token panel of five major cryptocurrencies, BTC, ETH, BNB, SOL, and XRP, selected for their continuous market and derivatives coverage across the sample period.¹ Each observation records one token on one calendar date from January 1, 2020 through December 31, 2025. Because tokens enter the panel as their series become available, the panel is mildly unbalanced and contains 10,556 token-date observations. After rolling-threshold burn-in, the one-day composite-instability frame reports 9,606 eligible token-date forecast observations. A row enters a target-horizon sample only when its complete forward outcome window is observable. The resulting sample is the evaluation sample for that cell.²

The descriptive exhibits report BTC, as the representative token, alongside the pooled panel. Token-level coverage and further raw-descriptive detail are reported in Online Appendix A.1, Online Appendix Tables A.1.1 and A.1.2. The raw records cover market activity, derivatives-positioning pressure, liquidity, GDELT news tone and coverage, Google search attention, and the Fear and Greed Index.³ Market and liquidity records are spot-market records, and the positioning block is measured in the perpetual-futures market (Alexander et al., 2020). The two markets form one monitoring state. Funding

¹Reported results are pooled-panel estimates unless explicitly labelled token-level. Token-level results beyond those reported are available from the authors upon request.

²BTC and ETH begin on January 1, 2020; XRP on February 4, 2020; BNB on April 23, 2020; and SOL on September 14, 2020. All series end on December 31, 2025. Online Appendix Figure A.12.1 plots the raw input series.

³All sources are aligned to a single UTC calendar-day information clock. Market and derivatives series were retrieved from Binance (<https://www.binance.com>) and from Coinalyze (<https://coinalyze.net>); news tone and coverage from the GDELT Project (<https://www.gdeltproject.org>); search attention from Google Trends (<https://trends.google.com>); and the Fear and Greed Index from Alternative.me (<https://alternative.me/crypto/fear-and-greed-index/>). Rolling thresholds and all predictors use information through day $t - 1$ only (Equation (1)), and forward targets begin at day t , so no source’s same-day value enters its own forecast window.

payments anchor perpetual-futures prices to spot, and liquidation engines mark margined positions against spot-based index prices, so the spot record carries the price leg of the forced-trading mechanism whose positioning leg the derivatives records measure (He et al., 2022). Trading in these markets takes place at frequencies down to the subsecond level (Shynkevich, 2026). The monitoring design aggregates to the daily clock, where positioning and liquidation records are complete. Timing follows one UTC calendar day. Predictors use values available through day $t - 1$ and forward targets begin at day t . Daily cutoffs differ across providers, so we evaluate a one-day source delay among the design perturbations rather than claiming that the monitor could run in real time without restriction.

Variable construction follows established measures in the literature. Daily Binance spot OHLCV records provide open, high, low, close, and stored volume in base-token units. The daily return, $r_{i,t}$, is the close-to-close percentage change for token i on date t . Realized volatility and Amihud illiquidity follow Andersen et al. (2003) and Amihud (2002), and the high-low spread (HLSpread) follows the range-based estimator of Corwin and Schultz (2012). Coinalyze supplies the derivatives-positioning block. It consists of cross-venue aggregate perpetual-futures funding (Fund) sampled at 00:00 UTC, aggregate open interest, and reported aggregate liquidation volume (LiqVol), denoted $liq_{i,t}$ for token i on date t , the raw outcome from which the liquidation shock is derived (Cheng et al., 2021). Over their overlapping dates, Coinalyze and Binance-native funding correlations range from 0.85 to 0.98. GDELT supplies news tone and coverage counts (NewsTone, NewsCnt), Google Trends supplies three dated attention series (Search, CrashSearch, and BroadCrash), following search-volume attention measures (Da et al., 2011), and Alternative.me supplies the daily Fear and Greed Index as an external sentiment index.

The raw-variable descriptives in Online Appendix Table A.12.1 summarize the statistical properties of the data. For BTC, returns exhibit high excess kurtosis, while trading volume, realized volatility, reported aggregate liquidation volume, liquidity proxies, and attention variables are right-skewed. The pooled token panel shows the same broad pattern, with large dispersion across scale-dependent variables and substantial tail behavior in derivatives, liquidity, and attention measures.

These distributional features motivate rolling within-token quantiles rather than fixed cutoffs (Cheng et al., 2021; Drehmann and Juselius, 2014). Variables enter as returns or in first-differenced and lagged form. Ljung-Box tests (Ljung and Box, 1978) show short-run dependence. Augmented Dickey-Fuller (ADF) tests (Dickey and Fuller, 1979) reject the unit-root null for most variables. Token prices and open interest remain persistent (Online Appendix Table A.12.1). Preprocessing is fitted on training observations and then held fixed. Training ends in 2022, validation covers 2023, and historical evaluation covers 2024-2025. A 240-observation boundary purge protects multi-day labels without reducing the 9,606-observation one-day frame. The reported 67/33 design and the alternative sample shares are documented in Online Appendix A.8 and A.9.

3.2 Event Variables

We convert raw daily market, derivatives, liquidity, news, and attention records into event variables. The construction flags a token as under abnormal stress relative to its own recent history, rather than relative to another token’s scale. For a row dated t , thresholds and predictor variables use information available before the forecast date. Forward targets are defined separately, so the realized event window does not enter the predictor set.

The measurement rests on a token-specific rolling historical threshold. For token i , date t , raw variable X , rolling window w , and quantile probability p , we measure the p -quantile as follows.

$$Q_{i,t-1}^{w,p}(X) = \text{Quantile}_p \{X_{i,s} : s = t - w, \dots, t - 1\}. \quad (1)$$

Equation (1) ends the window at $t - 1$. It controls for token-specific scale and time variation without using the observation being labelled.

The implementation uses a 365-day rolling window with at least 180 observations, estimates thresholds within token from prior data, and treats each series’ opening months as burn-in. Online Appendix A.3 documents event construction, Online Appendix A.5 documents history initialization and preprocessing, and Online Appendix Table A.9.2 reports the completed window perturbations.

The first constructed variables define the instability outcome. Crash events identify extreme downside price moves. Liquidation-volume shocks identify abnormal forced derivatives unwinding. Let $E_{i,t}^{comp}$ denote the composite-instability indicator for token i on date t . It records the occurrence of either channel as follows. Throughout, $1\{\cdot\}$ denotes the indicator function, equal to one when its condition holds and zero otherwise.

$$C_{i,t} = 1 \{r_{i,t} < Q_{i,t-1}^{w,0.05}(r)\} \quad (2)$$

$$L_{i,t} = 1 \{liq_{i,t} > Q_{i,t-1}^{w,0.95}(liq)\} \quad (3)$$

$$E_{i,t}^{comp} = \max \{C_{i,t}, L_{i,t}\} \quad (4)$$

C is the crash event and L is the liquidation-volume shock. Composite instability is a descriptive union, not a distinct mechanism or independent corroborating outcome. Online Appendix Table A.3.1 decomposes outcomes into four mutually exclusive states. They are crash only, liquidation only, joint, or neither. Of one-day contemporaneous composite events, 74.6% (795/1,065) involve liquidation (594 liquidation-only plus 201 joint), which motivates the liquidation-centered primary comparison. The two components nevertheless identify price-discontinuity and liquidation-pressure events consistent

with empirical work on cryptocurrency jumps, crash liquidity, and liquidation mechanics (Cheng et al., 2021; Donier and Bouchaud, 2015; Scaillet et al., 2020).

We next define dated market-state and information-state indicators. Derivatives-positioning pressure combines an abnormal open-interest change and signed upper-tail funding pressure. Here $\Delta OI_{i,t}$ is the first difference in aggregate open interest for token i on date t . The indicators are defined as follows.

$$E_{i,t}^{OI} = \mathbf{1} \{ \Delta OI_{i,t} > Q_{i,t-1}^{w,0.95}(\Delta OI) \} \quad (5)$$

$$E_{i,t}^{fund} = \mathbf{1} \{ fund_{i,t} > Q_{i,t-1}^{w,0.95}(fund) \} \quad (6)$$

$$E_{i,t}^{LEV} = \max \{ E_{i,t}^{OI}, E_{i,t}^{fund} \} \quad (7)$$

Equations (5) to (7) flag derivatives-positioning pressure when an open-interest increase or signed funding pressure exceeds its trailing 95th percentile. The Coinalyze cross-venue funding series is sampled at 00:00 UTC and correlates 0.85 to 0.98 with Binance-native funding over overlapping dates. The lagged indicator and its history enter as predictors, not contemporaneous outcomes.

Liquidity contraction captures abnormal trading frictions. It is triggered by high Amihud illiquidity or by negative-return low-volume pressure as follows.

$$E_{i,t}^{AMI} = \mathbf{1} \{ Ami_{i,t} > Q_{i,t-1}^{w,0.95}(Ami) \} \quad (8)$$

$$E_{i,t}^{LVOL} = \mathbf{1} \{ r_{i,t} < 0, Vol_{i,t} < Q_{i,t-1}^{w,0.05}(Vol) \} \quad (9)$$

$$E_{i,t}^{LIQ} = \max \{ E_{i,t}^{AMI}, E_{i,t}^{LVOL} \} \quad (10)$$

Economically, Equations (8) to (10) capture two complementary symptoms of strained liquidity. They are abnormal price impact, and negative returns arriving on unusually thin volume. In Equations (8) to (10), $Ami_{i,t}$ is Amihud illiquidity, $Vol_{i,t}$ is trading volume, $E_{i,t}^{AMI}$ flags high Amihud illiquidity, $E_{i,t}^{LVOL}$ flags negative returns on unusually low volume, and $E_{i,t}^{LIQ}$ is the resulting liquidity-contraction indicator. It combines an established price-impact proxy with a low-volume negative-return condition so that the liquidity channel captures both measured illiquidity and thin-market selling pressure.

Negative news and search attention convert information conditions into dated event indicators. Negative news requires unusually low tone and a minimum news count. Search attention records an abnormal spike across token-specific or crash-related Google Trends series as follows.

$$E_{i,t}^{NEWS} = \mathbf{1} \{Tone_{i,t} < Q_{i,t-1}^{w,0.10}(Tone), Count_{i,t} \geq 5\} \quad (11)$$

$$E_{i,t}^{ATTN} = \max_j \mathbf{1} \{A_{i,t}^j > Q_{i,t-1}^{w,0.95}(A^j)\} \quad (12)$$

Equation (11) treats a day as a negative-news event only when tone is unusually low relative to its trailing 10th percentile and coverage clears a minimum of five stories, so thin-coverage days cannot generate news events, while Equation (12) flags abnormal spikes in any of the token-specific or crash-related search series. The negative-news and search-attention indicators enter the forecasting design as information-state variables. The news measure captures negative public information, while the attention measure captures the intensity of investor search activity.

These indicators become forecasting inputs through event histories. Let q index the event channels and let k denote the half-life. The decayed history is defined as follows.

$$H_{i,t}^{q,k} = \sum_{s < t} E_{i,s}^q \exp \left(-\log(2) \frac{t-s}{k} \right) \quad (13)$$

H is a lagged state variable. It accumulates prior events and discounts older observations by the half-life k . The main histories use one-, three-, and seven-day half-lives, matching the short-horizon monitoring design used in Sections 4 and 5.

We also construct interaction histories and cross-token summaries. Interactions represent joint stress states, while cross-token summaries measure pressure from the rest of the pooled token panel and exclude the focal token. Let a, b denote two event channels. Their definitions follow.

$$J_{i,t}^{a,b,k} = H_{i,t}^{a,k} H_{i,t}^{b,k} \quad (14)$$

$$X_{i,t}^{-i,q} = \frac{1}{|I_{t-1} \setminus \{i\}|} \sum_{\substack{j \in I_t \\ j \neq i}} E_{j,t-1}^q \quad (15)$$

J is used for combinations such as derivatives-positioning pressure with liquidity contraction or the information block. X averages the previous-day event indicators of the other observed tokens. Here I_t is the set of tokens observed on date t , and $|I_t| - 1$ counts the observed tokens other than i , so the focal token does not mechanically enter its own cross-token summary.

Finally, the forward target separates prediction from realization. For event channel q and horizon h , we use the following definition.

$$Y_{i,t}^{q,h} = \max_{\ell=0,\dots,h-1} E_{i,t+\ell}^q \quad (16)$$

Y is the outcome variable, not a predictor. For a forecast row dated t , predictors use information through $t - 1$, while the target window begins at t . The one-day target is therefore the date- t event, and the final $h - 1$ observations for each token are excluded when the target cannot be observed.

Every forecasting model uses predictors dated through $t - 1$ and targets beginning at t , with token identity reported in the pooled panel. Because thresholds, histories, interactions, cross-token summaries, and lags are computed within token from prior information, no predictor depends on validation or historical evaluation observations.

The continuous histories are skewed, leptokurtic, persistent, and stationary under the reported diagnostics. These properties support decayed histories as lagged predictors; Online Appendix A.3 reports the interaction histories and cross-token summaries derived from them.

Online Appendix Table A.12.2 shows that longer half-lives retain more event mass while histories remain concentrated in short stress clusters. Ljung-Box p-values are below 0.001 and ADF tests reject the unit-root null for the reported histories.

Figure 2 plots the event imbalance over calendar time. Event rates are concentrated in stress episodes rather than evenly distributed across the sample, with visible clustering around major market dislocations. The pooled token panel smooths token-specific noise but preserves the same broad timing pattern. Online Appendix Figure A.3.1 reports the corresponding ETH, BNB, SOL, and XRP token-level event-rate panels. Online Appendix Tables A.3.3, A.3.4, and A.3.5 report the constructed-variable, event-balance, and continuous-history diagnostics.

[Figure 2 about here.]

4 Methodology

4.1 Empirical Framework

The monitoring framework builds on financial event-history studies that estimate event risk with changing covariates and assess forecasts outside the estimation sample (Campbell et al., 2008; Cox, 1972; Duffie et al., 2007; Shumway, 2001). This logic fits cryptocurrency derivatives markets because instability can develop through dated stress signals rather than through a single contemporaneous market state. The methodology therefore starts from the event histories defined in the data section and asks how they should enter a disciplined forecasting design.

The empirical design proceeds in ordered stages. Event history construction comes first. It provides the measurement layer. The sequence in Figure 3 fixes the information boundary before performance comparison. Histories are constructed before estimation, and validation choice is fixed before historical evaluation. This staging fixes information timing but cannot identify the economic mechanism behind a forecast gain. These histories are fixed before estimation and then enter the forecasting models as dated indicators of prior instability. We organize the candidate model set by role. Conditional binary event-probability models provide the interpretable event-risk baseline, volatility benchmarks test whether event histories add information beyond ordinary persistence, and nonlinear or fusion models allow richer interactions across channels when supported by validation evidence. Validation then selects the forecasting specification for each target and horizon before we examine the out-of-sample historical evaluation. This separation prevents the final performance assessment from being shaped by ex post model choice. Historical evaluation reports out-of-sample performance only after the forecasting rule has been fixed. The final stage applies timing, measurement, attribution, and model-form checks to assess whether the forecasts genuinely reflect dated stress information rather than artefacts of sample construction or specification search. The following subsections introduce the conditional binary event-probability model, candidate models, validation and historical-evaluation procedures, and credibility checks.

[Figure 3 about here.]

4.2 Forecasting Specification

Following the financial literature on conditional event probabilities on distress and default prediction (Bharath and Shumway, 2008; Campbell et al., 2008; Duffie et al., 2007; Shumway, 2001), we express the monitoring problem as a conditional probability. The data section defines the composite instability, crash event, and liquidation shock targets, together with the lagged histories available before each forecast window. This subsection explains how those variables enter the forecasting equation and how the regression-based conditional binary event-probability models connect to the validation-gated model used in the final evaluation.

For target family q , forecast window h , and candidate specification m , the conditional binary event-probability model maps the information available before the window begins into a fitted probability as follows.

$$p_{i,t}(q, h, m) = \Lambda(\eta_{i,t}(q, h, m)) \quad (17)$$

In Equation (17), p is the forecast probability for the event target defined in the data section, Λ is the logistic link, and η is the fitted linear index. The indices keep

the target family, forecast window, and candidate specification explicit because all model comparisons use the same target definition and timing convention.

We use the following linear index, with a token-specific intercept consistent with the pooled-logit token indicators.

$$\eta_{i,t}(q, h, m) = \alpha_i(q, h, m) + \gamma_m^\top x_{i,t-1} + \beta_m^\top H_{i,t}^{(m)} + \theta_m^\top J_{i,t}^{(m)} \quad (18)$$

In Equation (18), vector x contains lagged market-state controls. These variables absorb persistence in returns, volatility, and recent instability. The vector H contains the event-history variables that measure whether prior stress arrivals continue to predict later instability after those market states are included. The vector J contains joint-history terms that allow one stress channel to change the warning content of another channel, such as derivatives-positioning pressure under weak liquidity. The coefficients have direct forecasting interpretations. α_i is the token-specific intercept for candidate m and implements the pooled-logit token indicators, γ measures the contribution of lagged market states, β measures the incremental warning content of individual event histories, and θ captures the additional content of channel interactions.

Consistent with financial conditional event-probability models, we use time-varying information to forecast distress or default risk (Campbell et al., 2008; Duffie et al., 2007; Shumway, 2001). We estimate two conditional binary event-probability specifications. They are the fixed-memory event-history logit, which uses three-day event histories, and the dynamic multi-memory event-history logit, which allows persistence to be selected from the validation sample. This comparison evaluates whether event histories are better represented by a parsimonious short-memory rule or by a validation-selected memory horizon. The dynamic multi-memory specification follows a training-validation-historical-evaluation design. The memory horizon is chosen on the validation sample and is then fixed before historical evaluation. This separates model tuning from final out-of-sample assessment, in line with forecasting practice in empirical asset-pricing research (Gu et al., 2020).

Our design adds a validation gate to choose the model for each target and forecast window in the historical evaluation. The gate compares the candidate conditional binary event-probability models and the additional forecasting models defined in Section 4.3 on the validation sample. For each target and forecast window, it selects the candidate with the highest validation AP, subject to the tie-breaking rules in Section 4.4.

$$m_{q,h}^* = \arg \max_{m \in M_{q,h}} S^{\text{val}}(m; q, h) \quad (19)$$

where M is the candidate set and S is the validation score. The monitoring probability reported for the selected model is

$$p_{i,t}(q, h, \text{gate}) = p_{i,t}(q, h, m_{q,h}^*) \quad (20)$$

Equation (20) states that the reported probability is the forecast from the selected candidate. The gate is a specification-selection rule. It can retain a fixed-memory conditional binary event-probability model when the event-history specification performs best, or select a dynamic, nonlinear, or fusion candidate when validation supports the richer structure. Section 4.3 defines the pool, and Online Appendix Table A.4.1 summarizes validation-gated metrics by reporting family. The gate evaluates the candidate universe documented in Online Appendix A.4. The comparison set is fixed before historical evaluation. Section 4.4 describes validation, historical evaluation, and paired inference. The ranking-oriented gate selects among the regression-based conditional binary event-probability models and nonlinear and fusion forecasts. Raw HC-TEM is evaluated separately in the probability-monitoring stage described in Section 4.5 and reported in Section 5.5.

4.3 Forecasting Models

The model pool tests whether nonlinear interactions or forecast combinations add value once every candidate uses the same dated information boundary and chronological split. The fixed-memory logit supplies the interpretable benchmark, while validation may select dynamic histories, tree ensembles, or convex forecast combinations. Table 2 summarizes the candidates; Online Appendix A.4 and A.5 report their settings and preprocessing.

[Table 2 about here.]

The regression-based forecasts extend the conditional binary event-probability specification in Equation (18). Equation (21) keeps the same logistic form and changes only the model-specific information block as follows.

$$\begin{aligned} \eta_{i,t}(q, h, m) = & \alpha_i(q, h, m) + \rho(q, h, m)Y_{i,t-1}(q, 1) \\ & + x'_{i,t-1}\gamma(q, h, m) + s_{i,t-1}(m)'\beta(q, h, m) \end{aligned} \quad (21)$$

In Equation (21), α_i is the token-specific intercept and represents the pooled-logit token indicators. Relative to Equation (18), s is the block that varies across model classes. It contains persistence variables, fixed event histories, dynamic event histories, or cross-token event histories, while x reports the common lagged market controls.

Random forest, extremely randomized trees, and histogram gradient boosting map the common lagged information set to event probabilities (Breiman, 2001; Geurts et al., 2006; Gu et al., 2020).

$$p_{i,t}(q, h, r) = f_r(g_{i,t-1}), \quad r \in \{\text{RF}, \text{ET}, \text{HGB}\} \quad (22)$$

Fusion combines validation-eligible probabilities with nonnegative weights that sum to one (Bates and Granger, 1969).

$$\omega_{q,h}^* = \arg \min_{\omega} L_V^{(q,h)}(\omega), \quad \omega_r \geq 0, \quad \mathbf{1}^\top \omega = 1 \quad (23)$$

The weights minimize early-validation log loss and are fixed before the common selection window.

$$p_{i,t}(q, h, F) = \omega_{q,h}^{*\top} p_{i,t}(q, h) \quad (24)$$

where p is the eligible forecast vector and the weights are target- and horizon-specific. Online Appendix A.4 and A.6 report the fitting window, objective, weights, and pool concentration. HC-TEM applies a pretrained in-context tabular learner to the same dated representation. Validation fixes its context and probability mapping; Online Appendix A.7 reports the architecture, implementation, and probability diagnostics (Hollmann et al., 2025).

4.4 Forecast Evaluation

Each candidate is estimated before validation, which selects one model per target and horizon before historical evaluation. The gate ranks by AP, with top-decile hit rate and log loss as tie-breakers. AP is reported with prevalence and AP lift; ROC-AUC, Brier score, and log loss are secondary metrics. Calibration is also chosen on validation data and then fixed. Online Appendix A.6 and A.7 report the gate, calibration rules, and realized choices. The primary comparison treats the selected monitor as fixed, while shifted windows, token omissions, validation resampling, and alternative gate metrics assess selection uncertainty.

The benchmark comparisons are paired. For each target and horizon, forecasts are evaluated on the same dated token observations and realized outcomes. The generic paired performance difference is

$$\Delta_{\phi,c}^{(q,h)} = \phi(y, p^{\text{gate}}) - \phi(y, p^c) \quad (25)$$

where ϕ is the evaluation metric and c is the comparator. The B0 to B3 ladder isolates the dated representation under a fixed logit, while B3 to the selected monitor isolates added model selection. Both contrasts are re-estimated across the 25 completed timing,

threshold, window, and token specifications described in Section 4.6. The comparisons therefore keep information-set changes separate from model-form changes.

Paired inference matches target, horizon, date, token, and outcome and resamples circular date blocks (Efron and Tibshirani, 1993; Politis and Romano, 1992). The primary one-day liquidation comparison uses 50,000 draws and block-length checks; eight secondary comparisons use 2,000 draws with Holm (1979) and complementary Benjamini and Hochberg (1995) adjustments. Online Appendix A.2 reports the matching and finite-sample p-value rules.

4.5 Probability Calibration and Relative Decision Value

Ranking performance does not ensure reliable probabilities. Section 5.5 therefore evaluates the validation-fixed active monitor separately as a probability and decision object, with definitions and matched samples reported in Online Appendix A.7.

Raw HC-TEM is a secondary probability-quality and high-precision-alert comparator, outside the AP gate and distinct from the active monitor. It uses the same dated event-history table; Section 5.5 and Online Appendix A.7 report its bounded role (Hollmann et al., 2025).

Probability quality is measured by the Brier score on matched one-day samples (Gneiting and Raftery, 2007). Online Appendix A.7 reports the compared objects and sample alignment.

The Murphy (1973) decomposition separates reliability, resolution, and outcome uncertainty. Online Appendix Table A.7.3 reports these components, calibration intercepts and slopes, and the validation-fixed calibration choices for raw HC-TEM and the active monitor.

A certainty-equivalent return is not reported because the action is review rather than portfolio investment. Decision evidence instead uses relative decision value (REV) under a stylized cost-loss model (Elliott and Lieli, 2013; Granger and Pesaran, 2000). Review costs r , a missed one-day liquidation shock costs one, and the reference is the better constant action at realized prevalence. Validation fixes thresholds before historical evaluation. Section 5.5 reports REV with alert workload and policy baselines; Online Appendix A.7 reports the full mechanics and scale bounds.

4.6 Credibility Checks

The validation and historical-evaluation split separates selection from final assessment, but performance may still reflect timing, measurement, attribution, or model-form choices (Gu et al., 2020; White, 2000). The credibility design tests these four limitations without reopening selection. Online Appendix A.8 and A.9 report the corresponding inference, robustness, and sensitivity evidence.

Timing checks add a source lag and reorder the joint six-event history with refitting and reselection; a narrower fixed-date perturbation is supplementary. Measurement and attribution checks vary thresholds, windows, liquidity proxies, crisis exclusions, token coverage, evaluation origins, and event onset definitions.

The model-form check keeps the primary one-day liquidation comparison separate from the adjusted secondary family and evaluates validation-choice stability. Online Appendix A.8 and A.9 report inference, selection, and all declared perturbations.

5 Empirical Findings

We report the empirical evidence in the order of the hypotheses, using the pooled token panel, event histories, and instability targets defined in Sections 3 and 4. The evidence proceeds from event recurrence and the positioning and information channels to representation versus model form, pooled and token-level transfer, probability quality and decision evidence, and credibility checks.

5.1 Event Recurrence

We begin by testing whether recent event histories carry information about near-term instability, the recurrence claim of H1. Table 3 examines this claim in two ways. Panel A reports penalized pooled-panel coefficients that relate each event variable to its own recent history, and Panel B reports the change in out-of-sample ranking performance when the same histories are added to a benchmark containing only return and volatility information. The two panels therefore separate conditional association from monitoring value, evaluated on the same dated observations.

Panel A shows that recent event histories enter positively for all four event variables, so past stress raises the conditional likelihood of further stress under every event definition. Because the penalized estimator delivers point estimates without standard errors, we interpret these coefficients descriptively and focus on their relative magnitudes. The recurrence pattern is strongest for liquidation shocks, with a coefficient of 1.264, followed by the episode-start target at 1, composite instability at 0.910, and crash events at 0.662. The coefficient ordering is economically intuitive. Crash events capture severe downside price movements, whereas liquidation shocks capture forced deleveraging under margin pressure, so recurrence is most pronounced where balance-sheet pressure, rather than price movement alone, drives the event.

[Table 3 about here.]

Panel B provides the corresponding forecast evidence. Adding recent event histories to the return and volatility benchmark raises AP for liquidation shocks by 0.038 and ROC-AUC by 0.039, and both bootstrap intervals exclude zero. The gains for composite

instability, crash events, and episode starts are small and statistically indistinguishable from zero. Online Appendix Table A.8.2 reports the full nine-cell matched AP comparison across targets and horizons. Episode starts illustrate the distinction between the two panels. They show recurrence in the conditional association but yield no separable forecast gain, so the predictive content of recurrence lies in the continuation of ongoing stress rather than in the arrival of new episodes. This evidence supports H1 in its persistence form. Unlike the liquidation and margin analyses of [Cheng et al. \(2021\)](#) and the liquidation and hedging design of [Alexander et al. \(2023\)](#), our evidence comes from a dated short-horizon forecast evaluated out of sample, and its predictive content is concentrated in continuation and recurrence.

5.2 Positioning and Information Channels

The recurrence evidence establishes that a token’s own stress history matters. We next examine whether the derivatives-positioning record adds warning content beyond that history. To isolate the increment, we start from a baseline that already includes own-liquidation history and the lagged target, and we add the combined positioning histories to it. The addition raises AP by 0.025, with a 95% interval from 0.006 to 0.047, and the improvement is positive in seven of the eight pooled add-in and leave-out comparisons reported in Online Appendix Table A.8.3. This evidence supports H2.

The increment, however, is not uniform across evaluation windows. When the positioning block is removed over the 2024 evaluation alone, the sign reverses to a small decline that is statistically indistinguishable from zero. This variation is consistent with the competing forces discussed in Section 2. Anchoring, carry, and deleveraging can each move funding and open interest in different directions around stress, so the positioning record adds information on average without reflecting a single fragile-positioning mechanism. Unlike the no-arbitrage funding analysis of [Ackerer et al. \(2026\)](#) and the basis and momentum factor evidence of [Chi et al. \(2023\)](#), our object is the dated incremental warning content of these records in a monitoring design, and the mixed signs we find are consistent with the competing forces those studies identify.

The information channels behave differently. A fixed-memory logit fitted to the negative-news and search-attention histories alone ranks one-day liquidation shocks above the constant-prevalence benchmark, with an AP of 0.221 against 0.096, a paired difference of 0.125 with a 95% interval from 0.043 to 0.237 and $p = 0.001$. The two series do carry warning information on their own. The same holds for composite instability, where the paired difference is 0.110 with a 95% interval from 0.038 to 0.207 and $p = 0.001$. That signal does not survive conditioning on the market state. Once market-fragility histories are included, adding the joint news and search-attention block does not improve ranking performance in either matched direction, and both point estimates are negative in the July 2024 to June 2025 and the 2025 historical-evaluation windows. The increment is statistically indistinguishable from zero. At 80% power the design would have detected

an increment of 0.033, roughly the size of the paper’s own representation gain. Lagged attention events also associate with next-day liquidation shocks, at an odds ratio of 5.24 with a 95% interval from 2.95 to 8.80, which is the association the absorption reading predicts. On their own the information series predict, the conditional increment vanishes once observed fragility is included, and information events associate with the next day’s market state. Three results are consistent with the absorption reading. The evidence supports H3. Online Appendix Table A.9.8 reports the standalone comparison, the conditional increment, and the next-day associations. This reading is consistent with the price-discovery evidence of [Brauneis and Mestel \(2018\)](#) and [Makarov and Schoar \(2020\)](#), in which connected venues impound public information within the day, and it differs from the equity-market tone and attention effects in [Tetlock \(2007\)](#), [Da et al. \(2011\)](#), and [Engelberg and Parsons \(2011\)](#) and the cryptocurrency attention evidence in [Smales \(2022\)](#). In our design, no separable daily warning channel remains once the observed market state is conditioned on.

5.3 Representation versus Model Form

The channel evidence establishes which records carry daily warning content. We next evaluate how the dated record is read, in the order the histories arrive, in the representation they are given, and in the model fitted to them. Chronology comes first. A test of ordering must compare the observed history with alternatives that are themselves coherent histories, and it must let the models re-estimate on each alternative, because rescoreing a fitted model on disordered inputs cannot separate the value of ordering from input corruption. The primary order test therefore permutes the joint six-event vector over each token’s full history and rebuilds the decayed histories from the permuted record. The candidate library is then refit and validation selection is rerun for the active monitor and the dynamic comparator, so each of the 200 draws is one internally consistent permuted history rather than a rescoreing of the observed fit. No permuted history reaches the observed AP of either model, at a permutation p-value of 0.005 for the active monitor and the same for the dynamic comparator. Online Appendix Table A.9.7 reports the design and the permuted distributions. An order-invariant representation gives the same answer. Replacing the decayed histories with trailing-window event counts over the same records and re-estimating the design lowers AP by 0.008 in the primary count specification and by 0.003 in the sensitivity specification, differences that are statistically indistinguishable from zero, so the loss is attributable to the ordering that the decayed representation preserves. These two results support H4a in its stated form. Whichever of the two monitors reads it, chronology itself carries the monitoring information. A narrower order-perturbation diagnostic⁴ holds each forecast date’s available events fixed and

⁴An order-perturbation diagnostic holds each forecast date’s available events fixed. Within trailing windows of 30, 60, and 90 days, the six event histories are jointly reordered in time, the decayed histories

reaches the same conclusion at the level of the fitted scores, and Online Appendix Table A.9.6 reports it.

With order established, Table 4 takes up representation and model form along the ladder defined in Section 4.4. Panel A reports the pooled AP level of each rung and the change from the preceding rung, so the move from B1 to B3 measures what the dated event-history representation adds to a fixed estimator. Panel B compares the rungs directly with the selected monitor, and its B3 row isolates what model selection adds once the histories are in place.

[Table 4 about here.]

Model selection contributes little beyond the fixed-memory event-history logit (B3). The selected monitor improves AP by 0.005 against B3, and the interval runs from a decline of 0.019 to an improvement of 0.034. An equivalence test makes this reading precise. We apply the two-one-sided-tests procedure with two margins anchored to observed reference magnitudes in this paper, the representation-step gain of 0.049 and the primary gain of 0.017, and both margins were set before the equivalence computation was run (Lakens, 2017; Schuirmann, 1987). No external convention fixes a minimally important AP difference for this design, so the two magnitudes act as interpretable scales rather than externally derived thresholds. The 90% interval for the increment runs from -0.015 to +0.030. Any undetected model-selection increment is therefore bounded above by +0.030 AP at the 90% level. Equivalence holds within the representation-step margin ($p < 0.001$), and it is not established at the stricter primary-gain margin ($p = 0.193$). The sign of the improvement also flips across the completed specifications. Together, the equivalence result and the unstable sign weigh against H4b as a claim of stable added value. Online Appendix Table A.6.4 reports the same pattern for the fusion step, with gains that are small and inconsistent across targets and horizons.

Taken together, the evidence separates the paper’s three ranking claims. The full-history permutation establishes H4a. Chronology itself carries the information. The representation increment from B1 to B3 is positive in all 25 completed specifications, at 0.049, but its interval spans zero, so the panel identifies the direction of the representation effect more clearly than its size. For this contrast the approximate minimum difference detectable at 80% power is 0.086. The panel cannot resolve an increment of the observed size and cannot rule out an increment near the interval’s upper end. The model-selection increment under H4b is small, sign-inconsistent, and statistically equivalent to zero at the representation-step scale. H4a is therefore supported, and H4b is not supported at the available precision. This pattern is consistent with model form becoming less binding

are rebuilt, and the fitted active monitor and dynamic comparator are re-scored without re-estimation. None of 200 reorderings per window reaches the observed ranking performance of either model, and at the 60- and 90-day windows none reaches the observed gap between them. Online Appendix Table A.9.6 reports the design and results.

once the dated state is well represented, a reading we treat as an interpretation to be tested rather than a prediction the design confirms.

Table 5 places the validation selections in context. It identifies the operating monitor for each target and horizon on the common validation window, and the model-confidence set measures how far each winner separates from the alternatives it beat.

[Table 5 about here.]

Validation selects one operating monitor per cell. Rare-event AP is estimated with sampling variation, so near winners need not be statistically separable from the selected model. The model-confidence set makes that distinction explicit by retaining every candidate whose performance on the window cannot be distinguished from the best at the stated confidence level (Hansen et al., 2011). A narrow set would mark the winner as clearly superior, and a broad set would show that the window ranks the leading candidates only weakly. The composition of the set therefore tells the reader how much weight the identity of the selected model can bear. On this window the set retains every event-history candidate. In the main one-day liquidation comparison, only the constant-prevalence benchmark falls outside the 90% set, while the ladder rungs, the dynamic multi-memory event-history comparator (hereafter the dynamic comparator), and the selected monitor all remain inside it. B3 and the selected monitor are statistically indistinguishable on this window, so the selected model identifies the operating monitor and the ranking value continues to rest on the dated event-history representation.

5.4 Pooled and Token-Level Transfer

The ranking value rests on the dated representation. We next test how far that value transfers across tokens, the transfer claim of H4c. Table 6 reports the pooled gain against the dynamic comparator for each target and horizon, together with token-win counts and the weakest token-level difference, and its primary row sets the scope for the decision evidence in Section 5.5. Every comparison in the table is read against that single fixed benchmark. Holding one strong benchmark fixed follows established forecast-evaluation practice and keeps the paired differences interpretable across cells (Hansen et al., 2011; White, 2000).

[Table 6 about here.]

The primary pooled gain does not travel to every token. Four of the five tokens improve, XRP declines by 0.011, and excluding BTC reverses the pooled gain to a decline of 0.025. An equal-token-weight version of the primary comparison gives a difference of 0.022, with a 95% interval from 0.010 to 0.042, so the pooled gain does not rest on BTC's sample share. The composite rows at the one- and seven-day horizons are positive

in all five tokens, but they are secondary and none rejects under Holm control. The evidence therefore supports H4c as partial transfer. The pooled gain is positive, and the all-five-token rule is not met.

Two features of the design explain the pattern. With five tokens, each token-level comparison rests on roughly a fifth of the pooled events, and AP is estimated with wide sampling variation when positives are this rare (He and Garcia, 2009), so single-token signs can flip within a two-year evaluation. A pooled average can also be carried by one strongly informative token, which is what the BTC exclusion shows. Consistent with Franco and Laurini (2025), who model intraday systemic tail risk for the same five tokens, our daily design finds common cross-token structure. Unlike their copula tail-risk design, we measure transfer as a dated out-of-sample AP gain against a common comparator, and shared structure at the system level does not guarantee a uniform gain in every token. Partial transfer therefore reads as a token-level estimation limit rather than evidence against the common mechanism, and it motivates reading the monitor as a pooled system whose token-level behaviour is checked before use.

5.5 Probability Quality and Decision Evidence

The transfer evidence completes the ranking claims. What remains is whether the fitted probabilities and the implied review policy support monitoring decisions. For all three one-day targets, the active monitor has a higher Brier score than the constant-prevalence reference, so the ranking gain does not extend to a decision-ready probability claim. The decision evidence in Table 7 therefore measures what the ranking alone delivers. It is a reallocation of review effort under explicit costs. Table 7 evaluates that reallocation with REV, referenced to the better constant action at the historical-evaluation event prevalence, following the decision-theoretic framing of Granger and Pesaran (2000) and Elliott and Lieli (2013). The reference makes the bar explicit. The monitor must beat the better constant action rather than no action. Against that reference the active monitor delivers positive REV at every reported cost-loss ratio from 0.02 through 0.50, a range wide enough to cover most plausible review costs.

[Table 7 about here.]

Dominance over the incumbent, a multi-memory event-history monitor fixed before the historical evaluation (Online Appendix A.7), is a separate matter. At cost-loss ratios near the middle of the reported range the two policies are indistinguishable, with REV close to 0.35 for both, and at one intermediate cost-loss ratio the incumbent is marginally ahead. The tie has a straightforward explanation. Both policies read the same dated record and flag similar shares of token-dates at similar precision, so near the middle of the range they produce nearly the same allocation of review effort, and the active monitor’s advantage is concentrated in higher recall.

Panel B reports the review cost of that additional recall. At the lowest reported cost-loss ratio the active policy flags nearly nine of every ten token-date observations, closer to continuous review than selective monitoring. At the reported operating threshold, raising recall from 0.88 to 0.95 detects fourteen additional event episodes and costs about 0.2 additional false alarms per observed token-month. Whether that trade is worth taking depends on the cost of a missed episode relative to a wasted review, which is exactly what the cost-loss ratio encodes. A monitor that must displace a cheap constant action faces a harder bar than one measured against no action, and the incumbent sets that harder bar. The decision object remains next-day exposure or collateral review.

That verdict concerns the active monitor’s scores rather than the probability quality attainable within the model pool. On the same one-day held-out cells, raw HC-TEM produces better-scaled probabilities. Its Brier score is lower for all three targets, and its reliability error is close to zero where the active monitor’s is not (Online Appendix Table A.7.3). Online Appendix Figure A.7.2 plots the reliability diagnostic, and Online Appendix Table A.7.7 reports the alert-level operating characteristics. Its calibration slope sits below one, so its probabilities are compressed and under-react to the strongest signals, which is why raw HC-TEM serves as the pool’s probability-quality comparator while the active monitor carries the ranking claim. This division of roles follows the design.

5.6 Credibility Checks

The reported gains require inference discipline because the active monitor and its comparators are evaluated on the same dated observations. Following the model-confidence perspective of [Hansen et al. \(2011\)](#), Table 8 keeps the primary comparison distinct from the eight secondary target-horizon comparisons, which carry Holm and complementary Benjamini-Hochberg adjustment (Online Appendix Table A.8.5). This structure permits one focused claim and treats the remaining comparisons as multiplicity-controlled secondary evidence. Each comparison is read against that cell’s incumbent, the event-history model leading on validation AP in the cell.⁵ No secondary comparison rejects under Holm control, so the secondary family does not broaden the primary evidence.

[Table 8 about here.]

The primary comparison asks a single question, whether the selected monitor ranks one-day liquidation shocks better than the dynamic comparator on the same evaluation

⁵The primary one-day liquidation-shock row and the three-day and seven-day liquidation-shock and composite-instability rows use the incumbent dynamic multi-memory logit. The one-day composite-instability row uses the incumbent full-feature fixed-memory logit, the three-day and seven-day crash-event rows use the incumbent three-channel fixed-memory logit, and the one-day crash-event row uses the incumbent lagged-market-control logit.

dates. Its null is a zero paired AP difference under the block inference committed in Section 4.4, which resamples whole date blocks. The draws therefore respect the serial dependence of the daily panel. The estimate is read as the value of the deployed choice, not as the outcome of a search across alternatives. The gain the primary comparison returns is positive but narrow. The selected monitor improves AP by 0.017 over the dynamic comparator, conditional on the selected model and the documented candidate set, and the 95% interval runs from just above zero to 0.038. The inference supports this narrow claim and little more. Across shorter and longer blocks the p-value moves across the conventional 5% threshold, and the two shortest-block intervals include zero, so the result does not establish invariance to the block choice (Online Appendix Table A.8.6).

The remaining credibility component is robustness to the declared design choices. Section 4.6 commits these checks in advance, and the design-sensitivity grid executes them, guarding the reported comparisons against dependence on any single favourable configuration. The grid varies event thresholds, construction windows, the May 2021 exclusion, token omissions, alternative liquidity proxies, rolling evaluation windows, source delay, block length, and event-order reordering. Online Appendix Tables A.9.3 through A.9.6 report the completed robustness exercises, and every declared cell is complete. Alternative liquidity measures leave the monitor’s AP close to its baseline level, and only one of the three paired increments separates from zero, so the comparator gain remains specification-dependent. With an added one-day source lag the selected-minus-comparator difference reverses to a decline of 0.005, with a 95% interval from a decline of 0.019 to an improvement of 0.015 and $p = 0.817$. Monitor AP falls to 0.311 under that lag, 0.051 below the baseline monitor’s AP. The sign of the difference also changes across the five rolling evaluation windows. The completed exercises therefore establish the chronology result and show that the selected-monitor edge moves with specification detail. That variability is consistent with the model-selection evidence in Section 5.3. Where the representation carries the value, the identity of the fitted model matters less.

6 Conclusion

This paper shows that near-term instability in cryptocurrency derivatives is not best monitored by searching over forecasting models. The credible warning content is event-based and liquidation-centered, and it lies in the dated record of past stress and in how that record is represented rather than in which model reads it. That conclusion changes the monitoring problem. What matters is whether the relevant histories enter in dated form and preserved order, how far a pooled gain travels across tokens, and whether the output is read as a ranking, a probability, or both. The economic mechanism is continuation and recurrence. Recent stress leaves a dated trace, and that trace is most informative where instability includes liquidation pressure, the point where leverage, margin calls, and forced deleveraging meet. Positioning histories add monitoring content beyond a token’s own

liquidation record, and negative news and abnormal search attention operate through the observed market state rather than as a separable warning channel. The short-horizon warning therefore comes from ordered histories of market fragility rather than from price variation alone.

The framework turns that mechanism into a disciplined design. The measurement layer converts the fragmented market, derivatives, liquidity, news, and attention records into look-ahead-free event histories and forward targets on a single information clock. The evaluation layer keeps the monitoring objects separate. Comparators are fixed before evaluation, ranking and probability quality are scored as distinct claims, and the review workload implied by an alert policy is reported rather than left implicit. HC-TEM contributes on the probability side, showing what well-scaled monitoring probabilities look like on the same evidence the ranking monitors face.

These findings have direct implications for practice. Exchanges and risk managers should treat dated liquidation and positioning histories as core state variables for next-day review rather than as secondary descriptions of price volatility, and should not expect news or attention feeds to supply a separate daily warning once market fragility is measured. For supervisors and policy makers, the same evidence locates the monitoring-relevant information in the derivatives records themselves, so timely access to liquidation and positioning data matters more than the choice of forecasting technology. The caution for investors is different. A ranking gain does not imply decision-ready probabilities. Scores that order risky days well can still be poorly scaled, so collateral and exposure decisions require the probability layer to be checked separately, and a pooled gain should be verified across tokens before it is trusted.

This research sheds light on how dated event histories support daily predictive monitoring of cryptocurrency derivatives instability. As in related forecasting studies, several caveats apply. The evidence comes from a five-token daily historical evaluation, and the positioning measure proxies leverage rather than measuring it directly. Complete positioning and liquidation records do not exist before the modern perpetual-futures era, so the sample cannot be lengthened backwards. Providers do not archive real-time publication vintages, so real-time availability cannot be audited retroactively. We hope future research extends the design to intraday records, venue-level liquidity, and broader token sets.

References

- Ackerer, D., Hugonnier, J., and Jermann, U. (2026). Perpetual futures pricing. *Mathematical Finance*, 36:481–499.
- Adrian, T. and Shin, H. S. (2010). Liquidity and leverage. *Journal of Financial Intermediation*, 19:418–437.
- Alexander, C., Choi, J., Park, H., and Sohn, S. (2020). BitMEX bitcoin derivatives: Price

- discovery, informational efficiency, and hedging effectiveness. *Journal of Futures Markets*, 40:23–43.
- Alexander, C., Deng, J., and Zou, B. (2023). Hedging with automatic liquidation and leverage selection on bitcoin futures. *European Journal of Operational Research*, 306:478–493.
- Amihud, Y. (2002). Illiquidity and stock returns: Cross-section and time-series effects. *Journal of Financial Markets*, 5:31–56.
- Andersen, T. G., Bollerslev, T., Diebold, F. X., and Labys, P. (2003). Modeling and forecasting realized volatility. *Econometrica*, 71:579–625.
- Auer, R. and Claessens, S. (2018). Regulating cryptocurrencies: Assessing market reactions. *BIS Quarterly Review*, pages 51–65.
- Bacry, E., Mastromatteo, I., and Muzy, J.-F. (2015). Hawkes processes in finance. *Market Microstructure and Liquidity*, 1:1550005.
- Barber, B. M. and Odean, T. (2008). All that glitters: The effect of attention and news on the buying behavior of individual and institutional investors. *Review of Financial Studies*, 21:785–818.
- Bates, J. M. and Granger, C. W. J. (1969). The combination of forecasts. *Operational Research Quarterly*.
- Baur, D. G. and Smales, L. A. (2022). Trading behavior in bitcoin futures: Following the “smart money”. *Journal of Futures Markets*, 42:1304–1323.
- Benjamini, Y. and Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*.
- Bharath, S. T. and Shumway, T. (2008). Forecasting default with the Merton distance to default model. *Review of Financial Studies*, 21:1339–1369.
- Brauneis, A. and Mestel, R. (2018). Price discovery of cryptocurrencies: Bitcoin and beyond. *Economics Letters*, 165:58–61.
- Breiman, L. (2001). Random forests. *Machine Learning*, 45:5–32.
- Brunnermeier, M. K. and Pedersen, L. H. (2009). Market liquidity and funding liquidity. *Review of Financial Studies*, 22:2201–2238.
- Campbell, J. Y., Hilscher, J., and Szilagyi, J. (2008). In search of distress risk. *Journal of Finance*, 63:2899–2939.
- Catania, L., Grassi, S., and Ravazzolo, F. (2019). Forecasting cryptocurrencies under model and parameter instability. *International Journal of Forecasting*, 35:485–501.
- Chen, W., Xu, H., Jia, L., and Gao, Y. (2021). Machine learning model for bitcoin exchange rate prediction using economic and technology determinants. *International Journal of Forecasting*, 37:28–43.
- Cheng, Z., Deng, J., Wang, T., and Yu, M. (2021). Liquidation, leverage and optimal margin in bitcoin futures markets. *Applied Economics*, 53:5415–5428.
- Chi, Y., Hao, W., Hu, J., and Ran, Z. (2023). An empirical investigation on risk factors

- in cryptocurrency futures. *Journal of Futures Markets*, 43:1161–1180.
- Conlon, T., Corbet, S., and Oxley, L. (2024). Investor sentiment, unexpected inflation, and bitcoin basis risk. *Journal of Futures Markets*, 44:1807–1831.
- Corwin, S. A. and Schultz, P. (2012). A simple way to estimate bid-ask spreads from daily high and low prices. *Journal of Finance*, 67:719–760.
- Cox, D. R. (1972). Regression models and life-tables. *Journal of the Royal Statistical Society: Series B*, 34:187–220.
- Da, Z., Engelberg, J., and Gao, P. (2011). In search of attention. *Journal of Finance*, 66:1461–1499.
- Davis, J. and Goadrich, M. (2006). The relationship between precision-recall and roc curves. In *Proceedings of the 23rd International Conference on Machine Learning*, pages 233–240.
- De Blasis, R. and Webb, A. (2022). Arbitrage, contract design, and market structure in bitcoin futures markets. *Journal of Futures Markets*, 42:492–524.
- Dickey, D. A. and Fuller, W. A. (1979). Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association*.
- Donier, J. and Bouchaud, J.-P. (2015). Why do markets crash? bitcoin data offers unprecedented insights. *PLOS ONE*, 10:e0139356.
- Drehmann, M. and Juselius, M. (2014). Evaluating early-warning indicators of banking crises: Satisfying policy requirements. *International Journal of Forecasting*, 30:759–780.
- Duffie, D., Saita, L., and Wang, K. (2007). Multi-period corporate default prediction with stochastic covariates. *Journal of Financial Economics*, 83:635–665.
- Efron, B. and Tibshirani, R. J. (1993). *An Introduction to the Bootstrap*. Chapman & Hall.
- Elliott, G. and Lieli, R. P. (2013). Predicting binary outcomes. *Journal of Econometrics*, 174:15–26.
- Engelberg, J. E. and Parsons, C. A. (2011). The causal impact of media in financial markets. *Journal of Finance*, 66:67–97.
- Franco, J. P. M. and Laurini, M. P. (2025). Quantifying systemic risk in cryptocurrency markets: A high-frequency approach. *International Review of Economics and Finance*, 102:104214.
- Geanakoplos, J. (2010). The leverage cycle. *NBER Macroeconomics Annual*, 24:1–65.
- Geurts, P., Ernst, D., and Wehenkel, L. (2006). Extremely randomized trees. *Machine Learning*, 63:3–42.
- Giacomini, R. and White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74:1545–1578.
- Gneiting, T. and Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102:359–378.
- Granger, C. W. J. and Pesaran, M. H. (2000). Economic and statistical measures of

- forecast accuracy. *Journal of Forecasting*, 19:537–560.
- Gromb, D. and Vayanos, D. (2002). Equilibrium and welfare in markets with financially constrained arbitrageurs. *Journal of Financial Economics*, 66:361–407.
- Gu, S., Kelly, B., and Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*, 33:2223–2273.
- Hansen, P. R., Lunde, A., and Nason, J. M. (2011). The model confidence set. *Econometrica*, 79:453–497.
- He, H. and Garcia, E. A. (2009). Learning from imbalanced data. *IEEE Transactions on Knowledge and Data Engineering*, 21:1263–1284.
- He, S., Manela, A., Ross, O., and von Wachter, V. (2022). Fundamentals of perpetual futures. *arXiv: 2212.06888*.
- Hoang, L. T. and Baur, D. G. (2020). Forecasting bitcoin volatility: Evidence from the options market. *Journal of Futures Markets*, 40:1584–1602.
- Hollmann, N., Müller, S., Purucker, L., Krishnakumar, A., Körfer, M., Hoo, S. B., Schirrmeister, R. T., and Hutter, F. (2025). Accurate predictions on small data with a tabular foundation model. *Nature*, 637:319–326.
- Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*.
- Katsiampa, P. (2017). Volatility estimation for bitcoin: A comparison of GARCH models. *Economics Letters*, 158:3–6.
- Kim, T., Jo, H., Choi, W., and Jang, B.-G. (2025). Bitcoin price direction forecasting and market variables. *Journal of Futures Markets*, 45:1579–1600.
- Lakens, D. (2017). Equivalence tests: A practical primer for t tests, correlations, and meta-analyses. *Social Psychological and Personality Science*, 8:355–362.
- Lessmann, S., Baesens, B., Seow, H.-V., and Thomas, L. C. (2015). Benchmarking state-of-the-art classification algorithms for credit scoring: An update of research. *European Journal of Operational Research*, 247:124–136.
- Liu, Y. and Tsyvinski, A. (2021). Risks and returns of cryptocurrency. *Review of Financial Studies*, 34:2689–2727.
- Ljung, G. M. and Box, G. E. P. (1978). On a measure of lack of fit in time series models. *Biometrika*.
- Makarov, I. and Schoar, A. (2020). Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics*, 135:293–319.
- Morris, S. and Shin, H. S. (2002). Social value of public information. *American Economic Review*, 92:1521–1534.
- Murphy, A. H. (1973). A new vector partition of the probability score. *Journal of Applied Meteorology*, 12:595–600.
- Palaiokrassas, G., Scherrers, S., Makri, E., and Tassioulas, L. (2024). Machine learning in DeFi: Credit risk assessment and liquidation prediction. In *2024 IEEE International Conference on Blockchain and Cryptocurrency*, pages 650–654.

- Park, S. and Yang, J.-S. (2024). Machine learning models based on bubble analysis for bitcoin market crash prediction. *Engineering Applications of Artificial Intelligence*, 135:108857.
- Petersen, M. A. (2009). Estimating standard errors in finance panel data sets: Comparing approaches. *Review of Financial Studies*, 22:435–480.
- Politis, D. N. and Romano, J. P. (1992). A circular block-resampling procedure for stationary data. In LePage, R. and Billard, L., editors, *Exploring the Limits of Bootstrap*, pages 263–270. John Wiley, New York.
- Rudin, C. (2019). Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 1:206–215.
- Scaillet, O., Treccani, A., and Trevisan, C. (2020). High-frequency jump analysis of the bitcoin market. *Journal of Financial Econometrics*, 18:209–232.
- Schuurmann, D. J. (1987). A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability. *Journal of Pharmacokinetics and Biopharmaceutics*, 15:657–680.
- Schularick, M. and Taylor, A. M. (2012). Credit booms gone bust: Monetary policy, leverage cycles, and financial crises, 1870–2008. *American Economic Review*, 102:1029–1061.
- Shi, S. (2022). Bitcoin futures risk premia. *Journal of Futures Markets*, 42:2190–2217.
- Shumway, T. (2001). Forecasting bankruptcy more accurately: A simple hazard model. *Journal of Business*, 74:101–124.
- Shynkevich, A. (2026). Trading periodicity and algorithmic divide in cryptocurrency markets. *Journal of Futures Markets*, 46:904–930.
- Smales, L. A. (2022). Investor attention in cryptocurrency markets. *International Review of Financial Analysis*, 79:101972.
- Tetlock, P. C. (2007). Giving content to investor sentiment: The role of media in the stock market. *Journal of Finance*, 62:1139–1168.
- Timmermann, A. (2018). Forecasting methods in finance. *Annual Review of Financial Economics*, 10:449–479.
- White, H. (2000). A reality check for data snooping. *Econometrica*, 68:1097–1126.

Table 1: Closest Literature: Outcomes, Designs, and Decision Roles

Reference	Outcome	Data	Prediction design	Evaluation	Decision role
This paper	Crash, liquidation, and composite instability	Daily BTC, ETH, BNB, SOL, XRP	Ordered dated histories; target-horizon validation	AP/AP lift, paired blocks, probability scores	Cost-loss value and alert workload
Park and Yang (2024)	Crash classification	Bitcoin	Bubble-cycle and ML inputs	Area under the precision-recall curve (PR-AUC)	Investment interpretation
Kim et al. (2025)	Next-day price direction	Daily Bitcoin and cross-market variables	Deep-learning direction classifiers	Classification accuracy and backtests	Trading interpretation
Franco and Laurini (2025)	Systemic tail risk	Five-token intraday	GARCH margins and copulas	VaR/CoVaR backtests	Stress monitoring
Ackerer et al. (2026)	No-arbitrage pricing	Perpetual futures	Funding and no-arbitrage dynamics	Analytical	Contract design
Cheng et al. (2021)	Liquidation and margin risk	Bitcoin derivatives	Extreme-value tails and leverage	Horizon comparison	Margin choice
Alexander et al. (2023)	Liquidation and hedging	Crypto futures	Extreme returns and collateral	Cross-instrument	Hedge and leverage choice
Chi et al. (2023)	Futures premia	Crypto futures	Basis and momentum factors	Sorts and regressions	Return portfolios
Brauneis et al. (2021)	Liquidity measurement	Crypto spot markets	Low-frequency liquidity proxies	Accuracy against high-frequency measures	Proxy choice

Notes: The table compares this paper with the closest studies across outcome, data, prediction design, evaluation, and decision role. AP denotes average precision; AP lift is AP divided by prevalence; PR-AUC denotes area under the precision-recall curve.

Table 2: Forecasting Model Classes

Model	Information set
Persistence logit	Lagged instability, lagged return, lagged realized volatility, and HAR volatility terms over one, seven, and thirty days.
Fixed-memory event-history logit	Three-day histories of the three event indicators and their pairwise interactions.
Dynamic multi-memory event-history logit	Event histories with one-, three-, seven-, and fourteen-day half-lives, estimated with elastic-net logistic regression.
Cross-token event-history logit	Leave-one-token-out lagged event summaries that measure stress in other tokens before the focal forecast date.
Nonlinear tree forecasts	Lagged market controls, volatility summaries, event histories, joint histories, cross-token summaries, and token indicators.
Convex fusion	Validation-eligible base-model probability vector, combined with non-negative weights that sum to one.
HC-TEM (probability-quality comparator)	Pretrained in-context tabular prediction applied to the same 48 dated features as the full event-history pool; evaluated outside the AP ranking selection.

Notes: Each logit row implements the conditional binary event-probability model of Equations (17), (18), and (21) on its stated information set. HAR denotes heterogeneous autoregressive realized volatility. The nonlinear tree forecasts comprise random forest, extremely randomized trees, and histogram gradient boosting. Online Appendix A.5 documents event-history initialization, recursion, joint features, attention windows, missing-value rules, and classifier objects. Online Appendix A.4 documents the candidate universe, estimator settings, validation criteria, and fusion weights, with the validation-gated winner in Online Appendix Table A.4.3.

Table 3: Event Recurrence: Descriptive Coefficients and Historical Evaluation Forecast Gains

<i>Panel A. Descriptive Conditional Event-History Associations</i>		
Variable	Penalized coefficient	
Composite instability	0.910	
Crash event	0.662	
Liquidation shock	1.264	
Episode start	1.000	
<i>Panel B. Historical Evaluation Forecast Gains from Event Histories</i>		
Variable	Δ ROC-AUC [95% interval]	Δ AP [95% interval]
Composite instability	0.003 [-0.012, 0.017]	-0.001 [-0.028, 0.020]
Crash event	-0.003 [-0.023, 0.014]	-0.009 [-0.021, 0.001]
Liquidation shock	0.039 [0.019, 0.058]	0.038 [0.001, 0.064]
Episode start	0.003 [-0.020, 0.028]	0.004 [-0.007, 0.014]

Notes: Variable follows the event definitions in Section 3. Panel A reports penalized pooled-panel coefficients, read descriptively by sign and relative magnitude because the penalized estimator delivers point estimates without standard errors. Panel B reports paired historical evaluation performance differences from adding recent event histories to the return and volatility benchmark. Δ ROC-AUC and Δ AP equal the event-history metric minus the benchmark metric on the same dated observations. Intervals are paired block-bootstrap intervals. Episode start identifies the first event in a local episode. Panel B uses the 2024 and 2025 historical evaluation.

Table 4: Ordered Representation Ladder and Model-Selection Comparison

Rung	AP	AP lift	Δ vs. prior	95% interval	p	Positive specs	Clear of zero
<i>Panel A. Fixed-baseline ladder and preceding-rung contrasts</i>							
B0: constant prevalence	0.096	1	–	–	–	–	–
B1: persistence + market	0.308	3.203	+0.212	[+0.142, +0.271]	0.001	25/25	25/25
B2: + HAR volatility	0.319	3.321	+0.011	[-0.016, +0.040]	0.535	23/25	0/25
B3: + event histories	0.357	3.717	+0.038	[-0.0103, +0.083]	0.135	25/25	6/25
Selected monitor	0.362	3.769	+0.005	[-0.019, +0.034]	0.533	14/25	0/25
B1 to B3	–	–	+0.049	[-0.017, +0.104]	0.141	25/25	6/25
<i>Panel B. Main-specification selected-monitor comparisons</i>							
B1 to selected monitor	+0.054	–	–	[-0.0099, +0.110]	0.093	–	–
B2 to selected monitor	+0.043	–	–	[-0.008, +0.097]	0.090	–	–
B3 to selected monitor	+0.005	–	–	[-0.019, +0.034]	0.533	14/25	0/25

Notes: Panel A reports pooled main-specification AP levels and preceding-rung differences. The positive-specification count is the number of the 25 completed specifications defined in Section 4.4 whose difference is positive, and the clear-of-zero count is the number whose interval excludes zero. The B1 to B3 row reports the direct representation contrast between those two rungs. Panel B reports the distinct B3-to-selected-monitor comparison. The panels use different nulls and report separate comparisons. The two equivalence margins are anchored to the two observed reference magnitudes in this paper, the representation-step gain of 0.049 and the primary gain of 0.017, and both were set before the equivalence computation was run (Schuirmann, 1987; Lakens, 2017). AP levels and differences use the 2024 and 2025 historical evaluation.

Table 5: Validation-Selected Monitors on the Common Validation Window

Target	Selected monitor	N (events)	ROC-AUC	AP	Brier / log loss
<i>Panel A. 1-day horizon</i>					
Composite instability	Nonlinear-augmented fusion	700 (74)	0.795	0.451	0.144 / 0.462
Crash event	Fixed-memory event-history logit	700 (27)	0.690	0.106	0.212 / 0.618
Liquidation shock	Nonlinear-augmented fusion	700 (59)	0.796	0.454	0.120 / 0.395
<i>Panel B. 3-day horizon</i>					
Composite instability	Random forest with dynamic event histories	690 (161)	0.749	0.542	0.182 / 0.543
Crash event	Fixed-memory event-history logit	690 (77)	0.706	0.261	0.212 / 0.619
Liquidation shock	Random forest with dynamic event histories	690 (123)	0.745	0.500	0.153 / 0.473
<i>Panel C. 7-day horizon</i>					
Composite instability	Nonlinear-augmented fusion	670 (267)	0.799	0.734	0.190 / 0.564
Crash event	Fixed-memory event-history logit	670 (142)	0.723	0.500	0.210 / 0.611
Liquidation shock	Random forest with dynamic event histories	670 (206)	0.740	0.628	0.194 / 0.569

Notes: The table reports the monitors selected on the later common validation window. The fixed-memory event-history logit is the Table 2 candidate, the nonlinear-augmented fusion is the convex fusion whose validation-eligible base set adds the nonlinear tree forecasts, and random forest with dynamic event histories is the random-forest estimator fitted on the full dynamic event-history block. Model labels correspond to the model definitions in Section 4.3 and the candidate-universe discussion in Online Appendix A.4. Validation metrics govern selection only; historical evaluation comparisons follow in Sections 5.4 and 5.6 and in Online Appendix A.6. The common validation window runs from 14 August to 31 December 2023.

Table 6: Pooled and Token-Level Forecast Gains

Target	Active AP	Δ AP vs. dynamic	95% interval	Token wins	Weakest token Δ
<i>Panel A. 1-day horizon</i>					
Composite instability	0.362	+0.015	[0.0007, 0.033]	5/5	SOL +0.015
Crash event	0.099	+0.0001	[-0.026, 0.016]	2/5	SOL -0.035
Liquidation shock	0.362	+0.0167	[+0.0004, +0.038]	4/5	XRP -0.011
<i>Panel B. 3-day horizon</i>					
Composite instability	0.506	+0.005	[-0.012, 0.020]	3/5	SOL -0.008
Crash event	0.247	+0.005	[-0.026, 0.026]	2/5	SOL -0.029
Liquidation shock	0.469	-0.002	[-0.021, 0.015]	4/5	XRP -0.040
<i>Panel C. 7-day horizon</i>					
Composite instability	0.694	+0.016	[0.0013, 0.026]	5/5	BTC +0.012
Crash event	0.416	-0.018	[-0.052, 0.020]	1/5	SOL -0.032
Liquidation shock	0.597	-0.014	[-0.053, 0.020]	3/5	XRP -0.033

Notes: The table reports active-monitor AP and paired AP differences on the same historical-evaluation observations. All rows report paired differences against the dynamic comparator defined in Section 5.3. The primary one-day liquidation-shock row uses the 21-day inference reported in Section 5.6, and the secondary rows use 2,000 paired 21-day circular date-block draws. Token wins count positive token-level differences. Only the one-day liquidation-shock row is primary, and the remaining eight rows form the adjusted secondary family. Online Appendix Table A.8.4 reports the same nine comparisons against the same comparator with ROC-AUC inference and token-level agreement. All comparisons use the 2024 and 2025 historical evaluation.

Table 7: Relative Decision Value, Alert Workload, and Compared Forecast Objects

Panel A. Relative decision value under the stylized cost-loss model									
r	Active REV	[95%]	Incumbent REV	[95%]	Difference	[95%]	Active threshold	Active alerts (rate)	
0.02	0.038	[-0.073, 0.131]	-0.321	[-0.660, -0.012]	+0.360	[+0.109, +0.644]	0.130	3,179	(87.0%)
0.05	0.227	[0.097, 0.362]	0.157	[-0.006, 0.319]	+0.070	[-0.009, +0.171]	0.222	1,906	(52.1%)
0.06	0.267	[0.112, 0.418]	0.225	[0.083, 0.366]	+0.042	[-0.021, +0.098]	0.284	1,356	(37.1%)
0.10	0.350	[0.189, 0.432]	0.353	[0.194, 0.429]	-0.003	[-0.029, +0.023]	0.536	414	(11.3%)
0.14	0.312	[0.167, 0.425]	0.212	[0.082, 0.320]	+0.099	[+0.052, +0.140]	0.536	414	(11.3%)
0.20	0.212	[0.076, 0.320]	0.193	[0.070, 0.294]	+0.019	[-0.005, +0.040]	0.662	227	(6.2%)
0.30	0.150	[0.031, 0.247]	0.140	[0.031, 0.233]	+0.010	[-0.024, +0.042]	0.662	227	(6.2%)
0.40	0.097	[0.021, 0.164]	0.096	[0, 0.185]	+0.001	[-0.036, +0.042]	0.827	79	(2.2%)
0.50	0.071	[0.010, 0.129]	0.034	[-0.061, 0.124]	+0.037	[-0.016, +0.095]	0.827	79	(2.2%)
Panel B. Episode-level alert workload									
Policy	Alert days (rate)		Detected/event episodes		Recall [95%]	Precision [95%]	False alarms/observed token-month [95%], N = 120		
Active monitor	1,906 (52.1%)		208/220		0.945 [0.915, 0.985]	0.485 [0.422, 0.614]	1.18 [0.908, 1.54]		
Incumbent	1,777 (48.6%)		194/220		0.882 [0.845, 0.950]	0.489 [0.434, 0.623]	0.975 [0.758, 1.28]		
Active – incumbent	+129 (+3.5 pp)		+14/220		+0.064 [+0.019, +0.093]	-0.004 [-0.057, +0.035]	+0.2 [+0.033, +0.35]		
Panel C. Compared Forecast Objects									
Prediction object	Role		Cells		Cells resolved by fallback		Fallback share	Interpretation	
Raw HC-TEM	Secondary comparator		9		Not applicable		Not applicable	Probability-quality diagnostic	
Active monitor	Active validation-controlled monitor		9		9		100.0%	Monitor outcome, separate from HC-TEM	

Notes: Panel A reports nine representative ratios for the one-day liquidation-shock target. REV uses the better constant action at realized historical-evaluation prevalence and the cost of reviewing exactly the event dates as the upper bound of the REV scale. Intervals are paired two-sided 95% percentile intervals from 2,000 circular 21-day date-block draws. The $r = 0.05$ row is the operating point read in Panel B and Section 5.5. The incumbent in Panels A and B is the elastic-net multi-memory logistic model that led on validation AP, fixed before the historical evaluation and documented in Online Appendix A.7. Panel B reports absolute alert days and episode workload. Its false-alarm rates use 141 and 117 false alarms for the active monitor and incumbent, respectively, divided by 120 observed token-months. Positive false-alarm differences mean heavier active-monitor workload. Panel C separates raw HC-TEM from the active monitor. Online Appendix A.7 reports the complete curve (Online Appendix Table A.7.4), the matching rule, denominator, and workload detail (Online Appendix Table A.7.5), and the policy baselines (Online Appendix Table A.7.6). All quantities use the 2024 and 2025 historical evaluation.

Table 8: Primary and Secondary-Family AP Inference

Target	h	Δ AP	95% interval	Raw p	Holm p	BH q
<i>Panel A. Single primary comparison</i>						
Liquidation shock; conditional on the selected model and the documented candidate set	1	+0.0167	[+0.0004, +0.038]	0.046	—	—
<i>Panel B. Eight secondary comparisons, 2,000 draws</i>						
Composite instability	7	+0.016	[+0.002, +0.026]	0.030	0.240	0.240
Composite instability	1	+0.011	[-0.0014, +0.025]	0.079	0.553	0.316
Liquidation shock	7	+0.0097	[-0.019, +0.034]	0.512	1	0.990
Liquidation shock	3	0	[-0.021, +0.018]	0.990	1	0.990
Crash event	7	+0.002	[-0.032, +0.033]	0.869	1	0.990
Crash event	3	-0.0008	[-0.025, +0.027]	0.964	1	0.990
Composite instability	3	-0.0005	[-0.009, +0.011]	0.935	1	0.990
Crash event	1	-0.011	[-0.055, +0.024]	0.558	1	0.990

Notes: The primary row reports the selected-monitor inference described in Section 4.4. The eight secondary rows use 2,000 draws, a 21-day circular date block, two-sided finite-sample p -values, Holm family-wise adjustment, and complementary Benjamini-Hochberg q -values. Each row is compared with its own incumbent, the event-history model that led on validation AP in that target-horizon cell. All comparisons use the 2024 and 2025 historical evaluation.

MECHANISM	HYPOTHESIS	COMPARISON	CRITERION
MECHANISM 1 Persistence of liquidation stress	H1 Recent instability in cryptocurrency derivatives is associated with higher near-term instability risk.	Recent-instability histories vs persistence/market controls; ordered vs shuffled histories.	Positive recent-history gain; ordering must matter.
MECHANISM 2 Positioning pressure	H2 Derivatives-positioning histories add monitoring information beyond recent liquidation recurrence.	With vs without positioning histories, holding own-liquidation recurrence fixed.	Positive increment, interval clear of zero, in the matched comparison.
MECHANISM 3 Information absorption	H3 At the daily horizon, negative news and abnormal search attention operate through observed market fragility rather than as a separable warning channel.	With vs without the joint news-attention block in the matched fragility model.	No separable increment once market-fragility histories are included.
MECHANISM 4 Representation versus model form	H4a The temporal ordering of event histories carries information for daily instability monitoring.	Ordered-history performance vs refitted full-history joint permutations.	Both models: no refitted permutation reaches the observed ranking performance.
	H4b Forecast-model selection adds incremental ranking value beyond the event-history representation.	Selected monitor vs fixed-memory event-history logit, same dated state.	Clearly positive overall; stable across designs.
	H4c Event-history monitoring generates gains across pooled and token-level evaluations.	Selected monitor vs stated comparator, pooled and token by token.	Positive overall; non-negative for every token.

Figure 1: Mechanisms, Hypothesis-Specific Comparisons, and Support Criteria
Notes: Whether each hypothesis is supported is reported with the evidence in Section 5.

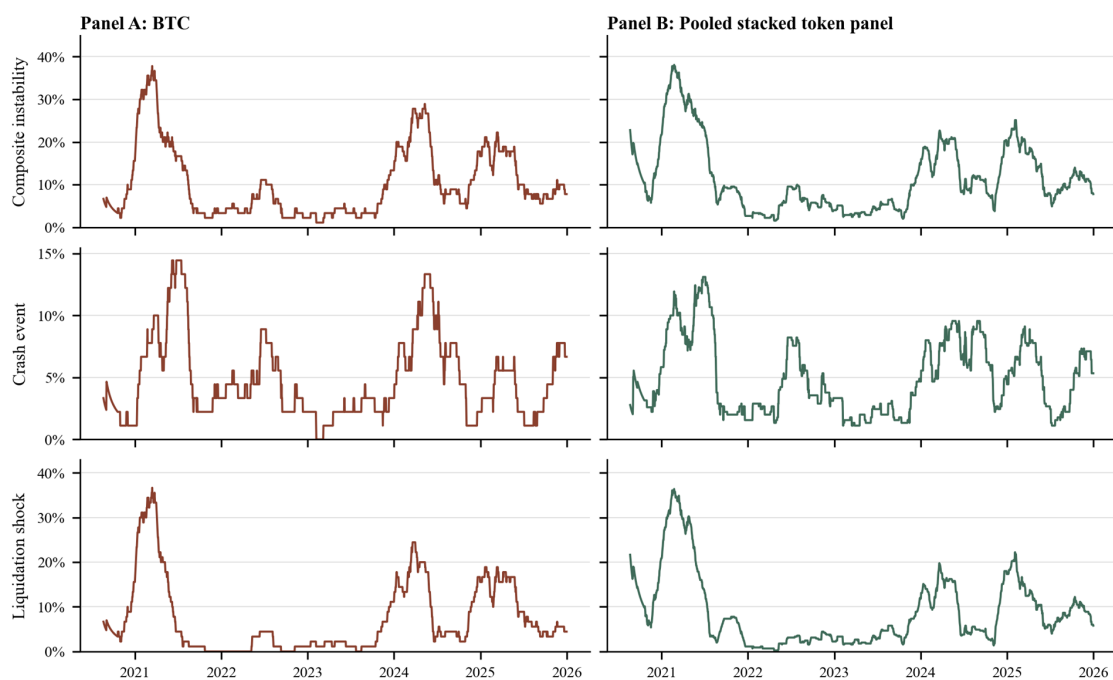


Figure 2: Event Rates

Notes: Panel A reports BTC and Panel B reports the pooled token panel. The figure reports 90-day rolling event rates for visualization only; binary event indicators and forward targets are fitted as 0/1 variables.

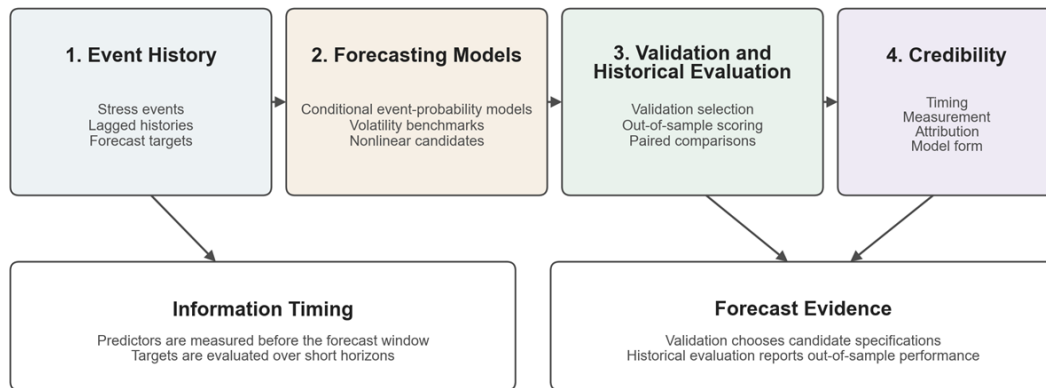


Figure 3: Empirical Framework

Notes: Figure 3 sets out the sequence of empirical design. The steps are event-history construction, model estimation, validation-based specification choice, out-of-sample evaluation, and subsequent credibility checks.

Event-History Monitoring of Cryptocurrency Derivatives Instability

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This online appendix provides the reader-facing data, matched-block bootstrap notation, event-variable diagnostics, parameter and candidate-model tables, feature, model, and preprocessing details, fusion, validation-gate, and calibration details, HC-TEM probability-monitoring evidence, extended forecast and inference evidence, robustness and sensitivity checks, forecast-metric formulas, descriptive attribution diagnostics, and supplementary descriptive and design exhibits needed to read the manuscript independently. References cited here follow the main manuscript's reference list.

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Online Appendix A.1 Data Coverage and Token-Level Raw Descriptives

Table A.1.1 shows that missingness is small within the period covered by each raw series, while distributional shape differs sharply across sources and tokens. Mean missing shares for GDELT news are 0.009 for BNB, BTC, ETH, and XRP and 0.01 for SOL. The corresponding values are zero for every Google-attention row and for most sentiment, liquidity, market, and derivatives rows, with maxima of 0.011 for BTC and ETH derivatives. The observed variables are nevertheless strongly non-Gaussian. Median absolute skewness reaches 12.2 for BTC Google attention and 8.95 for SOL, while median kurtosis reaches 167 and 104. These tails support the rolling within-token thresholds in Section 3.2 rather than a common cutoff applied to raw levels.

Table A.1.2 then identifies the observations available after the one-day composite-instability variables are aligned. BTC and ETH each contribute 1,988 eligible token date observations, followed by XRP with 1,978, BNB with 1,898, and SOL with 1,754, for a pooled total of 9,606. The validation and historical-evaluation partitions contain 365 and 731 observations per token. Training observations range from 658 for SOL to 892 for BTC and ETH because the token series begin on different dates. Longer forecast windows require the complete outcome window to remain inside its partition, so the usable count falls separately by target and horizon. Read with Section 3.1 and Table A.3.2, these counts explain why estimation is target-horizon specific and why comparisons within a cell use the same observed outcomes.

Table A.1.1. Raw-Variable Descriptives and Missingness

Family	Token	Median non-missing daily obser- vations across variables	Mean missing share	Maximum missing share	Median absolute skewness	Median kurtosis
GDELT news	BNB	2,060	0.009	0.009	6.7	156
GDELT news	BTC	2,173	0.009	0.009	1.44	5.36
GDELT news	ETH	2,173	0.009	0.009	1.37	4
GDELT news	SOL	1,916	0.01	0.01	1.67	12.4
GDELT news	XRP	2,139	0.009	0.009	4.09	68
Google attention	BNB	2,079	0	0	4.39	30
Google attention	BTC	2,192	0	0	12.2	167
Google attention	ETH	2,192	0	0	9.69	12
Google attention	SOL	1,935	0	0	8.95	104
Google attention	XRP	2,158	0	0	4.6	33.7
Derivatives	BNB	2,079	0	0	1.43	41.7
Derivatives	BTC	2,171.5	0.01	0.011	2.96	34.9
Derivatives	ETH	2,171.5	0.01	0.011	2.54	38.4
Derivatives	SOL	1,935	0	0.001	6.69	164
Derivatives	XRP	2,158	0	0	1.53	53.8
Fear and greed	BNB	2,078	0	0	0.037	-1.1
Fear and greed	BTC	2,191	0	0	0.023	-1.07
Fear and greed	ETH	2,191	0	0	0.023	-1.07
Fear and greed	SOL	1,934	0.001	0.001	0.05	-1.13
Fear and greed	XRP	2,157	0	0	0.026	-1.1
Liquidity	BNB	2,078.5	0	0	4.04	28.7
Liquidity	BTC	2,191.5	0	0	5.41	66.8
Liquidity	ETH	2,191.5	0	0	5.52	54
Liquidity	SOL	1,934.5	0	0.001	5.67	49.4
Liquidity	XRP	2,157.5	0	0	4.72	36.3
Market	BNB	2,078.5	0	0	3.19	32.6
Market	BTC	2,191.5	0	0	1.9	12.7
Market	ETH	2,191.5	0	0	1.41	10.4
Market	SOL	1,934.5	0	0.001	1.61	8.94
Market	XRP	2,157.5	0	0	3.87	39.5

Notes: The table combines raw-variable descriptives and missingness by token and variable family.

Table A.1.2. Data Coverage by Token

Token	Calendar span	Calendar dates with coverage	Raw token-date observations	Composite $h = 1$ forecast observations	Composite $h = 1$ eligible dates	Training observations	Validation observations	Historical-evaluation observations
BTC	2020-01-01 to 2025-12-31	2,192	2,192	1,988	1,988	892	365	731
ETH	2020-01-01 to 2025-12-31	2,192	2,192	1,988	1,988	892	365	731
BNB	2020-04-23 to 2025-12-31	2,079	2,079	1,898	1,898	802	365	731
SOL	2020-09-14 to 2025-12-31	1,935	1,935	1,754	1,754	658	365	731
XRP	2020-02-04 to 2025-12-31	2,158	2,158	1,978	1,978	882	365	731
Pooled	2020-01-01 to 2025-12-31	2,192	10,556	9,606	1,988	4,126	1,825	3,655

Notes: The table reports calendar coverage and eligible token-date observations by token. The forecast-observation and split columns refer to the one-day composite-instability target. Denominators differ by target and horizon, as Table A.3.2 shows. A complete forward outcome window must remain inside one partition. The split-boundary purge removes 240 observations across all targets and horizons, distributed over 135 target-horizon-token cells in the three splits (45 target-horizon-token combinations $\times$ 3 splits). All cells reconcile. The purge does not reduce the one-day frame of 9,606 observations.

Online Appendix A.2 Matched-Block Bootstrap Notation

Predictions are matched by target, horizon, date, token, and outcome. Inference uses circular date blocks, with 21 days as the primary length and 7, 14, and 42 days as sensitivity values. The primary and block-length comparisons use 50,000 paired draws, while eight secondary comparisons use 2,000 draws and Holm adjustment with complementary Benjamini-Hochberg q-values. Two-sided 95% percentile intervals retain all five tokens for each sampled date. This matched structure is what lets Section 5.6 interpret the block-bootstrap intervals as dependence-aware comparisons on the same dated outcomes. The sensitivity lengths bound inferential precision without selecting a uniquely correct dependence horizon.

The comparator-level bootstrap source schema pairs the target, horizon, selected monitor, comparator, and token-date cells, carries paired average precision (AP) and area under the receiver operating characteristic curve (ROC-AUC) differences, intervals, and p-values, records token win counts and weakest and strongest token differences, documents circular-block draws and the finite-sample rule, and separates one primary comparison from eight secondary comparisons with Holm and BH adjustments.

Online Appendix A.3 Event-Variable Diagnostics

Table A.3.1 reports component overlap, Figure A.3.1 shows token-level event rates, and Table A.3.2 reports forward target rates. Table A.3.3 reports constructed-variable diagnostics, Table A.3.4 reports event balance, and Table A.3.5 reports token-level time-series diagnostics. This section evaluates the measurement properties of the constructed event system. The remaining exhibits report component overlap, token-level event rates, forward target rates, event balance, and the persistence and stationarity of event histories. These diagnostics show where the pooled design is constrained and how the dated inputs behave, but they do not establish a causal mechanism or forecast gain. Taken together, the exhibits support the paper’s measurement layer by showing that histories retain dated clustering under the stated information clock. They do not by themselves support the ranking or probability claims in Section 5.

Event construction follows the notation in Section 3.2. For each token i and date t , the rolling quantile is estimated from the prior 365 calendar days once 180 observations are available. C marks a crash event from the lower return tail, L marks a liquidation shock from the upper liquidation-volume tail, and composite instability records either event. Derivatives-positioning pressure, liquidity contraction, negative news, and search attention use the corresponding one-sided thresholds and side conditions in Equations (5) to (12).

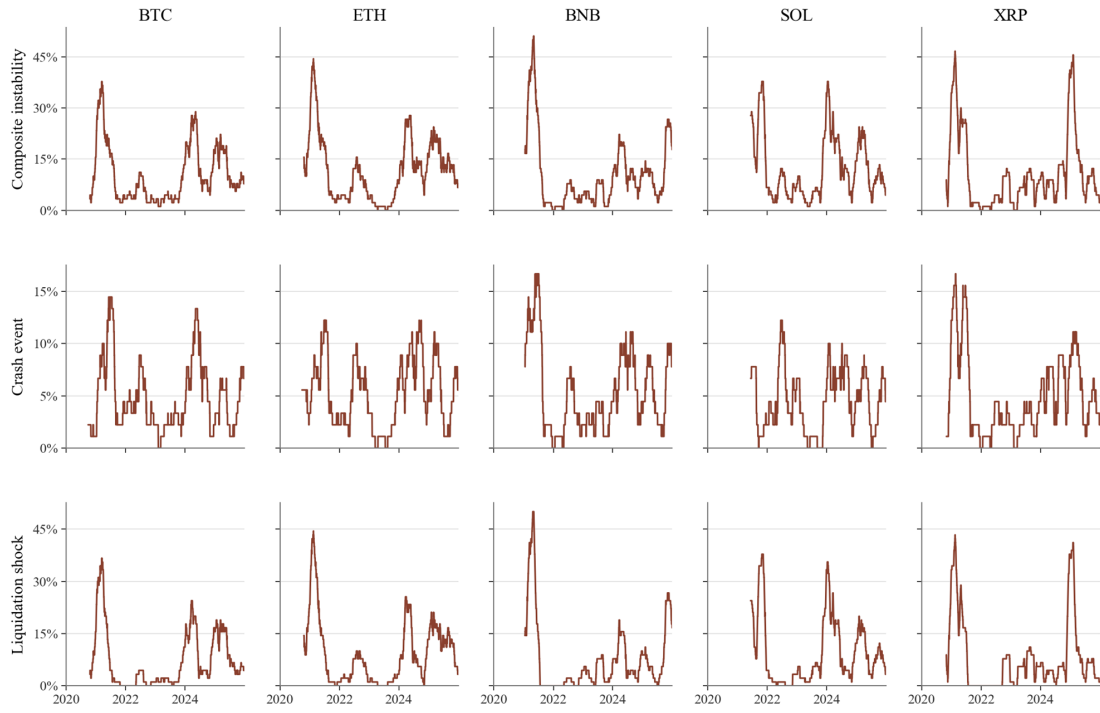
Each binary event is converted into event histories H using one-, three-, and seven-day half-lives. Joint histories J multiply three-day channel histories, and cross-token summaries X average the prior-day indicators of the other observed tokens while excluding the focal token. The forward target Y records whether event q occurs in the h -day window beginning at t , whereas every predictor ends at $t - 1$. This separation makes the constructed histories eligible inputs and prevents the realized outcome window from entering the information set.

Table A.3.1. Composite-Instability Component Overlap by Token and Split

Token	Split	Obs	Crash only	Liquidation only	Joint	Neither	Crash only rate	Liquidation only rate	Joint rate	Composite rate
BNB	All	1,898	53	94	48	1,703	0.028	0.05	0.025	0.103
BNB	Historical evaluation	731	24	39	21	647	0.033	0.053	0.029	0.115
BNB	Train	802	24	45	21	712	0.03	0.056	0.026	0.112
BNB	Validation	365	5	10	6	344	0.014	0.027	0.016	0.058
BTC	All	1,988	60	109	35	1,784	0.03	0.055	0.018	0.103
BTC	Historical evaluation	731	24	56	17	634	0.033	0.077	0.023	0.133
BTC	Train	892	29	41	16	806	0.033	0.046	0.018	0.096
BTC	Validation	365	7	12	2	344	0.019	0.033	0.005	0.058
ETH	All	1,988	53	134	51	1,750	0.027	0.067	0.026	0.12
ETH	Historical evaluation	731	23	65	24	619	0.031	0.089	0.033	0.153
ETH	Train	892	24	66	25	777	0.027	0.074	0.028	0.129
ETH	Validation	365	6	3	2	354	0.016	0.008	0.005	0.03
SOL	All	1,754	49	140	28	1,537	0.028	0.08	0.016	0.124
SOL	Historical evaluation	731	23	51	15	642	0.031	0.07	0.021	0.122
SOL	Train	658	24	55	8	571	0.036	0.084	0.012	0.132
SOL	Validation	365	2	34	5	324	0.005	0.093	0.014	0.112
XRP	All	1,978	55	117	39	1,767	0.028	0.059	0.02	0.107
XRP	Historical evaluation	731	28	49	14	640	0.038	0.067	0.019	0.124
XRP	Train	882	22	53	20	787	0.025	0.06	0.023	0.108
XRP	Validation	365	5	15	5	340	0.014	0.041	0.014	0.068

Notes: The table reports the mutually exclusive crash-only, liquidation-only, joint, and neither states inside one-day composite instability by token and split. This one-day contemporaneous decomposition supplies the liquidation-involvement share reported with Equation (4) in Section 3.2. Composite instability is therefore a descriptive union, not a distinct causal mechanism. Across the pooled one-day frame, 270 observations are crash only, 201 are joint, 594 are liquidation only, and 8,541 are neither, so liquidation-involved outcomes account for 74.6% of the 1,065 composite events.

Figure A.3.1. Token-Level Event-Rate Panels



Notes: The figure displays token-level event-rate patterns for constructed event variables.

Table A.3.2. Forward Target Rates by Horizon

Target	Horizon	Split	Observations	Events	Event rate
Composite instability	1	Historical evaluation	3,655	473	0.129
Composite instability	1	Train	4,126	473	0.115
Composite instability	1	Validation	1,825	119	0.065
Composite instability	3	Historical evaluation	3,645	1,008	0.277
Composite instability	3	Train	4,116	977	0.237
Composite instability	3	Validation	1,815	271	0.149
Composite instability	7	Historical evaluation	3,625	1,717	0.474
Composite instability	7	Train	4,096	1,595	0.389
Composite instability	7	Validation	1,795	497	0.277
Crash event	1	Historical evaluation	3,655	213	0.058
Crash event	1	Train	4,171	213	0.051
Crash event	1	Validation	1,825	45	0.025
Crash event	3	Historical evaluation	3,645	574	0.157
Crash event	3	Train	4,161	560	0.135
Crash event	3	Validation	1,815	130	0.072
Crash event	7	Historical evaluation	3,625	1,135	0.313
Crash event	7	Train	4,141	1,087	0.262
Crash event	7	Validation	1,795	265	0.148
Liquidation shock	1	Historical evaluation	3,655	351	0.096
Liquidation shock	1	Train	4,128	350	0.085
Liquidation shock	1	Validation	1,825	94	0.052
Liquidation shock	3	Historical evaluation	3,645	764	0.210
Liquidation shock	3	Train	4,118	671	0.163
Liquidation shock	3	Validation	1,815	203	0.112
Liquidation shock	7	Historical evaluation	3,625	1,327	0.366
Liquidation shock	7	Train	4,098	1,038	0.253
Liquidation shock	7	Validation	1,795	374	0.208

Notes: The table reports forward-target event rates by target, horizon, and split using target-horizon-specific denominators. At $h = 1$, denominators differ across targets because their constituent inputs have different availability. At longer horizons, forecast observations must also retain a complete forward outcome window inside the named partition. Historical evaluation denotes the 2024–2025 period.

Table A.3.3. Constructed-Variable Diagnostics

Construct family	Token	Mean nonzero share	Mean missing share	Median mean	Median standard deviation	Median skewness	Median kurtosis	LB 1% share	ADF 1% share
Cross-token summary	BNB	0.254	0	0.105	0.225	2.29	4.83	1	1
Cross-token summary	BTC	0.255	0	0.104	0.216	2.32	5.13	1	1
Cross-token summary	ETH	0.254	0	0.098	0.206	2.38	5.39	1	1
Cross-token summary	SOL	0.228	0	0.096	0.196	2.26	5.1	1	1
Cross-token summary	XRP	0.238	0	0.101	0.217	2.31	4.76	1	1
Event history	BNB	0.978	0	0.246	0.481	2.43	6.25	1	0.958
Event history	BTC	0.991	0	0.286	0.484	2.25	5.05	1	0.958
Event history	ETH	0.979	0	0.321	0.497	2.08	4.05	1	1
Event history	SOL	0.965	0	0.305	0.515	2.22	4.98	1	1
Event history	XRP	0.990	0	0.272	0.474	2.34	5.74	1	0.958
Interaction history	BNB	0.952	0	0.022	0.183	11.5	159	1	1
Interaction history	BTC	0.982	0	0.028	0.129	8.46	91.6	1	1
Interaction history	ETH	0.875	0	0.032	0.174	9.54	110	1	1
Interaction history	SOL	0.860	0	0.127	0.667	11.1	132	1	1
Interaction history	XRP	0.968	0	0.022	0.148	9.23	111	1	1

Notes: The table combines constructed-variable coverage, distribution, persistence, and stationarity diagnostics by token and construct family. Ljung–Box and ADF results are descriptive time-series properties only and are not construct-validation evidence.

The evidence in Table A.3.3 tests whether the constructed histories behave like persistent clustered states rather than smooth levels. Median event-history means range from 0.246 for BNB to 0.321 for ETH, while median standard deviations range from 0.474 for XRP to 0.515 for SOL. Every event-history panel has a Ljung–Box 1% share of 1. Interaction histories are rarer and much more heavy-tailed. Their median means range from 0.022 for BNB and XRP to 0.127 for SOL, and median kurtosis ranges from 91.6 for BTC to 160 for BNB. The ADF 1% share is 0.958 for the BNB, BTC, and XRP event histories and 1 for ETH and SOL. This pattern supports treating the histories as persistent but stationary predictors. It does not validate the constructions or show that any history improves a forecast. This is the diagnostic reason the forecasting design retains decayed histories while keeping construct validity separate from performance evaluation in Section 5.

Table A.3.4. Event Balance by Split

Event	Split	Observations	Events	Missing	Event rate
Composite instability	Historical evaluation	3,655	473	0	0.129
Composite instability	Train	4,126	473	0	0.115
Composite instability	Validation	1,825	119	0	0.065
Crash event	Historical evaluation	3,655	213	0	0.058
Crash event	Train	4,125	213	1	0.052
Crash event	Validation	1,825	45	0	0.025
Derivatives positioning pressure	Historical evaluation	3,655	391	0	0.107
Derivatives positioning pressure	Train	4,125	578	1	0.140
Derivatives positioning pressure	Validation	1,825	248	0	0.136
Liquidation shock	Historical evaluation	3,655	351	0	0.096
Liquidation shock	Train	4,126	350	0	0.085
Liquidation shock	Validation	1,825	94	0	0.052
Liquidity contraction	Historical evaluation	3,655	236	0	0.065
Liquidity contraction	Train	4,125	447	1	0.108
Liquidity contraction	Validation	1,825	142	0	0.078
Negative news	Historical evaluation	3,570	347	85	0.097
Negative news	Train	4,123	386	3	0.094
Negative news	Validation	1,820	47	5	0.026
Search attention	Historical evaluation	3,655	610	0	0.167
Search attention	Train	4,126	682	0	0.165
Search attention	Validation	1,825	124	0	0.068

Notes: The table reports contemporaneous constructed-event balance by sample split; rows retain each event variable's own observed denominator and missingness. Target- and horizon-specific forward denominators appear in Table A.3.1 and require complete outcome windows. Historical evaluation denotes the 2024–2025 period.

Table A.3.5. Token-Level Time-Series Diagnostic Summary

Token	Share LB $p < 0.05$	Share ADF $p < 0.05$
BNB	1	0.933
BTC	1	0.867
ETH	0.933	0.933
SOL	1	0.933
XRP	0.933	0.933

Notes: The table summarizes descriptive time-series diagnostic evidence by token. The tests do not validate causal constructs.

Online Appendix A.4 Parameter and Candidate-Model Tables

Table A.4.1 reports candidate metrics, Table A.4.2 shows fusion weights, and Table A.4.3 reports validation-gated winners. The source-family field groups candidates by the role defined in Section 4.3. Baseline library covers the fixed-memory event-history logit and the elastic-net dynamic comparator. Pool-weighted convex fusion combines the five linear and event-history forecasts, while nonlinear-augmented fusion adds random forest, extremely randomized trees, and histogram gradient boosting. Nonlinear tree pool contains those three single nonlinear forecasts. Full model definitions appear in Section 4.3. Family-level dispersion in Table A.4.1 shows that the strongest member and the typical member need not move together. That pattern supports a validation gate at the candidate level and cautions against treating a family label as evidence of uniform forecasting strength.

Table A.4.1. Validation-Gated Candidate Metrics Summary

Horizon	Source family	Tokens	Best AP	Median AP	Best ROC-AUC	Best log loss
<i>Panel A. Composite instability</i>						
1	Baseline library	6	0.560	0.329	0.857	0.396
1	Pool-weighted convex fusion	6	0.567	0.328	0.877	0.410
1	Nonlinear-augmented fusion	6	0.593	0.303	0.849	0.227
1	Nonlinear tree pool	6	0.592	0.316	0.853	0.226
3	Baseline library	6	0.606	0.474	0.790	0.478
3	Pool-weighted convex fusion	6	0.606	0.473	0.782	0.514
3	Nonlinear-augmented fusion	6	0.616	0.481	0.800	0.459
3	Nonlinear tree pool	6	0.611	0.455	0.798	0.454
7	Baseline library	6	0.715	0.649	0.775	0.528
7	Pool-weighted convex fusion	6	0.711	0.646	0.771	0.549
7	Nonlinear-augmented fusion	6	0.730	0.650	0.777	0.522
7	Nonlinear tree pool	6	0.755	0.638	0.781	0.522
<i>Panel B. Crash event</i>						
1	Baseline library	6	0.199	0.096	0.822	0.508
1	Pool-weighted convex fusion	6	0.190	0.091	0.830	0.489
1	Nonlinear-augmented fusion	6	0.134	0.065	0.699	0.127
1	Nonlinear tree pool	6	0.142	0.080	0.830	0.126
3	Baseline library	6	0.356	0.232	0.815	0.530
3	Pool-weighted convex fusion	6	0.358	0.215	0.809	0.526
3	Nonlinear-augmented fusion	6	0.311	0.177	0.735	0.290
3	Nonlinear tree pool	6	0.309	0.178	0.799	0.280
7	Baseline library	6	0.570	0.409	0.750	0.570
7	Pool-weighted convex fusion	6	0.540	0.386	0.746	0.567
7	Nonlinear-augmented fusion	6	0.526	0.380	0.757	0.447
7	Nonlinear tree pool	6	0.561	0.370	0.763	0.418
<i>Panel C. Liquidation shock</i>						
1	Baseline library	6	0.581	0.316	0.863	0.301
1	Pool-weighted convex fusion	6	0.593	0.324	0.880	0.317
1	Nonlinear-augmented fusion	6	0.575	0.308	0.877	0.137
1	Nonlinear tree pool	6	0.589	0.317	0.878	0.128
3	Baseline library	6	0.559	0.435	0.796	0.398
3	Pool-weighted convex fusion	6	0.559	0.428	0.802	0.416
3	Nonlinear-augmented fusion	6	0.558	0.424	0.798	0.310
3	Nonlinear tree pool	6	0.554	0.437	0.812	0.289
7	Baseline library	6	0.676	0.577	0.776	0.502
7	Pool-weighted convex fusion	6	0.670	0.567	0.765	0.511
7	Nonlinear-augmented fusion	6	0.703	0.595	0.772	0.481
7	Nonlinear tree pool	6	0.706	0.576	0.770	0.461

Notes: The table reports reader-facing source families and validation metrics. Baseline library contains the fixed-memory event-history logit and dynamic comparator, the two fusion families contain their stated linear or augmented member sets, and nonlinear tree pool contains random forest, extremely randomized trees, and histogram gradient boosting. The token column counts the pooled sample and five token samples.

Table A.4.2. Fusion Weight Summary

Target	Horizon	Active models	Base models	Top base model	Top weight
Composite instability	1	8	8	All members (equal token weights)	0.125
Composite instability	1	5	5	All members (equal token weights)	0.2
Composite instability	3	1	8	Extremely randomized trees (dynamic memory)	1
Composite instability	3	2	5	Fixed-memory event-history logit (B3)	0.974
Composite instability	7	8	8	All members (equal token weights)	0.125
Composite instability	7	5	5	All members (equal token weights)	0.2
Crash event	1	8	8	All members (equal token weights)	0.125
Crash event	1	1	5	Fixed-memory event-history logit (B3)	1
Crash event	3	8	8	All members (equal token weights)	0.125
Crash event	3	1	5	Fixed-memory event-history logit (B3)	1
Crash event	7	8	8	All members (equal token weights)	0.125
Crash event	7	5	5	All members (equal token weights)	0.2
Liquidation shock	1	8	8	All members (equal token weights)	0.125
Liquidation shock	1	5	5	All members (equal token weights)	0.2
Liquidation shock	3	2	8	Histogram gradient boosting (dynamic memory)	0.704
Liquidation shock	3	3	5	Fixed-memory event-history logit (B3)	0.572
Liquidation shock	7	2	8	Extremely randomized trees (dynamic memory)	0.653
Liquidation shock	7	4	5	Elastic-net multi-memory logit	0.481

Notes: The table reports active and base-model counts, the leading base member, and its weight. Weights use early-validation out-of-sample predictions and remain unchanged before the later common selection window.

Weight rows in Table A.4.2 show that the convex label covers both equal-weight pools and solutions dominated by a leading member. This is why the concentration evidence in Tables A.6.1 to A.6.3 is needed before a fusion result is read as diversified support

Table A.4.3. Validation-Gated Winner

Target	h	Source family	Model	Token	N	Events	ROC	AP	Brier score	Log loss
Composite instability	1	Nonlinear-augmented fusion	Nonlinear-augmented fusion	Pooled	700	74	0.795	0.451	0.144	0.462
Composite instability	3	Nonlinear model pool	Random forest (dynamic memory)	Pooled	690	161	0.749	0.542	0.182	0.543
Composite instability	7	Nonlinear-augmented fusion	Nonlinear-augmented fusion	Pooled	670	267	0.799	0.734	0.190	0.564
Crash event	1	Baseline library	Fixed-memory event-history logit (B3)	Pooled	700	27	0.690	0.106	0.212	0.618
Crash event	3	Baseline library	Fixed-memory event-history logit (B3)	Pooled	690	77	0.706	0.261	0.212	0.619
Crash event	7	Baseline library	Fixed-memory event-history logit (B3)	Pooled	670	142	0.723	0.500	0.210	0.611
Liquidation shock	1	Nonlinear-augmented fusion	Nonlinear-augmented fusion	Pooled	700	59	0.796	0.454	0.120	0.395
Liquidation shock	3	Nonlinear model pool	Random forest (dynamic memory)	Pooled	690	123	0.745	0.500	0.153	0.473
Liquidation shock	7	Nonlinear model pool	Random forest (dynamic memory)	Pooled	670	206	0.740	0.628	0.194	0.569

Notes: The table reports validation-selected winners from the later common selection window by target and horizon; AP is the selection metric. Here h is the forecast horizon, and the receiver-operating-characteristic entry denotes ROC-AUC. Interpretation is conditional on the AP-based selection rule; shifted sub-windows, token omissions, and validation resampling assess selection uncertainty rather than imply selection invariance.

Table A.4.3 separates ranking from probability quality in the winner rows. At the one-day horizon, composite instability and liquidation shock have similar ROC-AUC and AP, while their Brier scores and log losses differ. Across horizons, higher AP does not consistently bring lower proper-score loss. This pattern is consistent with treating ROC-AUC and AP as ranking metrics and Brier score and log loss as probability-quality metrics rather than interchangeable summaries (Davis and Goadrich, 2006; Gneiting and Raftery, 2007).

The weaker crash-event rows in Table A.4.3 are supportive evidence for the target-specificity claim, not a contradiction of the monitoring design. The stronger liquidation pattern matches Section 5.1’s liquidation-centered recurrence result, while the metric separation anticipates Section 5.5’s finding that ranking strength does not establish decision-ready probabilities. Because these are validation-window winners chosen by AP, the evidence remains conditional on the selection rule and candidate set.

The candidate universe keeps a common dated information boundary. B0 is validation-prevalence constant-prevalence benchmark, B1 adds lagged market persistence, B2 adds HAR volatility, B3 adds fixed three-day event histories, and B4 allows one-, three-, and seven-day histories through elastic-net selection. B3, B4, the three nonlinear tree forecasts, and the two fusions enter the active validation gate. Attention, graph, transformer,

and raw HC-TEM objects remain secondary comparison objects, and the graph placebo is excluded from selection, calibration, and fusion.

The elastic-net candidate evaluates five penalty values and mixing ratios of 0.1, 0.5, and 0.9 with three-fold cross-validation. The attention model compares maximum lags of 7, 14, 30, and 60 days. Random forest uses 400 trees with depth 8, extremely randomized trees uses 500 trees with depth 10, and histogram gradient boosting uses 250 iterations with 15 terminal leaves. The two fusion objects constrain nonnegative weights to sum to one and require at least two complete member forecasts.

Eligibility is evaluated separately for comparison, calibration, fusion, and active-gate selection. Candidate fitting uses the outer-training sample. The gate then selects the highest validation AP, followed by top-decile hit rate, lower log loss, and model name. Fusion weights use early-validation predictions from 1 January to 7 August 2023, and the completed fusion enters the later common selection window from 14 August to 31 December 2023. Historical evaluation begins only after these choices are fixed.

Online Appendix A.5 Feature, Model, and Preprocessing Details

Section 4.2 defines the conditional event-probability specification, while Section 4.3 and Table 2 define the model classes. This appendix follows that structure by describing the dated predictor block, the regression and nonlinear models, the forecast combinations, the secondary comparison objects, and the preprocessing applied before estimation. Every candidate receives information dated no later than the day before the forecast window. Tuning and model choice use validation data, and historical evaluation begins only after those choices are fixed. The common timing rule allows the analysis to compare model form without changing the information available to each model.

Dated Predictor Block

The predictor block represents the market state available before each forecast window. Lagged returns, realized volatility, HAR volatility summaries, and recent instability provide the common market controls. The block then adds one-, three-, and seven-day event histories, three-day joint histories, focal-token-excluded cross-token summaries, and token indicators. Equation (13) in Section 3.2 defines each event history as an exponentially decayed sum of prior binary events. A short half-life gives more weight to recent stress, while a longer half-life retains a slower-moving record. The current-day event enters only the next update, so the history used for a forecast contains no event from the forecast window.

Each token-event history begins at zero on the first observed token date, with no event mass assigned before the sample. Decay follows elapsed calendar days, so an irregular gap reduces the stored history by the correct amount rather than by one observation. Equation (14) forms joint histories from the three-day derivatives-positioning, liquidity-contraction, negative-news, and search-attention histories. Equation (15) averages the prior-day indicators of the other observed tokens and excludes the token being forecast. A missing deterministic history or cross-token summary is set to zero before the next valid state is written. These rules implement the dated information sets summarized in Section 4.2.

Conditional Event-Probability Logits

The regression candidates use the logistic link in Equations (17), (18), and (21) to convert the dated predictor block into a conditional event probability. The persistence specification uses lagged instability, returns, realized volatility, and HAR volatility terms. The fixed-memory event-history logit adds three-day histories of the three event indicators and their pairwise interactions. This model is retained as the transparent benchmark because its coefficients map directly into changes in conditional log odds, consistent with the event-probability tradition in financial distress forecasting (Shumway, 2001; Duffie et

al., 2007; Campbell et al., 2008; Bharath and Shumway, 2008).

The dynamic multi-memory logit tests whether a fixed three-day representation is too restrictive. It supplies one-, three-, seven-, and fourteen-day histories to an elastic-net logistic regression, which can shrink weak coefficients and retain histories that improve the training objective. The candidate grid uses five penalty values, mixing ratios of 0.1, 0.5, and 0.9, and three-fold cross-validation. Numeric predictors are standardized after training-sample median imputation, and token indicators enter through one-hot encoding. The resulting model is a flexible event-history comparator, not a replacement for the fixed-memory benchmark. Its tuning and evaluation follow the out-of-sample discipline described in Sections 4.2 and 4.4 (Gu et al., 2020; Timmermann, 2018).

Random Forest

Random forest represents nonlinear thresholds and interactions by averaging many decision trees. Each tree is fitted to a bootstrap sample and considers a random subset of predictors when choosing a split. The individual trees can respond to different portions of the feature space, while averaging their event probabilities reduces the variance of any one tree (Breiman, 2001). The implementation uses 400 trees with maximum depth 8. It receives the same lagged market controls, volatility summaries, event histories, joint histories, cross-token summaries, and token indicators as the other full-information candidates. Its role is to test whether nonlinear partitioning improves forecasts after the dated state has already been constructed.

Extremely Randomized Trees

Extremely randomized trees use the same ensemble principle as random forest but introduce more randomization when selecting split points. Candidate splits are drawn at random for the selected features, and the fitted tree probabilities are averaged. This added randomization can lower correlation across trees, although it may also increase approximation bias (Geurts et al., 2006). The model uses 500 trees with maximum depth 10 and the same dated feature block as random forest. It is included to distinguish the value of stronger split randomization from the value of changing the underlying information set.

Histogram Gradient Boosting

Histogram gradient boosting builds an additive probability model in stages. Continuous predictors are grouped into ordered bins, and each shallow tree is fitted to improve the current ensemble under the classification loss. Later trees therefore concentrate on patterns that earlier trees did not capture. The implementation uses 250 boosting iterations and 15 terminal leaves per tree. Like the other tree candidates, it receives the common

dated controls and event-history features without numeric scaling. The model provides a sequential nonlinear alternative to the two averaging ensembles and follows the use of tree-based classification and boosting in forecasting applications (Lessmann et al., 2015; Gu et al., 2020).

Convex Forecast Fusion

The fusion models combine forecast probabilities rather than refitting the underlying feature block. Their weights are nonnegative and sum to one, which keeps the combined probability inside the range generated by eligible members. The weights minimize log loss on early-validation out-of-sample predictions and require at least two complete member forecasts. Once fitted, they remain fixed during the later common selection window and historical evaluation. The pool-weighted fusion combines the eligible regression candidates, while the nonlinear-augmented fusion also admits the three tree forecasts. This construction follows standard forecast-combination practice (Bates and Granger, 1969). Equations (23) and (24) in Section 4.3 define the combination, Table A.4.2 reports the fitted weights, and Tables A.6.1 to A.6.3 describe their concentration.

Attention, Graph, and Transformer Comparisons

The attention comparison uses an ordered window of prior binary events rather than the exponentially decayed state. It learns how strongly each position in that window contributes to the forecast. Candidate maximum lags are 7, 14, 30, and 60 days. A forecast observation enters only when its complete prior window is available, and a missing binary event inside an otherwise eligible window is set to zero. The graph comparison treats tokens as connected nodes and allows a token’s representation to incorporate information from linked tokens. Its adjacency matrix is estimated from training observations, and the graph placebo is excluded from selection, calibration, and fusion. The transformer comparison supplies a trainable sequence mapping of the same ordered history. These models test whether learned sequence or cross-token structure adds value, as distinct from changing the dated inputs, and retain the secondary comparison role stated in Section 4.3 (Kim et al., 2025).

HC-TEM Probability Comparator

HC-TEM applies a pretrained in-context tabular learner to the same 48 dated features used by the full event-history pool. Instead of estimating a separate parametric model for each target-horizon cell, it forms an event probability from a labelled context table and the current query row in one forward pass. Validation selects the context configuration and probability mapping before historical evaluation. Raw HC-TEM remains outside the

AP-based validation gate and is used as a probability-quality and high-precision-alert comparator, so it does not change the identity of the ranking-selected monitor. This role follows the in-context tabular-learning framework of Hollmann et al. (2025). Section 4.3 states the main-text role, while Online Appendix A.7 provides the architecture, context rules, implementation settings, and probability diagnostics.

Preprocessing and Sample Discipline

Preprocessing is fitted separately within the outer-training sample for each target and forecast horizon. Logistic and elastic-net candidates use training-sample median imputation, numeric standardization, and token encoding. Tree candidates use training medians but do not scale numeric predictors. Sequence candidates require a complete prior window, and graph adjacency is estimated only from training observations. The fitted transformations are then applied unchanged to validation and historical-evaluation data. No outcome from either later partition is used to estimate an imputer, scaler, encoding, adjacency matrix, model parameter, or fusion weight. The reported HC-TEM path uses its training context without a separate numeric imputer. These restrictions preserve the chronological split in Section 3.1 and the common information boundary in Sections 4.2 and 4.4.

Online Appendix A.6 Fusion, Validation-Gate, and Calibration Details

This section separates four implementation questions. Tables A.6.1 to A.6.3 report concentration separately for composite instability, crash events, and liquidation shocks. The calibration paragraphs state how validation probabilities are mapped, and Table A.6.4 reports whether combination changes AP.

Fusion remains nested inside validation. Early-validation predictions determine weights, the later common window compares the completed candidates, and historical evaluation begins after both choices are fixed. Realized calibration methods remain in the probability-monitoring section. Separate target tables make the concentration diagnostic part of the target-specific design rather than a pooled claim about fusion in general.

Table A.6.1. Fusion Pool Concentration for Composite Instability

Target	Horizon	Top constituent	Top share	Top-three concentration
Composite instability	1	BTC, ETH, SOL, XRP (tie)	0.225	0.675
Composite instability	1	BTC, ETH, SOL (tie)	0.267	0.801
Composite instability	1	BNB, ETH, SOL, XRP (tie)	0.225	0.675
Composite instability	1	BNB, BTC, SOL, XRP (tie)	0.225	0.675
Composite instability	1	BNB, BTC, ETH, XRP (tie)	0.225	0.675
Composite instability	1	BNB, BTC, ETH, SOL (tie)	0.225	0.675
Composite instability	1	All five tokens (tie)	0.2	0.6
Composite instability	1	Highest event-balanced token	0.243	0.628 to 0.636
Composite instability	1	BNB	0.3	0.65
Composite instability	1	BNB and XRP (tie)	0.3	0.733
Composite instability	1	BTC	0.3	0.65
Composite instability	1	ETH	0.3	0.65
Composite instability	1	SOL	0.3	0.65
Composite instability	1	XRP	0.3	0.65
Composite instability	3	BTC, ETH, SOL, XRP (tie)	0.225	0.675
Composite instability	3	BTC, ETH, SOL (tie)	0.267	0.801
Composite instability	3	BNB, ETH, SOL, XRP (tie)	0.225	0.675
Composite instability	3	BNB, BTC, SOL, XRP (tie)	0.225	0.675
Composite instability	3	BNB, BTC, ETH, XRP (tie)	0.225	0.675
Composite instability	3	BNB, BTC, ETH, SOL (tie)	0.225	0.675
Composite instability	3	All five tokens (tie)	0.2	0.6
Composite instability	3	Highest event-balanced token	0.261	0.648 to 0.658
Composite instability	3	BNB	0.3	0.65
Composite instability	3	BNB and XRP (tie)	0.3	0.733
Composite instability	3	BTC	0.3	0.65
Composite instability	3	ETH	0.3	0.65
Composite instability	3	SOL	0.3	0.65
Composite instability	3	XRP	0.3	0.65
Composite instability	7	BTC, ETH, SOL, XRP (tie)	0.225	0.675
Composite instability	7	BTC, ETH, SOL (tie)	0.267	0.801
Composite instability	7	BNB, ETH, SOL, XRP (tie)	0.225	0.675
Composite instability	7	BNB, BTC, SOL, XRP (tie)	0.225	0.675
Composite instability	7	BNB, BTC, ETH, XRP (tie)	0.225	0.675
Composite instability	7	BNB, BTC, ETH, SOL (tie)	0.225	0.675
Composite instability	7	All five tokens (tie)	0.2	0.6
Composite instability	7	Highest event-balanced token	0.263	0.653 to 0.668
Composite instability	7	BNB	0.3	0.65
Composite instability	7	BNB and XRP (tie)	0.3	0.733
Composite instability	7	BTC	0.3	0.65
Composite instability	7	ETH	0.3	0.65
Composite instability	7	SOL	0.3	0.65
Composite instability	7	XRP	0.3	0.65

Notes: The table reports top constituent, top share, and top-three concentration for composite instability. Event-balanced rows retain the source-bounded concentration range because the underlying summary does not name the top token.

Table A.6.1 shows whether the composite-instability fusion behaves as a diversified pool or as a near-single-constituent forecast. Equal-token rows distribute mass evenly, while the event-balanced and token-specific rows tilt toward a leading constituent without collapsing to one source. This qualifies the pooled composite results because a fusion label alone does not show that several signals contribute independent warning content.

Table A.6.2. Fusion Pool Concentration for Crash Events

Target	Horizon	Top constituent	Top share	Top-three concentration
Crash event	1	BTC, ETH, SOL, XRP (tie)	0.225	0.675
Crash event	1	BTC, ETH, SOL (tie)	0.267	0.801
Crash event	1	BNB, ETH, SOL, XRP (tie)	0.225	0.675
Crash event	1	BNB, BTC, SOL, XRP (tie)	0.225	0.675
Crash event	1	BNB, BTC, ETH, XRP (tie)	0.225	0.675
Crash event	1	BNB, BTC, ETH, SOL (tie)	0.225	0.675
Crash event	1	All five tokens (tie)	0.2	0.6
Crash event	1	Highest event-balanced token	0.229	0.638 to 0.698
Crash event	1	BNB	0.3	0.65
Crash event	1	BNB and XRP (tie)	0.3	0.733
Crash event	1	BTC	0.3	0.65
Crash event	1	ETH	0.3	0.65
Crash event	1	SOL	0.3	0.65
Crash event	1	XRP	0.3	0.65
Crash event	3	BTC, ETH, SOL, XRP (tie)	0.225	0.675
Crash event	3	BTC, ETH, SOL (tie)	0.267	0.801
Crash event	3	BNB, ETH, SOL, XRP (tie)	0.225	0.675
Crash event	3	BNB, BTC, SOL, XRP (tie)	0.225	0.675
Crash event	3	BNB, BTC, ETH, XRP (tie)	0.225	0.675
Crash event	3	BNB, BTC, ETH, SOL (tie)	0.225	0.675
Crash event	3	All five tokens (tie)	0.2	0.6
Crash event	3	Highest event-balanced token	0.234	0.642 to 0.702
Crash event	3	BNB	0.3	0.65
Crash event	3	BNB and XRP (tie)	0.3	0.733
Crash event	3	BTC	0.3	0.65
Crash event	3	ETH	0.3	0.65
Crash event	3	SOL	0.3	0.65
Crash event	3	XRP	0.3	0.65
Crash event	7	BTC, ETH, SOL, XRP (tie)	0.225	0.675
Crash event	7	BTC, ETH, SOL (tie)	0.267	0.801
Crash event	7	BNB, ETH, SOL, XRP (tie)	0.225	0.675
Crash event	7	BNB, BTC, SOL, XRP (tie)	0.225	0.675
Crash event	7	BNB, BTC, ETH, XRP (tie)	0.225	0.675
Crash event	7	BNB, BTC, ETH, SOL (tie)	0.225	0.675
Crash event	7	All five tokens (tie)	0.2	0.6
Crash event	7	Highest event-balanced token	0.236	0.641 to 0.694
Crash event	7	BNB	0.3	0.65
Crash event	7	BNB and XRP (tie)	0.3	0.733
Crash event	7	BTC	0.3	0.65
Crash event	7	ETH	0.3	0.65
Crash event	7	SOL	0.3	0.65
Crash event	7	XRP	0.3	0.65

Notes: The table reports top constituent, top share, and top-three concentration for crash events. Event-balanced rows retain the source-bounded concentration range because the underlying summary does not name the top token.

The crash-event concentration in Table A.6.2 has the highest upper end of the reported top-three ranges among the three targets. That is the clearest design fragility in the exhibit. A crash fusion can inherit the estimation error or missing coverage of a narrow constituent set, so its weak validation metrics in Table A.4.3 are consistent with target-specific signal limits rather than evidence that the full monitoring design fails.

Table A.6.3. Fusion Pool Concentration for Liquidation Shocks

Target	Horizon	Top constituent	Top share	Top-three concentration
Liquidation shock	1	BTC, ETH, SOL, XRP (tie)	0.225	0.675
Liquidation shock	1	BTC, ETH, SOL (tie)	0.267	0.801
Liquidation shock	1	BNB, ETH, SOL, XRP (tie)	0.225	0.675
Liquidation shock	1	BNB, BTC, SOL, XRP (tie)	0.225	0.675
Liquidation shock	1	BNB, BTC, ETH, XRP (tie)	0.225	0.675
Liquidation shock	1	BNB, BTC, ETH, SOL (tie)	0.225	0.675
Liquidation shock	1	All five tokens (tie)	0.2	0.6
Liquidation shock	1	Highest event-balanced token	0.259	0.648 to 0.674
Liquidation shock	1	BNB	0.3	0.65
Liquidation shock	1	BNB and XRP (tie)	0.3	0.733
Liquidation shock	1	BTC	0.3	0.65
Liquidation shock	1	ETH	0.3	0.65
Liquidation shock	1	SOL	0.3	0.65
Liquidation shock	1	XRP	0.3	0.65
Liquidation shock	3	BTC, ETH, SOL, XRP (tie)	0.225	0.675
Liquidation shock	3	BTC, ETH, SOL (tie)	0.267	0.801
Liquidation shock	3	BNB, ETH, SOL, XRP (tie)	0.225	0.675
Liquidation shock	3	BNB, BTC, SOL, XRP (tie)	0.225	0.675
Liquidation shock	3	BNB, BTC, ETH, XRP (tie)	0.225	0.675
Liquidation shock	3	BNB, BTC, ETH, SOL (tie)	0.225	0.675
Liquidation shock	3	All five tokens (tie)	0.2	0.6
Liquidation shock	3	Highest event-balanced token	0.296	0.655 to 0.68
Liquidation shock	3	BNB	0.3	0.65
Liquidation shock	3	BNB and XRP (tie)	0.3	0.733
Liquidation shock	3	BTC	0.3	0.65
Liquidation shock	3	ETH	0.3	0.65
Liquidation shock	3	SOL	0.3	0.65
Liquidation shock	3	XRP	0.3	0.65
Liquidation shock	7	BTC, ETH, SOL, XRP (tie)	0.225	0.675
Liquidation shock	7	BTC, ETH, SOL (tie)	0.267	0.801
Liquidation shock	7	BNB, ETH, SOL, XRP (tie)	0.225	0.675
Liquidation shock	7	BNB, BTC, SOL, XRP (tie)	0.225	0.675
Liquidation shock	7	BNB, BTC, ETH, XRP (tie)	0.225	0.675
Liquidation shock	7	BNB, BTC, ETH, SOL (tie)	0.225	0.675
Liquidation shock	7	All five tokens (tie)	0.2	0.6
Liquidation shock	7	Highest event-balanced token	0.3	0.669 to 0.68
Liquidation shock	7	BNB	0.3	0.65
Liquidation shock	7	BNB and XRP (tie)	0.3	0.733
Liquidation shock	7	BTC	0.3	0.65
Liquidation shock	7	ETH	0.3	0.65
Liquidation shock	7	SOL	0.3	0.65
Liquidation shock	7	XRP	0.3	0.65

Notes: The table reports top constituent, top share, and top-three concentration for liquidation shocks. Event-balanced rows retain the source-bounded concentration range because the underlying summary does not name the top token. The equal-token-weight difference for the primary comparison is +0.022 [+0.01, +0.042], $p = 0.003$, from 2,000 paired draws with 21-day blocks.

Rows in Table A.6.3 combine diffuse equal-token allocations with learned allocations that put more weight on a leading constituent. The equal-token comparison reported in the note shows that the Section 5.1 pooled gain is not only a sample-share result, while the concentration ranges still caution against reading every fusion as diversified evidence. This supports the paper’s liquidation-centered conclusion within the stated five-token design and does not establish uniform token-level transfer.

The calibration set contains native probabilities, Platt scaling, beta calibration, and regularized isotonic regression. Platt, beta, and isotonic candidates are compared through a five-fold stratified split with seed 17, so every validation observation is scored out of fold. The selected nonnative mapping is then refitted on all validation observations and applied without reselection in historical evaluation.

Cells with fewer than 25 events or 25 non-events retain native probabilities because the fitted methods are ineligible. Eligible methods are ranked by validation log loss, and method name resolves an exact tie. The HC-TEM positive-enrichment isotonic candidate remains separate because its validation rows are used for both fitting and comparison and the generic sparse-cell guard is not stated for that path. This exception does not alter the active monitor or the other calibration methods. These rules implement the probability-quality separation discussed in Section 5.5 by fixing the mapping before historical evaluation and by leaving sparse cells native.

Table A.6.4. Fusion Evaluation Summary

Target	Horizon	Model	Delta pooled AP	Delta pooled ROC-AUC	Delta macro token AP	Token AP wins	Decision gate
Composite instability	1	Equal token weights-weight fusion	-0.005	-0.009	-0.006	2	Retain incumbent
Composite instability	1	Pool-weighted convex fusion	-0.005	-0.010	-0.005	2	Retain incumbent
Composite instability	3	Equal token weights-weight fusion	-0.004	-0.010	-0.007	1	Retain incumbent
Composite instability	3	Pool-weighted convex fusion	-0.005	-0.012	-0.009	1	Retain incumbent
Composite instability	7	Equal token weights-weight fusion	0.002	-0.002	-0	4	Support in some tokens
Composite instability	7	Pool-weighted convex fusion	-0	-0.002	-0.001	3	Support in some tokens
Crash event	1	Equal token weights-weight fusion	0.002	-0.003	0	4	Support Equal token weights-weight fusion
Crash event	1	Pool-weighted convex fusion	0.002	-0.003	-0	3	Support in some tokens
Crash event	3	Equal token weights-weight fusion	-0.001	-0.012	-0.003	3	Support in some tokens
Crash event	3	Pool-weighted convex fusion	-0.001	-0.013	-0.004	3	Support in some tokens
Crash event	7	Equal token weights-weight fusion	0.008	0.004	-0.002	4	Support in some tokens
Crash event	7	Pool-weighted convex fusion	-0.006	0.001	-0.009	2	Retain incumbent
Liquidation shock	1	Equal token weights-weight fusion	0.001	-0.010	0.001	2	Support in some tokens
Liquidation shock	1	Pool-weighted convex fusion	0.005	-0.007	0.004	4	Support pool-weighted fusion
Liquidation shock	3	Equal token weights-weight fusion	-0.007	0.004	-0.007	0	Retain incumbent
Liquidation shock	3	Pool-weighted convex fusion	-0.008	0.004	-0.009	0	Retain incumbent
Liquidation shock	7	Equal token weights-weight fusion	-0.017	-0.003	-0.015	0	Retain incumbent
Liquidation shock	7	Pool-weighted convex fusion	-0.016	-0.010	-0.014	0	Retain incumbent

Notes: The table reports compact, descriptive fusion-evaluation evidence using AP.

Table A.6.4 shows small AP changes that change sign across targets and horizons, while token-win patterns also vary. This is supportive evidence for Section 5.3’s conclusion that fusion does not add stable value on top of the dated representation. It does not turn a target-specific improvement into a general fusion claim.

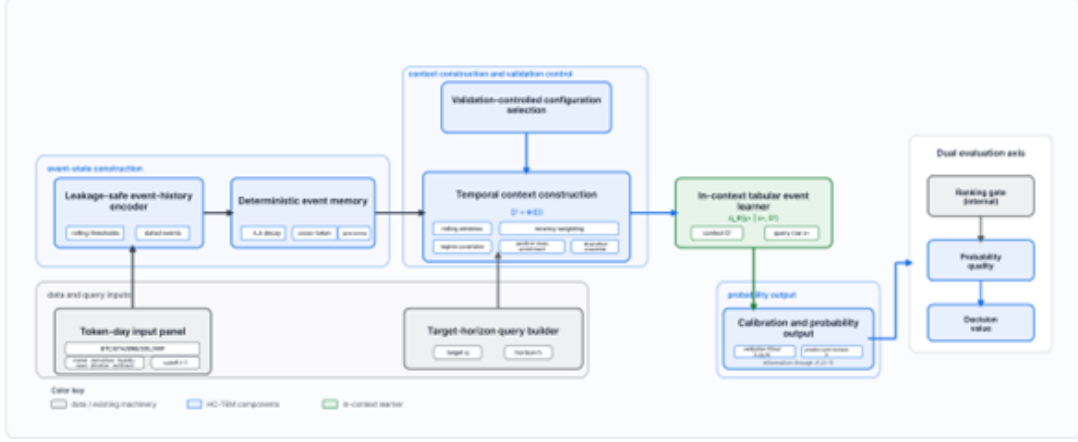
Online Appendix A.7 HC-TEM Probability-Monitoring Evidence

This section specifies equations, selection rules, implementation settings, and architecture for the probability-monitoring evidence. Figure A.7.1 reports the architecture, Table A.7.1 reports parameter settings, Table A.7.2 reports realized calibration methods, Table A.7.3 reports pooled calibration diagnostics, Table A.7.4 reports relative decision value (REV), Table A.7.5 reports episode-level workload, Table A.7.6 reports policy baselines, Figure A.7.2 reports raw reliability diagnostics, Table A.7.7 reports onset and continuation operating characteristics, and Table A.7.8 reports sample alignment. The deployed object is reported as the active monitor, which remains distinct from raw HC-TEM and realizes its fallback in all 9 of 9 target-horizon cells. The exhibit sequence therefore separates architecture, calibration, probability quality, decision value, workload, operating characteristics, and sample alignment rather than treating them as one performance claim.

HC-TEM Architecture and Implementation

Figure A.7.1 shows the HC-TEM architecture for dated recurrent-event data with predictors observed through $t - 1$ and targets beginning at t . The architecture is organized around five designed mechanisms. The event-history encoder represents each token date using predictors observed through $t - 1$. Targets begin at t . A deterministic event-history component summarizes recurrent stress through decayed, cross-token, and joint-stress histories without any trainable recurrence. A temporal context constructor, together with a target-horizon query, assembles the conditioning table supplied to the learner. Only rows satisfying $s + h \leq t$ are admitted, so equality is included. An in-context tabular event learner forms the monitoring probability from that table in a single posterior-predictive forward pass. A validation-controlled stage selects the context configuration and an available calibration method on validation data alone. Figure A.7.1 shows the high-level architecture, with each component described in detail in the paragraphs that follow.

Figure A.7.1. HC-TEM Architecture.



Notes: The panel traces the path from the token date inputs and target-horizon query to the calibrated monitoring probability and its dual evaluation axis. Shading distinguishes data and the existing pipeline (grey), HC-TEM’s designed components (blue), and the adopted in-context tabular learner (green). Only rows satisfying $s + h \leq t$ are admitted, so equality is included.

We formalize the probability object first. For token i on date t , target q , and horizon h , the monitoring probability is

$$\Pr(Y_{i,t}^{q,h} = 1 \mid \mathcal{F}_{t-1}) = p_{i,t}^{q,h} \quad (\text{A1})$$

where Y is the forward event target and F denotes the information set available through $t-1$. The event-history encoder maps predictors observed through $t-1$ into target-horizon probabilities.

$$p_{i,t}^{q,h} = f_{\Theta}(Z_{i,t-1}, Q_{i,t-1}; q, h) \quad (\text{A2})$$

Here Z denotes the lagged token date inputs, Q denotes rolling threshold information, and Θ collects the fixed encoder settings defined in Section 3.2.

The encoder is designed against the failure mode that most damages recurrent-event forecasting, namely information leakage, in which contemporaneous or look-ahead features make out-of-sample performance illusory, a problem repeatedly documented in financial machine learning (López de Prado, 2018; Gu et al., 2020). Admitting only information measurable through $t-1$ keeps every encoded feature measurable with respect to the prior information set, so the conditioning set cannot embed future outcomes and the historical-evaluation-period assessment remains a valid out-of-sample test.

The deterministic event-history component aggregates the decayed event histories, cross-token summaries, and joint-stress interactions constructed in Section 3.2.

This deterministic memory is a deliberate alternative to a trainable neural recurrence. It carries zero estimation variance, spans short- to long-horizon persistence through its half-lives (Corsi, 2009), and avoids the gradient pathologies of trained recurrence by construction. The following discussion gives the design rationale. The temporal context constructor then assembles the rows supplied to the in-context learner as follows.

$$\mathcal{D}_t^{q,h,c} = \Phi_c \left(\left\{ \left(Z_{j,s-1}, Q_{j,s-1}, Y_{j,s}^{q,h} \right) : s + h \leq t \right\} \right) \quad (\text{A3})$$

The operator Φ covers rolling windows, recency weighting, time and regime covariates, positive-class enrichment, and diversified ensemble. It changes the conditioning table, not the target definition or the leakage rule. Under the Equation (16) convention, a label dated s covers the target window s through $s+h-1$. Only rows satisfying $s+h \leq t$ are admitted, so equality is included.

The context operators trade bias for variance in the conditioning set. A regime-shifting panel violates the learner’s exchangeability assumption and rare positives are sparse in any window (He and Garcia, 2009), so windowing, recency weighting, and positive-class enrichment reshape the conditioning table, with the operating point chosen on the validation sample alone.

The in-context tabular event learner is the central probability-forming core described by the following expression.

The map q_θ conditions on the constructed context D and the query row x to produce a posterior-predictive event probability. This is the architectural analogue of a forecasting core. It is the block that forms the probability, while Validation-controlled configuration selection plays the gating role over candidate contexts.

The map in Equation (A4) approximates a Bayesian posterior-predictive integral in a single forward pass with fixed weights (Müller et al., 2022; Hollmann et al., 2025), regularizing sparse cells toward the learned tabular prior without per-cell gradient training or per-cell hyperparameter search. The discussion below gives the estimator-level rationale.

Validation selects the context configuration and calibration method separately as follows.

$$c_{q,h}^* = \arg \min_{c \in \mathcal{G}_{q,h}} \mathcal{L}_{\text{val}}(c) \quad (\text{A5})$$

$$r_{q,h}^* = \arg \min_{r \in \mathcal{R}} \mathcal{L}_{\text{val}} \left(g_r \left(p^{q,h,c_{q,h}^*} \right) \right) \quad (\text{A6})$$

where $\mathcal{G}$ is the finite set of context/configuration candidates, $L_{\text{val}}^{(q,h)}$ denotes the log-loss component of the validation criteria, and g_κ is the method-specific calibration map. Each candidate context is constructed from training observations and compared on the validation year at the target-horizon level. Within each target-horizon-token-prediction-object

cell, non-native maps are fit on validation calibration folds and all methods are compared on out-of-fold validation predictions.

The implementation is as follows. The in-context learner is the tabular foundation model of Hollmann et al. (2025) with its published pretrained weights. The configuration set $\mathcal{G}_{q,h}$ contains nine context constructions spanning rolling context windows, recency-weighted subsampling, regime covariates, positive-class enrichment, and a diversified-window ensemble. The method set R contains native probabilities, Platt scaling, beta calibration, and cross-fitted regularized isotonic calibration. A cell with fewer than 25 positive or 25 negative validation outcomes remains native. The selected HC-TEM configuration and mapping are then fixed before historical evaluation. The candidate-specific positive-enrichment isotonic path was unselected in all nine target-horizon cells, so its same-sample branch lies outside the general cross-fitted calibration statement and no reported quantity depends on it.

Restricting configuration and calibration choices to the validation sample under these validation rules guards against data snooping, since selecting a context configuration on the historical-evaluation sample would overfit the evaluation itself (White, 2000; Gu et al., 2020). Because the reported quantity is a probability that is consumed as a probability, the model is assessed with proper scoring rules and utility-based value, the decision-theoretically appropriate yardsticks (Gneiting and Raftery, 2007; Elliott and Lieli, 2013), separately from the discrimination axis that governs gate selection.

For the ranking-oriented evaluation, the active monitor is the candidate selected by the validation gate. Raw HC-TEM and the active monitor are distinct prediction objects. Raw HC-TEM is evaluated as a secondary probability-quality and high-precision-alert comparator on the same historical-evaluation target-horizon-token-date cells used by the active model pool. The active monitor is reported separately with its realized fallback frequency in the probability-monitoring results and is not relabelled HC-TEM. This section collects the posterior-predictive notation, context-construction details, proper-score definitions, and decision-value equations.

This section separates raw HC-TEM from the active monitor and reports the posterior-predictive and temporal-context notation referenced in Section 4.3. Equation (A4) defines the probability object for target q , horizon h , date t , and context c together with its conditioning set. The query row is measurable before the forecast window. Only context rows whose outcomes are observed by date t are admitted, including rows observed exactly on that date. The validation-fitted calibration map is fixed before historical evaluation. Probability diagnostics, relative decision value under a stylized cost-loss model, onset/continuation operating characteristics, and workload are then reported for their distinct roles.

Decision-Layer Specification

Decision policy. The decision layer converts each one-day liquidation shock score into a date- t review action using information through $t - 1$. The active monitor and incumbent are fixed before historical evaluation. The incumbent is the elastic-net multi-memory logistic model that led on validation AP, trained through 31 December 2022 and compared on the common 14 August to 31 December 2023 validation window of 700 token date observations and 59 events. At $r = 0.05$, the fixed thresholds are 0.222 for the active monitor and 0.232 for the incumbent. A score at or above its threshold triggers next-day exposure or collateral review, not an automatic trade.

Episode construction. Within each token, observations are sorted by target date. An event episode is a maximal run of positive event outcomes on consecutive calendar days. An alert episode is constructed identically from positive review actions. A zero observation or a gap longer than one day closes a run. Token boundaries always close an episode, whereas month and year boundaries do not when dates remain consecutive. The evaluation boundaries truncate open runs. The seven-day rule is not an episode-merging rule, and one observed inactive day separates neighboring episodes.

Matching. Within a token, an event and an alert episode are linked when the alert occurs from seven days before the event begins through the event’s final day. Thus the estimand is a directional seven-day leading-or-overlap window. An event is detected if an alert occurs from seven days before its first event day through its last event day, while an alert is matched if it overlaps an event or ends no more than seven days before the event begins. Links are many-to-many, with no assignment or precedence rule. The active monitor has 277 links covering 208 detected event episodes and 133 matched alert episodes. 56 detected events link to more than one alert and 51 matched alerts link to more than one event.

Denominators and alert units. A token-month is one token crossed with one observed calendar year-month in the eligible historical-evaluation panel. It is calm when its one-day liquidation shock count is zero. No additional completeness screen is applied. The panel contains 120 observed token-months and 23 calm token-months. Dividing 141 unmatched active-monitor alert episodes by the 23 calm token-months gives 6.13 false alarms per calm token-month. The corresponding 117 incumbent episodes give 5.09. An operational alert day is one alerted token date, so simultaneous alerts for several tokens count separately. **Cost and reference.** Review is a binary action and the realized one-day liquidation shock is a binary outcome. The normalized cost charges r whenever review is undertaken and a unit loss only when an event occurs without review. The remaining action-outcome combination carries zero cost. With realized historical-evaluation prevalence denoted by $\bar{p}$, the reference cost is the lesser of review cost and prevalence, corresponding to the better of always review and never review. The exact-review policy reviews exactly the event dates and sets the upper bound of the relative-decision-value scale, so its cost is

review cost multiplied by prevalence. Relative decision value divides the reference-to-policy cost reduction by the reference-to-exact-review cost reduction. This is realized-sample constant-prevalence benchmark, distinct from a validation-prevalence policy.

Threshold schedule. For each policy and each displayed r , validation searches every distinct validation score and the next floating-point value above the maximum, which represents never review. A score at or above the candidate threshold triggers review and is scored by mean validation cost. A tie reports the largest threshold, and the selected threshold is fixed for historical evaluation. The $r = 0.05$ workload threshold is not reused at other ratios.

Paired resampling. Two-sided 95% percentile intervals use 2,000 paired circular date-block draws of length 21 over 731 sorted unique historical-evaluation dates. Each draw samples 35 starting positions with replacement, keeps all five tokens together, concatenates wrapped 21-position blocks, and truncates to 731 positions. The same positions are applied to outcomes and both policies. Thresholds remain fixed. Episode statistics are reconstructed on a consecutive synthetic daily calendar beginning 1 January 2024, including a draw-specific calm token-month denominator. Relative decision value recomputes draw-specific prevalence, reference cost, and exact-review cost. All intervals use 2,000 valid draws and zero discarded draws. Paired differences use the two-sided finite-sample p-value with the plus-one correction.

Table A.7.1 links the candidate settings to the context selected in each target-horizon cell. The selected context varies across targets and horizons, so the exhibit does not support a single universally preferred context. Instead, it supports the validation-controlled configuration step and the target-specific reading of the monitoring evidence.

Table A.7.1. HC-TEM Parameter Settings

Panel A. Candidate parameter set			
Component	Parameter	Candidate set	
In-context learner	Model and weights	TabPFN tabular foundation model (Hollmann et al., 2025), published pretrained weights	
In-context learner	Ensemble size (full-context configuration)	8	
Temporal context	Rolling context windows	180, 365, 730 days	
Temporal context	Recency-weighted subsampling half-lives	90, 180 days	
Temporal context	Context size cap	1,000 rows	
Temporal context	Regime covariates	Time-as-covariate features	
Temporal context	Class treatment	Positive-class enrichment	
Temporal context	Ensemble variant	Diversified-window ensemble	
Selection	Rule	Validation AP, with ties resolved by top-decile hit rate and then log loss; one configuration per target-horizon cell	
Calibration	Map and fitting sample	Native, Platt, beta, or cross-fitted regularized isotonic method; one validation-year method per target-horizon-token-prediction-object cell; sparse cells remain native	
Panel B. Selected configuration by target-horizon cell			
Target	1-day	3-day	7-day
Composite instability	Full context	Regime covariates	Full context
Crash event	Recency, 90-day half-life	Full context	Full context
Liquidation shock	Regime covariates	Regime covariates	Recency, 180-day half-life

Notes: Panel A reports the HC-TEM candidate parameter set; Panel B reports the context configuration selected on the validation year for each target-horizon cell. Realized probability-calibration methods are reported separately in Table A.7.2. Parameter tables for other pool members are in Online Appendix A.4.

Table A.7.2 reports the selected calibration methods directly. Validation log loss selects among native probabilities, Platt scaling, beta calibration, and cross-fitted regularized isotonic calibration (Platt, 1999; Niculescu-Mizil and Caruana, 2005; Kull et al., 2017). Sparse cells with fewer than 25 positive or 25 negative validation outcomes remain native.

The method counts and diagnostic coverage are reported together so calibration choice is not confused with the later historical-evaluation diagnostic. Method heterogeneity across the reported cells supports cell-specific calibration checks, while the separate held-out diagnostic prevents method selection from being mistaken for probability-quality evidence.

Table A.7.2. Realized Calibration Methods and Diagnostic Coverage

Diagnostic component	Realized count	Coverage	Interpretation
Native probabilities	141	360 calibration cells	Selected by validation log loss
Platt scaling	80	360 calibration cells	Selected by validation log loss
Beta calibration	57	360 calibration cells	Selected by validation log loss
Cross-fit isotonic	82	360 calibration cells	Selected by validation log loss
Diagnostic summaries	36	$2 \text{ objects} \times 3 \text{ targets} \times$ (pooled + 5 token samples) $= 36$	Intercept, slope, Brier decomposition
Fixed 10-bin reliability rows	360	$2 \text{ objects} \times 18 \text{ groups} \times 10$ bins	Empty bins reported under the fixed 10-bin rule

Notes: Method counts are validation selections across 360 target-horizon-token-prediction-object cells. The one-day diagnostic is a separate held-out historical-evaluation assessment of raw HC-TEM and the active monitor; it does not replace or reselect predictions. The 360 calibration cells (method selection, eight objects, all horizons) and the 360 reliability-bin rows (two objects, one-day horizon) are distinct counts that coincide numerically.

Pooled diagnostics in Table A.7.3 show that ranking and probability quality remain separate objects. Raw HC-TEM has a lower Brier score and smaller reliability error than the active monitor for each one-day target, while neither object's slope and intercept pattern supports a universal calibration claim. The table therefore supplies the probability-quality evidence used in Section 5.5 without reselecting the AP winner.

Table A.7.3. One-Day Pooled Calibration Diagnostics

Object	Target	h	N	Intercept [95% interval]	Slope [95% interval]	Brier	Reliability (10 bins)	Resolution (10 bins)	Uncertainty
Raw HC-TEM	Composite	1	3,655	-0.445 [-0.711, -0.179]	0.626 [0.511, 0.748]	0.101	0.002	0.011	0.113
Raw HC-TEM	Crash	1	3,655	-1.97 [-2.77, -1.31]	0.278 [0.047, 0.493]	0.058	0.003	0	0.055
Raw HC-TEM	Liquidation	1	3,655	-0.254 [-0.596, 0.046]	0.745 [0.634, 0.864]	0.073	0.001	0.012	0.087
Active monitor	Composite	1	3,655	-1.35 [-1.52, -1.2]	0.854 [0.717, 1.03]	0.142	0.042	0.011	0.113
Active monitor	Crash	1	3,655	-2.72 [-3.02, -2.51]	0.722 [0.425, 1.03]	0.220	0.166	0.001	0.055
Active monitor	Liquidation	1	3,655	-1.43 [-1.61, -1.23]	0.965 [0.831, 1.1]	0.107	0.033	0.011	0.087

Notes: The table reports pooled rows for both prediction objects. Fixed ten-bin reliability and resolution components are reported alongside outcome uncertainty. The active monitor's one-day Brier score exceeds the constant-prevalence reference for composite instability (0.142 versus 0.113), crash events (0.22 versus 0.055), and liquidation shocks (0.107 versus 0.087); its output is therefore not established as a decision-ready probability. Both prediction objects cover 9 cells.

Relative decision value in Table A.7.4 divides the reference-to-policy cost reduction by the reference-to-exact-review cost reduction, as specified above. The reference is the less expensive constant action under realized historical-evaluation prevalence, and the exact-review policy reviews exactly the event dates. Cost follows the review-cost and missed-event-loss definition above. Each policy searches all distinct validation scores plus never review at each r . Equal costs select the largest threshold, which is fixed for historical evaluation. The certainty-equivalent-return check fails under the target-compatible payoff and is not used as a decision metric or return result. No policy dominates throughout the displayed cost range, so Table A.7.4 supports a cost-conditional review interpretation rather than an unrestricted deployment claim.

Table A.7.4. Relative Decision Value Under the Stylized Cost-Loss Model

r	Active REV	[95% interval]	Incumbent REV	[95% interval]	Active – incumbent	[95% interval]	Active threshold	Active alerts (rate)
0.02	0.038	[-0.073, 0.131]	-0.321	[-0.660, -0.012]	+0.360	[+0.109, +0.644]	0.130	3,179 (87%)
0.05	0.227	[0.097, 0.362]	0.157	[-0.006, 0.319]	+0.070	[-0.009, +0.171]	0.222	1,906 (52.1%)
0.06	0.267	[0.112, 0.418]	0.225	[0.083, 0.366]	+0.042	[-0.021, +0.098]	0.284	1,356 (37.1%)
0.10	0.350	[0.189, 0.432]	0.353	[0.194, 0.429]	-0.003	[-0.029, +0.023]	0.536	414 (11.3%)
0.14	0.312	[0.167, 0.425]	0.212	[0.082, 0.320]	+0.099	[+0.052, +0.140]	0.536	414 (11.3%)
0.20	0.212	[0.076, 0.320]	0.193	[0.070, 0.294]	+0.019	[-0.005, +0.040]	0.662	227 (6.2%)
0.30	0.150	[0.031, 0.247]	0.140	[0.031, 0.233]	+0.010	[-0.024, +0.042]	0.662	227 (6.2%)
0.40	0.097	[0.021, 0.164]	0.096	[0, 0.185]	+0.001	[-0.036, +0.042]	0.827	79 (2.2%)
0.50	0.071	[0.010, 0.129]	0.034	[-0.061, 0.124]	+0.037	[-0.016, +0.095]	0.827	79 (2.2%)

Notes: The table reports the nine cost-loss ratios 0.02, 0.05, 0.06, 0.1, 0.14, 0.2, 0.3, 0.4, and 0.5 of the grid in Manuscript Table 7. The incumbent is the elastic-net multi-memory logistic model that led on validation AP, fixed before the historical evaluation. The $r = 0.05$ row is the reported operating point, and its intervals follow the same paired design as every other row. All intervals use 2,000 circular 21-day date-block draws.

Table A.7.5 reports episode-level workload under many-to-many and one-to-one matching. Episode-level workload uses a directional seven-day leading-or-overlap estimand with many-to-many links and no precedence assignment. An event is detected by an alert from seven days before its start through its end. An alert is matched if it overlaps an event or ends no more than seven days before the event begins. Under the seven-day matching rule, the median lead from first matched alert start to episode start is 11 days for both policies. The historical evaluation has 120 observed token-months and 23 calm token-months. Among unmatched episodes for the active monitor and incumbent, respectively, 42 and 34 begin inside one of the 23 zero-event token-months, and 44 and 36 do so when any overlap counts. The active monitor detects more event episodes and carries about 0.2 additional false alarms per observed token-month. Always-on, never-on, and persistence baselines are evaluated below with nondegenerate alert-day metrics. The episode comparison tests whether higher event recall comes with a manageable review burden. The active monitor detects 208 of 220 event episodes, compared with 194 for the incumbent, and raises recall from 0.882 to 0.945. The paired increase is 0.064 with an interval from 0.019 to 0.093. That gain requires 274 rather than 229 alert episodes and 141 rather than 117 false alarms. Under the same many-to-many rule, 133 alert episodes are matched for the active monitor and 112 for the incumbent. The paired differences are +14 detected episodes, +45 alert episodes, and +21 matched episodes. False alarms per observed token-month rise from 0.98 to 1.18, a paired increase of 0.2 with an interval from 0.033 to 0.35, while precision remains 0.485 versus 0.489 and its paired change spans a decline of 0.057 to an improvement of 0.035. The monitor therefore buys recall with heavier workload

rather than better episode precision. The table cannot establish a net economic benefit outside the stated review rule. Panel B repeats the comparison with one-to-one matching. Detection falls for both policies because long alert episodes span several events, the recall advantage narrows to ten episodes and is no longer distinguishable from zero, and the heavier false-alarm load remains at 35 additional unmatched episodes. The workload conclusion is unchanged, and the size of the recall advantage depends on the matching rule. This directly supplies the workload trade-off summarized in Section 5.5. The resampled intervals concatenate circular date blocks into a synthetic calendar, so block seams can create artificial adjacency between unrelated days. We therefore measure that effect. Excluding seam-spanning episodes lowers detected-event and false-alarm episode counts by less than four percent for both policies and leaves the direction of every paired difference unchanged.

Table A.7.5. Episode-Level Alert Workload and Paired Uncertainty

<i>Panel A. Seven-day leading-or-overlap matching</i>					
Policy role	Event episodes	False alarms	Recall [95%]	Precision [95%]	Alarm rate
Active monitor	220	141	0.945 [0.915, 0.985]	0.485 [0.422, 0.614]	1.18 [0.908, 1.54]
Incumbent	220	117	0.882 [0.845, 0.95]	0.489 [0.434, 0.623]	0.975 [0.758, 1.28]
Active – incumbent	Not applicable	+24	+0.064 [+0.019, +0.093]	-0.004 [-0.057, +0.035]	+0.2 [+0.033, +0.35]
<i>Panel B. One-to-one matching</i>					
Policy role	Detected events	False alarms	Recall	Precision	Alarm rate
Active monitor	103	171	0.468	0.376	1.43
Incumbent	93	136	0.423	0.406	1.13
Active – incumbent	+10 [-4, +20]	+35 [+11, +54]	+0.045 [-0.017, +0.096]	-0.03 [-0.072, +0.008]	+0.292 [+0.091, +0.45]

Notes: Alarm rate is false-alarm episodes per observed token-month, with 141 for the active monitor and 117 for the incumbent across all 120 observed token-months. The paired p -values for Panel A recall, precision, and alarm rate are 0.004, 0.62, and 0.023. In Panel B, maximum-cardinality and first-alert assignment give identical one-to-one assignments, detected events equal matched alerts, and difference-row intervals are paired. The paired p -values are 0.182 for detection and recall, 0.129 for precision, and 0.003 for false alarms. Excluding seam-spanning episodes lowers episode counts by less than four percent for both policies and leaves the direction of every paired difference unchanged.

The policy comparison in Table A.7.6 asks whether the active monitor adds value without inheriting the mechanical recall of an always-on rule. At the reported cost-loss ratio, the active monitor has an alert rate of 0.521, balanced accuracy of 0.686, and relative decision value of 0.227, compared with 0.486, 0.677, and 0.157 for the incumbent. Always-on reaches recall of 1 but balanced accuracy of 0.5 and relative decision value approximately zero. Simple persistence is more selective, with precision of 0.373 and specificity of 0.933,

but its relative decision value is negative by 0.332. The active monitor’s gain is therefore not an artifact of permanent alerting, yet it still generates 274 alert episodes and 15.9 alert days per observed token-month, versus 229 and 14.8 for the incumbent. The result supports a bounded review interpretation, not unrestricted deployment value. Balanced accuracy and relative decision value are therefore read beside recall and workload, which prevents the higher-detection result from being mistaken for automatic policy dominance.

Table A.7.6. Alert Intensity and Nondegenerate Policy Baselines

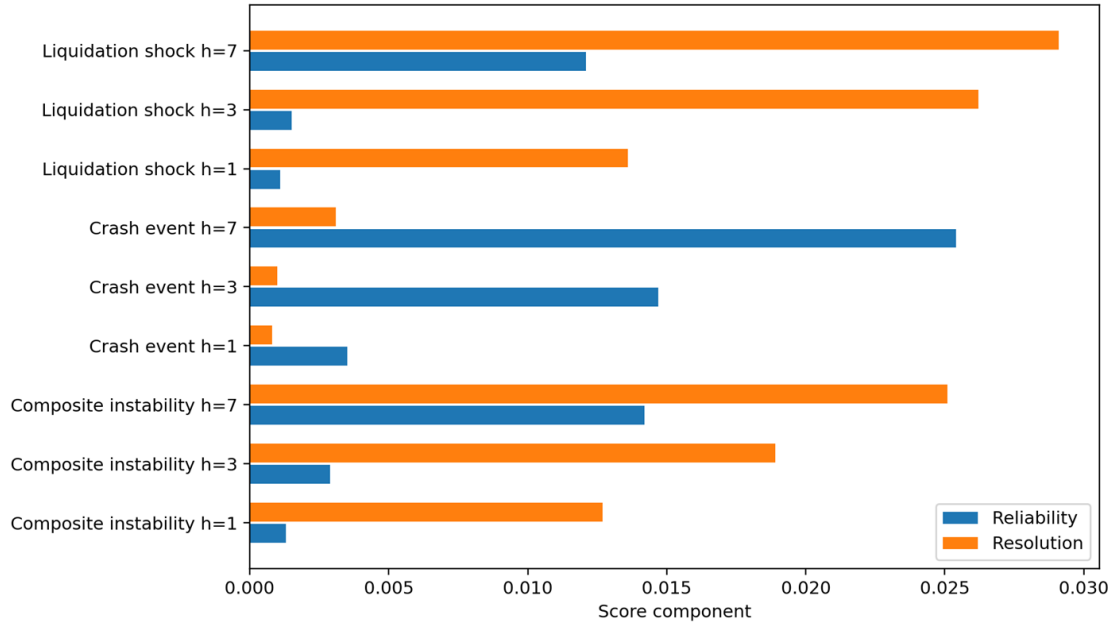
Policy	Alert days	Alert rate	Alert-day precision	Alert-day recall	Specificity	Balanced accuracy	REV at $r = 0.05$
Active monitor	1,906	0.521	0.158	0.858	0.514	0.686	0.227
Incumbent	1,777	0.486	0.159	0.806	0.548	0.677	0.157
Always-on	3,655	1	0.096	1	0	0.5	0
Never-on	0	0	0	0	1	0.5	-1.02
Simple persistence	351	0.096	0.373	0.373	0.933	0.653	-0.332

Panel B. Operational alert intensity at $r = 0.05$

Policy	Alert episodes	Mean duration	Median duration	P90 duration	P95 duration	Maximum duration	Alert days/observed token-month
Active monitor	274	6.96	2	15.4	37.4	120	15.9
Incumbent	229	7.76	2	17.2	39	112	14.8

Notes: Always-on and never-on have balanced accuracy 0.5. Simple persistence reviews date t when the same token’s date- $t-1$ liquidation shock outcome is one, carrying the terminal validation state into the first historical-evaluation date. Maximum concurrency is five of five tokens for both threshold policies.

Figure A.7.2. Raw HC-TEM Reliability Diagnostics.



Notes: The figure visualizes the raw HC-TEM reliability diagnostic. Table A.7.3 reports pooled diagnostics for both raw HC-TEM and the active monitor; Table A.7.2 reports the complete 360 fixed 10-bin reliability rows.

Reliability patterns in Figure A.7.2 make the probability-quality separation visible. Raw HC-TEM is evaluated as a diagnostic beside the active monitor, so the plot supports Section 5.5’s calibration discussion without altering the AP-based winner.

Table A.7.7 asks whether raw HC-TEM alerts provide usable onset timing without excessive false alarms. A median lead is estimable only for BTC and XRP, and both values are 7 days, exactly at the quiet-period bound. The BNB, ETH, and SOL medians are not estimable. First alarms range from 7 for SOL to 12 for ETH, against onset-episode counts from 18 for XRP to 30 for ETH, while false alarms per calm period range from 0.079 for BNB to 0.233 for XRP. The cross-token pattern therefore does not establish general onset timing and remains within a continuation-or-recurrence interpretation. Because first alarms are not necessarily matched onset alerts, the table cannot support an onset-superiority claim. This is the operating-characteristic counterpart to Section 5.1’s continuation finding and weighs against an onset-superiority interpretation.

Table A.7.7. Raw HC-TEM Onset and Continuation Operating Characteristics

Token	Quiet period	Onset episodes	Continuation episodes	First alarms	Median lead	False alarms	Calm periods	False alarms/calm period
BNB	7	21	39	10	Not estimable	3	38	0.079
BTC	7	26	47	11	7	8	51	0.157
ETH	7	30	59	12	Not estimable	12	58	0.207
SOL	7	24	42	7	Not estimable	6	48	0.125
XRP	7	18	45	10	7	7	30	0.233

Notes: The table reports the one-day liquidation-shock raw-HC-TEM alert profile at the reported high-precision operating point (fixed probability threshold 0.5), with a seven-day quiet period. A first alarm is the first alert after a quiet period, whether or not it falls within an onset's matching window; a matched onset-alert pair is required to estimate lead time. First alarms are reported descriptively and are not evidence of an onset-superiority test. A median lead is estimable only when at least one onset has an alert dated from seven days before through the onset date. The summary has no such matched onset-alert pair for BNB, ETH, or SOL, so their medians are Not estimable. The two estimable medians, for BTC and XRP, are both 7 days and sit at the seven-day quiet-period bound.

Table A.7.8. Raw HC-TEM Sample Alignment

Target	Horizon	Historical- evaluation observations	Historical- evaluation dates	Events	Event rate	Model- pool members	Same held-out keys
Composite instability	1	3,655	731	473	12.9%	8	Yes
Composite instability	3	3,645	729	1,008	27.7%	8	Yes
Composite instability	7	3,625	725	1,717	47.4%	8	Yes
Crash event	1	3,655	731	213	5.8%	8	Yes
Crash event	3	3,645	729	574	15.7%	8	Yes
Crash event	7	3,625	725	1,135	31.3%	8	Yes
Liquidation shock	1	3,655	731	351	9.6%	8	Yes
Liquidation shock	3	3,645	729	764	21%	8	Yes
Liquidation shock	7	3,625	725	1,327	36.6%	8	Yes

Notes: The table verifies that raw HC-TEM uses the same held-out target-horizon-token-date keys and outcomes as the evaluation sample; this is a sample-alignment statement, not a parity claim.

Table A.7.8 confirms that raw HC-TEM and the established model pool use the same held-out keys and outcomes in every target-horizon cell. That alignment supports a like-for-like diagnostic comparison, not a parity claim. Raw HC-TEM remains the secondary comparator, while the active monitor returns to the established pool in all nine recorded cells under both fallback margins.

Online Appendix A.8 Extended Forecast and Robustness Details

This section reports the atomic positioning-block contrasts, multiplicity treatment, block-length design sensitivity, outcome composition, validation-rule checks, comparator-level inference, source delay, and the descriptive matched ladder. Each display is tied to a named object in the stated design. The conclusions are check-specific. They do not imply that one model is invariant to every validation sample or metric. Table A.8.1 reports the gate-stability analysis. Table A.8.2 retains the descriptive fixed-estimator ladder without using it as evidence for H2. Table A.8.3 reports the atomic H2 evidence. Tables A.8.4 through A.8.6 report comparator-level inference, multiplicity treatment, and block-length design sensitivity.

Composite instability is the union of crash and liquidation events. Aggregating the one-day mutually exclusive states across the three splits gives 270 crash-only, 201 joint, and 594 liquidation-only observations. Thus 795 of 1,065 composite events, or 74.6%, involve liquidation. The composite target is therefore a liquidation-centered descriptive union, while crash and liquidation remain separately scored outcomes.

The mutually exclusive one-day outcome-state interpretation is summarized in the event-variable section.

Table A.8.1 reports validation-selection uncertainty for every target-horizon cell. The active one-day liquidation model is recovered in two of five shifted sub-windows and three of five token omissions. The modal winner under 500 date-block resamples is different, and a log-loss gate also selects a different model. These perturbations do not establish winner invariance. This directly qualifies the validation-winner metrics in Table A.4.3 and supports reading them as evidence conditional on the AP rule rather than as an invariant model ranking.

Table A.8.1. Gate-Stability Analysis Under Validation Perturbations

Target	h	Reported winner	Shifted- window recoveries	Token- omission recoveries	500-draw modal winner	Log-loss-gate winner
Composite instability	1	Nonlinear-augmented fusion	1/5	4/5	Pool-weighted convex fusion	Histogram gradient boosting
Composite instability	3	Random forest	2/5	4/5	Random forest	Histogram gradient boosting
Composite instability	7	Nonlinear-augmented fusion	2/5	3/5	Nonlinear-augmented fusion	Nonlinear-augmented fusion
Crash event	1	Fixed-memory event-history logit	2/5	4/5	Fixed-memory event-history logit	Histogram gradient boosting
Crash event	3	Fixed-memory event-history logit	2/5	4/5	Fixed-memory event-history logit	Histogram gradient boosting
Crash event	7	Fixed-memory event-history logit	2/5	3/5	Fixed-memory event-history logit	Histogram gradient boosting
Liquidation shock	1	Nonlinear-augmented fusion	2/5	3/5	Extremely randomized trees	Histogram gradient boosting
Liquidation shock	3	Random forest	3/5	4/5	Random forest	Nonlinear-augmented fusion
Liquidation shock	7	Random forest	3/5	5/5	Random forest	Nonlinear-augmented fusion

Notes: Shifted-window and token-omission entries are exact recovery counts. The five shifted sub-windows are 2023-03-01..2023-08-31, 2023-04-01..2023-09-30, 2023-05-01..2023-10-31, 2023-06-01..2023-11-30, and 2023-07-01..2023-12-31; leave-one-token-out variants omit BNB, BTC, ETH, SOL, or XRP; the resampling design uses 500 circular 21-day date-block draws; and the alternative gate uses the same candidate set and validation window but ranks by validation log loss, then Brier score and model name. The resampling column reports the modal selected model in the 500-draw summary row. All variants use stored validation predictions; historical evaluation predictions do not enter selection.

Every matched-logit row in Table A.8.2 gains AP when within-token event histories replace raw controls, with positive token counts in most cells. Because the estimator is held fixed, the table is supportive evidence for the representation result in Section 5.3. The order dependence stated in the note prevents reading the ladder as an additive decomposition or as the H2 positioning test.

Table A.8.2. Nine-Cell Matched Fixed-Estimator Event-History Ladder					
Target	h	Raw-controls AP	Within- history AP	Δ AP	Positive tokens
Composite instability	1	0.314	0.367	+0.053	5/5
Composite instability	3	0.458	0.520	+0.062	5/5
Composite instability	7	0.667	0.689	+0.022	4/5
Crash event	1	0.081	0.099	+0.018	4/5
Crash event	3	0.192	0.238	+0.046	5/5
Crash event	7	0.379	0.414	+0.035	4/5
Liquidation shock	1	0.327	0.365	+0.038	4/5
Liquidation shock	3	0.423	0.485	+0.062	5/5
Liquidation shock	7	0.567	0.618	+0.051	5/5

Notes: Each row holds the matched-logit estimator fixed and compares raw persistence-and-market controls with constructed within-token event histories on the same historical-evaluation observations. The ladder is descriptive and order-dependent. It is not an additive component of the B3-to-selected-monitor comparison and does not replace the order-shuffle permutation test.

Positioning contrasts in Table A.8.3 leave a broadly positive but not specification-stable trace when the matched logistic estimator is held fixed. Adding the positioning block to a baseline that already contains own-liquidation history and the lagged target raises pooled AP from 0.341 to 0.366, a difference of +0.025 [+0.006, +0.047]. Across four specifications and two directions, seven of eight pooled comparisons are positive and three intervals exclude zero, but removing the block over the 2024 evaluation reverses the sign to -0.013 ($p=0.358$). The pooled increment is positive and its interval is clear of zero, consistent with positioning pressure leaving monitoring information beyond recent liquidation recurrence. The 2024 removal is the single negative comparison among the eight, and this variation reflects the competing anchoring, carry, and deleveraging readings of positioning pressure. The table therefore supports H2 as an average dated increment while the 2024 reversal keeps the claim conditional on the evaluation window.

Table A.8.3. Atomic Positioning-Block Contrasts

Historical-evaluation period	Direction	N	AP with	AP without	Δ AP	95% interval	p
Reported design (2024–2025)	Add	3,655	0.366	0.341	+0.025	[+0.006, +0.047]	0.017
Reported design (2024–2025)	Remove	3,655	0.365	0.356	+0.009	[-0.009, +0.032]	0.277
Calendar 2024	Add	1,830	0.422	0.408	+0.014	[-0.011, +0.042]	0.233
Calendar 2024	Remove	1,830	0.440	0.453	-0.013	[-0.033, +0.013]	0.358
July 2024–June 2025	Add	1,825	0.394	0.373	+0.021	[0, +0.053]	0.051
July 2024–June 2025	Remove	1,825	0.391	0.378	+0.012	[-0.001, +0.039]	0.064
Calendar 2025	Add	1,825	0.299	0.256	+0.043	[+0.022, +0.064]	0.001
Calendar 2025	Remove	1,825	0.266	0.228	+0.038	[+0.021, +0.048]	0.001

Notes: Each row holds the matched logistic estimator, target, horizon, training sample, and evaluation keys fixed. The first column identifies the reported 2024–2025 design and the three additional historical-evaluation periods. Adding the block compares the recurrence baseline with and without positioning; removing it compares the full feature set with and without positioning. Seven of eight pooled comparisons are positive and three intervals exclude zero. The 2024 removal comparison is negative; the seven remaining comparisons are positive.

Table A.8.4 reports forecast comparisons not already single-sourced in the main fixed-baseline ladder. For the one-day liquidation target, the removed autoregressive, HAR-volatility, and fixed-memory AP occurrences point to the baseline ladder in Manuscript Table 4. The primary one-day liquidation row compares the active monitor with the dynamic multi-memory event-history logit under the 21-day inference, and its ROC-AUC difference is reported in Panel B. AP is read with prevalence and AP lift in the main exhibits. The liquidation and composite rows carry the positive pooled evidence, while the crash rows remain weak. That cross-target pattern supports the paper’s target-specificity argument and preserves the primary-versus-secondary distinction used in Section 5.6.

Table A.8.4. Comparator-Level Historical Evaluation Inference

Panel A. AP inference

Target	h	N	Events	AP difference	[95% interval]	AP p
Liquidation shock	1	3,655	351	+0.0167	[+0.0004, +0.038]	0.046
Composite instability	7	3,625	1,717	+0.016	[0.0013, 0.026]	0.029
Composite instability	1	3,655	473	+0.015	[0.0007, 0.033]	0.045
Liquidation shock	7	3,625	1,327	-0.014	[-0.053, 0.020]	0.405
Liquidation shock	3	3,645	764	-0.002	[-0.021, 0.015]	0.881
Crash event	7	3,625	1,135	-0.018	[-0.052, 0.020]	0.312
Crash event	3	3,645	574	+0.005	[-0.026, 0.026]	0.850
Composite instability	3	3,645	1,008	+0.005	[-0.012, 0.020]	0.486
Crash event	1	3,655	213	+0.0001	[-0.026, 0.016]	0.925

Panel B. ROC-AUC inference and token-level agreement

Target	h	ROC-AUC difference	[95% interval]	ROC-AUC p	Token wins / weakest token
Liquidation shock	1	+0.026	[0.011, 0.044]	0.002	4 / XRP (-0.011)
Composite instability	7	+0.022	[0.008, 0.036]	0.002	5 / BTC (+0.012)
Composite instability	1	+0.018	[0.008, 0.031]	0.002	5 / SOL (+0.015)
Liquidation shock	7	+0.008	[-0.021, 0.035]	0.597	3 / XRP (-0.033)
Liquidation shock	3	+0.017	[-0.004, 0.040]	0.119	4 / XRP (-0.040)
Crash event	7	-0.011	[-0.044, 0.021]	0.507	1 / SOL (-0.032)
Crash event	3	+0.003	[-0.028, 0.032]	0.810	2 / SOL (-0.029)
Composite instability	3	+0.008	[-0.010, 0.024]	0.374	3 / SOL (-0.008)
Crash event	1	0	[-0.031, 0.029]	0.986	2 / SOL (-0.035)

Notes: The two panels report the primary comparison and the eight secondary-family comparisons named in Table A.8.5. Every difference in both panels is the active monitor for that cell minus one and the same comparator, the dynamic multi-memory event-history logit, evaluated on the same historical-evaluation observations. The table therefore carries a single comparator identity and does not supply the cell-specific comparator identities used in Manuscript Table 8, which are named in Table A.8.5.

Multiplicity in Table A.8.5 is separated by inferential role. The single primary comparison reports 50,000 draws. The eight secondary target-horizon AP comparisons use 2,000 draws and form one family under Holm family-wise error control. Benjamini-Hochberg q -values are complementary. No secondary row survives Holm control, so Table A.8.5 supports one narrow primary claim without expanding it across targets or horizons.

Table A.8.5. Primary and Secondary-Family Multiplicity

Role	Target	h	Draws	Δ AP	95% interval	Raw p	Holm p	BH q
Primary	Liquidation shock	1	50,000	+0.0167	[+0.0004, +0.038]	0.046	Not applicable	Not applicable
Secondary	Composite instability	7	2,000	+0.016	[+0.002, +0.026]	0.030	0.240	0.240
Secondary	Composite instability	1	2,000	+0.011	[-0.0014, +0.025]	0.079	0.553	0.316
Secondary	Liquidation shock	7	2,000	+0.0097	[-0.019, +0.034]	0.512	1	0.990
Secondary	Liquidation shock	3	2,000	0	[-0.021, +0.018]	0.990	1	0.990
Secondary	Crash event	7	2,000	+0.002	[-0.032, +0.033]	0.869	1	0.990
Secondary	Crash event	3	2,000	-0.0008	[-0.025, +0.027]	0.964	1	0.990
Secondary	Composite instability	3	2,000	-0.0005	[-0.009, +0.011]	0.935	1	0.990
Secondary	Crash event	1	2,000	-0.011	[-0.055, +0.024]	0.558	1	0.990

Notes: The primary row reports the stated result. Each row's comparator is that cell's incumbent, the event-history model that led on validation AP, fixed before historical evaluation. The incumbent dynamic multi-memory logit serves the three liquidation-shock rows and the composite-instability rows at $h = 3$ and $h = 7$, the incumbent full-feature fixed-memory logit serves composite instability at $h = 1$, the incumbent three-channel fixed-memory logit serves the crash-event rows at $h = 3$ and $h = 7$, and the incumbent lagged-market logit serves the crash-event row at $h = 1$. Comparators are cell-specific, so each difference is read against the model named on that row. Every secondary row uses 2,000 21-day circular date-block draws and a two-sided finite-sample p -value. Holm p -values control the eight-comparison family, and Benjamini-Hochberg q -values are complementary. No secondary row rejects under Holm control.

Table A.8.6 treats block length as a design-sensitivity axis alongside event threshold, construction window, token set, rolling evaluation window, and source delay. Across 7-, 14-, 21-, and 42-day circular blocks, the finite-sample p -value ranges from 0.014 to 0.069. The 7-day and 14-day intervals include zero, so two of the four block lengths do not separate the selected-monitor comparison from zero. The table therefore motivates reporting the primary block result beside shorter and longer alternatives, since dependence assumptions change precision even when point performance is fixed.

The observed average-precision difference remains 0.017 in every row of Table A.8.6, so block length changes uncertainty rather than point performance. Moving from a 7-day to a 42-day block reduces the lower-tail count from 1,732 to 343 and raises the monitor share from 0.965 to 0.993. Over the same comparison, the interval's lower end moves from a decline of 0.002 to an improvement of 0.004, while the finite-sample p -value falls from 0.069 to 0.014. The ranking difference is therefore clearer under longer dependence blocks but unresolved under the shorter alternatives. This sensitivity bounds the precision of the selected-monitor comparison. It cannot identify the correct dependence horizon.

Table A.8.6. Fixed-Length Circular Date-Block Sensitivity

Block days	Draws	Lower-tail count	Δ AP	95% interval	Finite p	Monitor share	Benchmark share
7	50,000	1,732	+0.0167	[-0.002, +0.038]	0.069	0.965	0.035
14	50,000	1,397	+0.0167	[-0.0005, +0.038]	0.056	0.972	0.028
21	50,000	1,150	+0.0167	[+0.0004, +0.038]	0.046	0.977	0.023
42	50,000	343	+0.0167	[+0.004, +0.039]	0.014	0.993	0.007

Notes: Each row uses 50,000 draws and the two-sided finite-sample p -value. The observed AP difference is unchanged across rows; only the resampled interval and tail counts vary. These are block-length checks around the selected-monitor comparison, not additional primary tests.

Online Appendix A.9 Robustness and Sensitivity Checks

The specification grid contains 228 declared cells, all of which complete. Tables A.9.1 through A.9.6 report, respectively, in-sample proportions, construction windows and rolling evaluation windows, an added source lag with a crisis-window exclusion and alternative liquidity measures, event thresholds, token omissions, and order sensitivity.

Table A.9.1 varies the in-sample/out-of-sample proportion inside the fixed construction. Each alternative configuration regenerates the forward-target grid, refits the candidate pool, applies the same validation gate, and runs the same block bootstrap, and only the proportion moves. Table A.9.1 reports the 67 percent design used throughout the paper alongside re-estimates at 40 and 50 percent.

Table A.9.1. Primary Comparison Under Alternative In-Sample Proportions

In-sample share	Selection window	Evaluation N	Selected monitor	Δ AP	95% CI	p
40 percent	2022-01-06–2022-05-25	6,580	Extremely randomized trees	+0.0242	[+0.0076, +0.0404]	0.006
50 percent	2022-08-14–2022-12-31	5,480	Extremely randomized trees	+0.039	[+0.018, +0.0548]	0.002
67 percent (reported)	2023-08-14–2023-12-31	3,655	Nonlinear-augmented fusion	+0.0167	[+0.0001, +0.0385]	0.047

Notes: Each row re-estimates the primary comparison, holding event construction, forward targets, candidate pool, gate rule, and inference fixed and varying only the in-sample proportion. That comparison is the one-day liquidation shock, the validation-gated monitor against that configuration’s best validation benchmark, 21-day circular date blocks, 2,000 draws, and the two-sided finite-sample p -value. Partitions are purged at their own boundaries, so no forward outcome window crosses a split. The 67 percent row is the reported design and reproduces the primary result of Table 8. The table reports the proportions listed; it is not an exhaustive grid and does not establish that the comparison holds at proportions outside it.

The three rows in Table A.9.1 are not three draws of one quantity. Changing the proportion moves the selection window as well as the evaluation window, and the validation gate does not return the same candidate each time. At 40 and 50 percent it selects extremely randomized trees with dynamic event histories, where the reported design selects nonlinear-augmented fusion. Each reported difference therefore combines the event-history signal with the gate’s choice among candidates. That is the selection dependence Table A.8.1 already documents, reached a second way. There the reported winner is recovered in two of five shifted validation sub-windows, the 500-draw modal winner differs, and a log-loss gate selects differently. The proportion belongs with the selection rule as part of the reported design, not as a nuisance parameter the result is invariant to. Table A.9.1 therefore supports the result only over the displayed design shares and reinforces the need to treat sample proportion as part of the selection design.

Construction-window and rolling-window results in Table A.9.2 complete three choices and five rolling evaluation windows, respectively. Rolling-window monitor-minus-comparator AP differences are positive at the first two windows and negative at the final three, so the sign is not stable even though AP lift remains above prevalence. This sign change supports a time-sensitive warning interpretation and weighs against treating the primary comparison as stable across historical evaluation windows.

Table A.9.2. Construction-Window and Rolling-Window Grid

Perturbation	Status	Monitor AP	Comparator AP	Δ AP	95% interval	p
Construction window 180 days, minimum history 89 days	PASS	0.325	0.345	-0.020	[-0.048, +0.009]	0.188
Construction window 365 days, minimum history 180 days	PASS	0.362	0.345	+0.0167	[+0.0004, +0.038]	0.046
Construction window 730 days, minimum history 360 days	PASS	0.379	0.412	-0.033	[-0.072, +0.014]	0.210
Rolling window 1	PASS	0.436	0.419	+0.016	[-0.009, +0.050]	0.157
Rolling window 2	PASS	0.417	0.403	+0.014	[-0.012, +0.044]	0.300
Rolling window 3	PASS	0.395	0.396	-0.001	[-0.057, +0.030]	0.993
Rolling window 4	PASS	0.369	0.370	-0.002	[-0.040, +0.020]	0.883
Rolling window 5	PASS	0.243	0.252	-0.008	[-0.028, +0.021]	0.655

Notes: The first three rows vary construction windows, and the final five vary the historical-evaluation window. All eight displayed cells complete.

Table A.9.3 reports these additional design perturbations. Across the additional perturbations, a one-day source lag gives monitor AP 0.311 and an AP difference of -0.005, excluding the May 2021 window gives 0.356 and +0.009, and the Abdi-Ranaldo, Corwin-Schultz, and dollar-volume Amihud liquidity proxies give monitor AP of 0.372, 0.37, and 0.365 with paired differences of +0.024, +0.011, and +0.018. The source-lag reversal identifies information timing as a binding design condition, while the liquidity substitutions show that the positive comparison is not tied to only one proxy. Because several intervals include zero, the table supports sensitivity analysis rather than proxy invariance.

Table A.9.3. Additional Design Perturbations

Perturbation	Status	Monitor AP	Comparator AP	AP difference	95% interval	<i>p</i>
One-day added source lag	PASS	0.311	0.316	-0.005	[-0.019, +0.015]	0.817
May 2021 crisis-window exclusion	PASS	0.356	0.347	+0.009	[-0.009, +0.031]	0.270
Abdi-Ranaldo spread proxy	PASS	0.372	0.348	+0.024	[+0.006, +0.048]	0.011
Corwin-Schultz spread proxy	PASS	0.370	0.359	+0.011	[-0.005, +0.031]	0.145
Dollar-volume Amihud proxy	PASS	0.365	0.347	+0.018	[+0.0001, +0.041]	0.051

Notes: The first row adds one source-lag day to seven dated fields, the second excludes 1 May through 15 June 2021, and the final three replace the liquidity proxy. All rows repeat model selection and paired historical evaluation.

Online Appendix Table A.9.3 reports the result. The one-day added-source-lag exercise adds one day to seven dated source fields and repeats the comparison. Monitor AP is 0.311 and comparator AP is 0.316, giving -0.005 [-0.019, +0.015], $p=0.817$. Monitor AP is 0.051 lower than in the baseline analysis.

The threshold grid in Table A.9.4 crosses three liquidation thresholds with three crash thresholds. All nine cells complete. Table A.9.4 reports monitor and comparator AP, paired differences, intervals, and p-values under the same evaluation rule. Differences change in size and sign across the threshold pairs, and only the reported pair separates from zero. Table A.9.4 therefore supports the stated threshold design as one documented specification rather than a threshold-invariant result.

Table A.9.4. Event-Threshold Specification Grid

Perturbation	Status	Monitor AP	Comparator AP	Δ AP	95% interval	p
Liquidation 90%, crash 2.5%	PASS	0.489	0.489	+0.001	[-0.013, +0.023]	0.717
Liquidation 90%, crash 5%	PASS	0.490	0.489	+0.002	[-0.012, +0.025]	0.614
Liquidation 90%, crash 10%	PASS	0.491	0.487	+0.004	[-0.011, +0.031]	0.512
Liquidation 95%, crash 2.5%	PASS	0.357	0.346	+0.011	[-0.007, +0.033]	0.192
Liquidation 95%, crash 5%	PASS	0.362	0.345	+0.0167	[+0.0004, +0.038]	0.046
Liquidation 95%, crash 10%	PASS	0.361	0.360	+0.001	[-0.016, +0.022]	0.714
Liquidation 97.5%, crash 2.5%	PASS	0.296	0.293	+0.003	[-0.019, +0.036]	0.575
Liquidation 97.5%, crash 5%	PASS	0.294	0.294	0	[-0.023, +0.036]	0.706
Liquidation 97.5%, crash 10%	PASS	0.297	0.307	-0.009	[-0.036, +0.025]	0.867

Notes: Each threshold-pair row repeats selection and paired historical evaluation under the stated thresholds.

Table A.9.5 omits each token in turn. All five cells complete. Dropping SOL selects HC-TEM, while the other four omissions return nonlinear-augmented fusion. The monitor-minus-comparator AP difference varies across omissions. The winner change after omitting SOL and the varying paired differences show that token composition affects both selection and incremental performance.

The token-omission check in Table A.9.5 asks whether the pooled ranking increment depends on a single token. Monitor AP remains between 0.314 when XRP is omitted and 0.392 when ETH is omitted, but the comparison with the benchmark ranges from a decline of 0.025 after dropping BTC to an improvement of 0.026 after dropping XRP. Only the XRP-omission interval lies wholly above zero, from an improvement of 0.006 to 0.043 with $p=0.007$. The other intervals include zero, and the BTC-omission reversal has $p=0.059$. The active monitor therefore retains its observed absolute ranking range across omissions, while the sign and precision of its incremental advantage depend materially on token composition. The grid bounds transfer and does not establish robustness to a broader token universe.

Table A.9.5. Token-Omission Specification Grid

Perturbation	Status	Monitor AP	Comparator AP	Δ AP	95% interval	p
Omit BNB	PASS	0.360	0.343	+0.017	[-0.003, +0.042]	0.100
Omit BTC	PASS	0.339	0.365	-0.025	[-0.058, +0.001]	0.059
Omit ETH	PASS	0.392	0.387	+0.005	[-0.019, +0.038]	0.527
Omit SOL	PASS	0.386	0.369	+0.017	[-0.012, +0.040]	0.240
Omit XRP	PASS	0.314	0.289	+0.026	[+0.006, +0.043]	0.007

Notes: Each row omits one token and repeats model selection and paired historical evaluation.

Table A.9.6 reports the order-perturbation diagnostic. It examines whether the value of the dated record depends on the order in which events arrived at each forecast date, holding the events themselves fixed. Within trailing windows of 30, 60, and 90 days before each forecast date, the six event histories are jointly reordered in time, the decayed histories are rebuilt, and the fitted active monitor and the fitted dynamic comparator are re-scored without re-estimation. With 200 reorderings per window and independent draws per token, none reaches the observed AP of either model at any window. The cost of reordering grows with the window, from 0.149 to 0.213 for the active monitor. The observed gap of 0.017 between the active monitor and the dynamic comparator also exceeds every reordered gap at the 60- and 90-day windows and 197 of 200 at the 30-day window. Panel B replaces the decayed histories with trailing-window event counts over the same records and repeats estimation and model selection under the unchanged design. AP falls to 0.354 under 7-, 30-, and 90-day windows and to 0.359 under 3-, 14-, and 60-day windows, against 0.362 for the decayed representation. Ordering carries warning content beyond the collection of events, and the decayed representation retains a small advantage over order-invariant counts.

Table A.9.7 asks whether the value of the dated record survives when order is destroyed over each token's entire history rather than inside a trailing window. Each of the 200 draws permutes the joint six-event vector by row position, rebuilds the decayed histories, refits the candidate library, and reruns validation selection for the active monitor and for the dynamic comparator, so every draw delivers one internally consistent permuted history. Observed active-monitor AP is 0.362 against a permuted mean of 0.269, and observed dynamic-comparator AP is 0.345 against 0.248. No draw reaches either observed value, at a permutation p-value of 0.005 for the active monitor and 0.005 for the dynamic comparator. The order costs are 0.093 and 0.098 AP. Because both models are refitted and reselected on every draw, this exercise holds the design fixed rather than the fitted scores.

For the chronology claim, Table A.9.7 supplies the primary evidence. Table A.9.6 supplies two corroborating readings, because the within-window reorderings and the order-invariant count replacement both weaken the dated representation under their stated designs. The design-sensitivity evidence varies event thresholds, construction windows, a named crisis window, token inclusion, alternative liquidity measures, historical evaluation windows, source timing, block length, and event order within the observed five-token daily panel. It establishes the chronology result and shows that the active-minus-comparator sign is not stable across rolling evaluation windows. That evidence does not convert the different AP comparisons into additive components.

Table A.9.6. Order Sensitivity of the Dated Event Record

<i>Panel A. Within-window reordering (200 draws per window)</i>							
Window (days)	Mean AP cost, monitor	Mean AP cost, comparator	Max reordered AP, monitor	Max reordered AP, comparator	Permutation p , monitor	Permutation p , comparator	Permutation p , gap
30	0.149	0.135	0.244	0.232	0.005	0.005	0.020
60	0.205	0.192	0.177	0.174	0.005	0.005	0.005
90	0.213	0.200	0.173	0.164	0.005	0.005	0.005

<i>Panel B. Order-invariant windowed counts (full re-estimation)</i>							
Windows (days)			Monitor AP		Decayed-representation AP		
7, 30, 90			0.354		0.362		
3, 14, 60			0.359		0.362		

Notes: Reordering permutes the six event histories jointly within each forecast date's trailing window, keeping window composition fixed. This is the order-perturbation diagnostic, which holds the fitted models fixed. Table A.9.7 reports the full-history permutation that refits and reselects on every draw. Observed AP is 0.362 for the active monitor and 0.345 for the dynamic comparator, and the observed gap is 0.017. Permutation p -values equal $(1 + b)/201$, where b counts reordered draws at or above the observed value. Panel B replaces the three decay half-lives with trailing-window event counts. Estimation and model selection repeat under the unchanged design.

Table A.9.7. Full-History Joint Permutation of the Dated Event Record

Object	Observed AP	Permuted mean AP	Permuted 2.5th	Permuted 97.5th	Permuted max	Draws at or above observed	Permutation p
Active monitor	0.362	0.269	0.231	0.298	0.306	0 of 200	0.005
Dynamic comparator	0.345	0.248	0.230	0.258	0.263	0 of 200	0.005

Notes: Each of the 200 draws applies one row-position permutation jointly to the six daily event channels over each token's full available predictor history, rebuilds the decayed histories, refits the candidate library, and reruns validation selection for both objects. A draw is therefore one internally consistent permuted history rather than a rescoring of the observed fit. The permutation p -value equals $(1 + b)/201$, where b counts draws at or above the observed value. This exercise does not separate the observed active-minus-comparator difference of 0.017 from its permuted distribution. Table A.9.6 reports the within-window order-perturbation diagnostic that addresses the gap. A dependence-preserving variant applies one circular shift to each token's joint six-channel event history in each of 200 draws and again refits and reselects both objects. No shifted draw reaches either observed AP, and the active-minus-comparator difference again remains inside its shifted distribution.

Table A.9.8 separates two questions about the information channels that Section 5.2 reads together. Panel A asks whether negative news and abnormal search attention rank near-term instability at all when nothing else is in the model. They do. The two decayed histories on their own reach an AP of 0.221 against a constant-prevalence benchmark of 0.096 for one-day liquidation shocks, and 0.239 against 0.129 for one-day composite instability, and both differences are clear of zero. The standalone signal is therefore present rather than absent, so the conditional null is not a statement that news and attention carry no warning content. It is a statement about what remains once the represented market state is in the model, and the design would have detected a conditional increment of 0.033 AP at 80 percent power, which is below the representation-step scale this paper reports elsewhere. That is the absorption reading. The content is present in the raw series and is already carried by the represented state.

Panel B shows the association that reading predicts, though unevenly. Abnormal search attention on the previous day is associated with all four market-state events, most strongly with liquidation shocks, where the odds ratio is 5.243 and the average marginal effect is 0.200. Negative news carries little. Of its four odds-ratio intervals, three span one, and only the derivatives-positioning association is clear of it. Two of the eight associations are negative, search attention with liquidity contraction and negative news with derivatives-positioning pressure, so the block is not a uniform positive lead indicator for every component of the represented state. Taken together, attention carries the next-day association, news tone contributes little at this sample size, and the absorption reading rests mainly on the attention channel.

Table A.9.8. Standalone News and Attention Signal and Next-Day Association with Market-State Events

<i>Panel A. Standalone news and attention signal</i>						
Target	h	Standalone AP	Constant-prevalence AP	Difference	[95% interval]	p
Liquidation shock	1	0.221	0.096	+0.125	[+0.043, +0.237]	0.001
Composite instability	1	0.239	0.129	+0.110	[+0.038, +0.207]	0.001
<i>Panel B. Next-day association between information events and market-state events</i>						
Outcome on date t	Lagged predictor on date $t - 1$		Odds ratio [95% interval]		Average marginal effect [95% interval]	
Crash event	Negative news		1.374 [0.756, 2.170]		+0.020 [-0.014, +0.056]	
Crash event	Search attention		1.686 [1.046, 2.643]		+0.034 [+0.002, +0.069]	
Liquidation shock	Negative news		0.986 [0.424, 1.906]		-0.001 [-0.054, +0.063]	
Liquidation shock	Search attention		5.243 [2.955, 8.802]		+0.200 [+0.111, +0.283]	
Liquidity contraction	Negative news		1.037 [0.514, 1.663]		+0.002 [-0.031, +0.035]	
Liquidity contraction	Search attention		0.366 [0.143, 0.700]		-0.044 [-0.063, -0.021]	
Derivatives-positioning pressure	Negative news		0.488 [0.255, 0.839]		-0.055 [-0.096, -0.015]	
Derivatives-positioning pressure	Search attention		2.662 [1.372, 4.595]		+0.117 [+0.029, +0.219]	

Notes: Panel A evaluates a standalone model, a pooled fixed-memory logistic fitted on the two decayed negative-news and search-attention histories only and trained through 2022, against a constant-prevalence score equal to the historical-evaluation event rate. The conditional increment is a separate quantity. Once the market-fragility histories are present, adding the same block leaves an increment that is statistically indistinguishable from zero, with a minimum detectable difference of 0.033 AP at 80 percent power. Panel B reports pooled token-fixed-effect logistic associations on the historical evaluation window, with both lagged event indicators entered jointly and token fixed effects included. Marginal effects are average discrete changes from 0 to 1 for the named lagged event. Of 3,655 rows, 85 are excluded because the lagged negative-news indicator is unobservable there. All intervals use paired fixed-length circular 21-day date blocks and 2,000 draws.

Statistical significance is assessed within each exercise's own table

Online Appendix A.10 Forecast-Metric Formulas

Table A.10.1 reports the forecast-metric register. AP is the principal rare-event ranking metric and is reported with each cell's prevalence and AP lift. Raw AP is not compared across horizons (Davis and Goadrich, 2006). Online Appendix Table A.10.1 summarizes metric direction and role. The register explains why the validation-winner evidence in Table A.4.3 and the probability-quality evidence in Section 5.5 are read as distinct claims. It also prevents a higher raw AP at a longer horizon from being treated as a like-for-like gain without prevalence.

Table A.10.1. Forecast-Metric Register

Metric	Direction	Role	Reference
AP	Higher	Primary rare-event ranking	Davis and Goadrich (2006)
Prevalence and AP lift	Higher	Within-cell normalization	Davis and Goadrich (2006)
ROC-AUC	Higher	Secondary ranking	Davis and Goadrich (2006)
Log loss and Brier score	Lower	Probability quality	Gneiting and Raftery (2007)
Top-decile hit rate	Higher	Risk concentration	Not applicable
Relative decision value	Higher	Decision value under a stylized cost-loss model	Granger and Pesaran (2000)

Notes: The table defines ranking, probability-quality, and decision-value metrics. AP is AP, AP lift is AP divided by prevalence, and raw AP is not compared across target horizons without the corresponding event rate.

Online Appendix A.11 Descriptive Attribution Diagnostics

This section reports descriptive model-behavior diagnostics across crisis windows. The summaries show whether attribution records are stable enough to describe model behavior during stress episodes. The evidence is descriptive only and should not be read as causal evidence, proof of mechanisms, or evidence of structural drivers. Table A.11.1 reports the descriptive attribution fidelity summary.

Attribution patterns in Table A.11.1 show that the composite-instability event history is largest for BTC, search attention for ETH, derivatives-positioning pressure for SOL, and the intercept for BNB and XRP by mean absolute contribution. The mean-absolute ranking includes the intercept, while the most-frequent top component displayed in the table excludes it by construction, so the intercept leads the ranking for BNB and XRP while the displayed column names a fitted component for those two tokens. The cross-token differences show that the fitted monitor does not rely on one attribution pattern across crisis windows. They support a descriptive account of model behaviour in Section 5.2, while the fidelity summary and non-causal scope prevent the leading components from being read as structural drivers.

Table A.11.1. Descriptive Attribution Fidelity Summary

Token	Crisis window	Dates	Mean predicted probability	Mean fidelity share	Most frequent top component
BNB	Aug-Sep 2023 correction	10	0.333	0.215	Negative news
BNB	FTX Nov 2022	10	0.369	0.408	Derivatives-positioning pressure
BNB	May 2021 selloff	10	0.851	0.205	Instability decay
BNB	Terra/Luna May 2022	10	0.481	0.418	Joint lev $\times$ liq $\times$ attn
BTC	Aug-Sep 2023 correction	10	0.453	0.175	Instability decay
BTC	COVID March 2020	10	0.317	0.140	Lagged market
BTC	FTX Nov 2022	10	0.661	0.277	Derivatives-positioning pressure
BTC	May 2021 selloff	10	0.788	0.455	Instability decay
BTC	Terra/Luna May 2022	10	0.582	0.370	Instability decay
ETH	Aug-Sep 2023 correction	10	0.255	0.062	Joint lev $\times$ liq $\times$ attn
ETH	COVID March 2020	10	0.294	0.180	Liquidity contraction
ETH	FTX Nov 2022	10	0.423	0.163	Derivatives-positioning pressure
ETH	May 2021 selloff	10	0.868	0.288	Search attention
ETH	Terra/Luna May 2022	10	0.579	0.206	Search attention
SOL	Aug-Sep 2023 correction	10	0.237	0.166	Joint lev $\times$ liq $\times$ news
SOL	FTX Nov 2022	10	0.650	0.544	Derivatives-positioning pressure
SOL	May 2021 selloff	10	0.928	0.202	Search attention
SOL	Terra/Luna May 2022	10	0.659	0.351	Search attention
XRP	Aug-Sep 2023 correction	10	0.196	0.114	Negative news
XRP	COVID March 2020	10	0.181	0.258	Lagged market
XRP	FTX Nov 2022	10	0.512	0.285	Liquidity contraction
XRP	May 2021 selloff	10	0.799	0.275	Search attention
XRP	Terra/Luna May 2022	10	0.457	0.361	Search attention

Notes: The table summarizes descriptive attribution fidelity across tokens and crisis windows. Table A.11.1 reports token-by-window mean fidelity shares and most frequent top components.

Online Appendix A.12 Raw Inputs and Event Histories

This section reports the raw-input series and the diagnostic evidence for the continuous event histories used in the manuscript. Figure A.12.1 shows that the raw inputs evolve through distinct episodes rather than a common smooth path. Returns fluctuate around zero, while realized volatility, liquidation volume, illiquidity, and search attention display clustered spikes. Open-interest changes become more variable later in the sample, and news tone and the Fear and Greed Index move through persistent regimes. The pooled token panel preserves the broad timing of these shifts while smoothing some BTC-specific variation. This heterogeneity motivates a common daily information clock that retains the timing of each channel. The standardized rolling paths are descriptive only; the forecasting models use the lagged raw controls defined in Section 3.

Table A.12.1 complements the time-series plots with distributional and diagnostic evidence. Returns exhibit high excess kurtosis, while volume, realized volatility, liquidation volume, liquidity measures, and attention variables are strongly right-skewed. The pooled token panel retains these broad patterns and shows wide dispersion in scale-dependent variables. Ljung-Box tests indicate short-run dependence, and the ADF results reject the unit-root null for most transformed inputs. Price and open-interest levels remain persistent, which supports the first differences and lagged inputs used in the forecasting design. These diagnostics connect the raw records to the within-token thresholds and preprocessing choices described in Section 3.

Table A.12.1. Raw Input Diagnostics

Variable	N	Mean	SD	Skewness	Kurtosis	LB(5) p -value	ADF p -value
<i>Panel A. BTC</i>							
Close	2,192	47,182.88	31,418.23	0.731	-0.515	< 0.001***	0.732
Ret	2,191	0.002	0.032	-0.579	13.189	< 0.001***	< 0.001***
TrdVol	2,192	73,884.17	87,038.28	3.069	12.177	< 0.001***	0.055*
RV	2,191	0.022	0.024	3.520	31.005	< 0.001***	< 0.001***
OI	2,172	6.84×10^9	5.86×10^9	0.983	-0.110	< 0.001***	0.796
Δ OI	2,171	6.86×10^6	4.01×10^8	-2.709	54.523	< 0.001***	< 0.001***
Fund	2,172	0.012	0.022	3.210	15.313	< 0.001***	< 0.001***
LiqVol	2,168	4.52×10^7	6.63×10^7	6.019	66.513	< 0.001***	0.024**
Amihud	2,191	0	0	5.362	57.773	< 0.001***	0.039**
HLSpread	2,192	0.046	0.036	5.455	75.811	< 0.001***	< 0.001***
NewsTone	2,173	-0.309	0.865	-0.748	1.408	< 0.001***	0.001***
NewsCnt	2,173	530.112	363.851	2.124	9.302	< 0.001***	0.006***
Search	2,192	17.613	9.379	2.323	8.649	< 0.001***	0.002***
CrashSearch	2,192	0.002	0.022	12.205	166.662	< 0.001***	< 0.001***
BroadCrash	2,192	0.003	0.034	14.353	247.802	< 0.001***	< 0.001***
FGI	2,191	49.963	22.128	-0.023	-1.071	< 0.001***	0.002***
<i>Panel B. Pooled token panel</i>							
Close	10,556	10,343.63	23,696.77	2.780	7.321	< 0.001***	0.539
Ret	10,551	0.003	0.048	1.226	20.931	< 0.001***	< 0.001***
TrdVol	10,556	8.76×10^7	2.78×10^8	8.758	148.573	< 0.001***	< 0.001***
RV	10,551	0.030	0.037	4.602	44.389	< 0.001***	< 0.001***
OI	10,516	2.56×10^9	4.12×10^9	2.480	6.173	< 0.001***	0.396
Δ OI	10,511	2.73×10^6	2.42×10^8	-2.906	126.187	< 0.001***	< 0.001***
Fund	10,516	0.009	0.046	-18.058	865.092	< 0.001***	< 0.001***
LiqVol	10,508	1.78×10^7	3.89×10^7	8.652	152.324	< 0.001***	< 0.001***
Amihud	10,551	0	0	17.449	373.236	< 0.001***	< 0.001***
HLSpread	10,556	0.065	0.055	4.711	46.688	< 0.001***	< 0.001***
NewsTone	10,461	0.415	1.236	0.225	2.467	< 0.001***	< 0.001***
NewsCnt	10,461	157.458	277.167	3.019	14.970	< 0.001***	< 0.001***
Search	10,556	11.498	10.447	2.408	9.693	< 0.001***	< 0.001***
CrashSearch	10,556	0.167	0.734	9.203	114.689	< 0.001***	< 0.001***
BroadCrash	10,556	0.856	3.853	14.671	283.295	< 0.001***	< 0.001***
FGI	10,551	50.275	22.175	-0.032	-1.093	< 0.001***	< 0.001***

Notes: The table reports descriptive statistics for the raw variables used to construct model inputs. It covers token closing price, daily return, trading volume, realized volatility, open interest, open-interest change, funding rate, reported aggregate liquidation volume, Amihud illiquidity, high-low spread, GDELT news tone and count, token search attention, token crash-search attention, broad crypto crash-search attention, and the Fear and Greed Index. The funding-rate series is the Coinalyze daily series described in Section 3.2. Ljung–Box p -values are through five lags, and ADF p -values test the unit-root null. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively. For the pooled token panel, p -values combine token-level tests using Fisher’s method (Fisher, 1932). ETH, BNB, SOL, and XRP token-level raw-descriptive tables are reported in Online Appendix A.1.

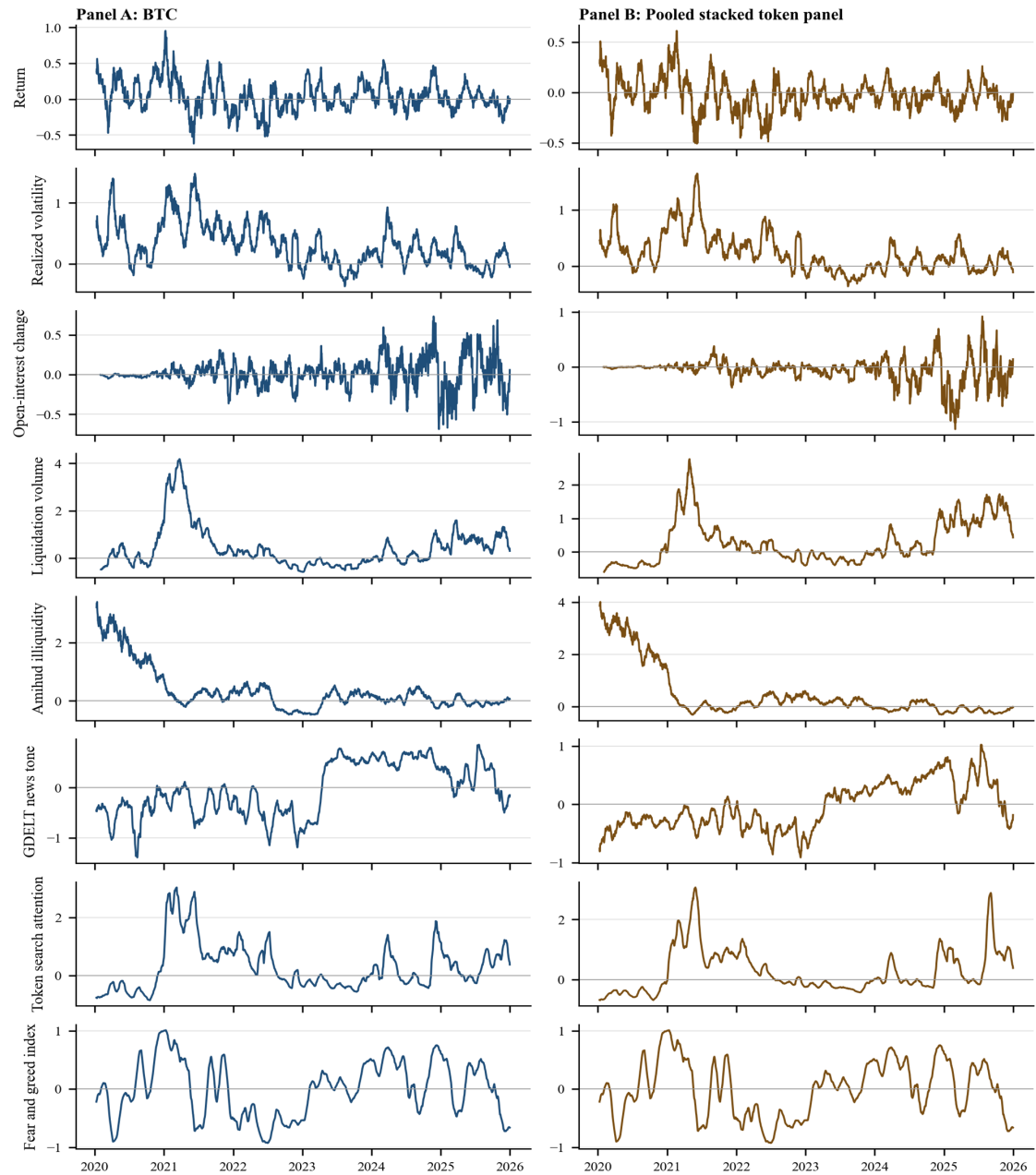
Table A.12.2 shows that the continuous event histories retain the episodic shape of the underlying stress signals. Every reported history is positively skewed in both panels, so elevated states remain concentrated around relatively infrequent episodes. For composite instability, longer half-lives increase both the mean and dispersion because earlier events remain in the state variable for more days. The similarity between the BTC and pooled panels indicates that the construction behaves consistently across the two reporting levels. Ljung-Box tests reject serial independence throughout, which confirms that the histories preserve short-run persistence. The ADF tests reject the unit-root null for every series, so the histories preserve persistence without behaving as nonstationary levels and remain suitable as dated forecasting inputs.

Table A.12.2. Continuous Event-History Diagnostics

Variable	N	Mean	SD	Skewness	Kurtosis	LB(5) p -value	ADF p -value
<i>Panel A. BTC</i>							
$H(\text{CI}, 1)$	1,988	0.103	0.201	2.246	4.356	$< 0.001^{***}$	$< 0.001^{***}$
$H(\text{CI}, 3)$	1,988	0.395	0.544	1.907	3.875	$< 0.001^{***}$	$< 0.001^{***}$
$H(\text{CI}, 7)$	1,988	0.985	1.074	1.714	3.248	$< 0.001^{***}$	$< 0.001^{***}$
$H(\text{CR}, 3)$	1,988	0.184	0.312	2.245	5.367	$< 0.001^{***}$	$< 0.001^{***}$
$H(\text{LS}, 3)$	1,988	0.279	0.503	2.510	6.930	$< 0.001^{***}$	$< 0.001^{***}$
$H(\text{LEV}, 3)$	1,988	0.528	0.687	2.090	4.729	$< 0.001^{***}$	$< 0.001^{***}$
$H(\text{LIQ}, 3)$	1,988	0.294	0.464	1.932	3.608	$< 0.001^{***}$	0.002^{***}
$H(\text{NEWS}, 3)$	1,988	0.419	0.689	2.032	3.788	$< 0.001^{***}$	$< 0.001^{***}$
$H(\text{ATTN}, 3)$	1,988	0.339	0.790	2.779	7.177	$< 0.001^{***}$	$< 0.001^{***}$
<i>Panel B. Pooled token panel</i>							
$H(\text{CI}, 1)$	9,606	0.111	0.223	2.284	4.464	$< 0.001^{***}$	$< 0.001^{***}$
$H(\text{CI}, 3)$	9,606	0.427	0.647	2.214	5.423	$< 0.001^{***}$	$< 0.001^{***}$
$H(\text{CI}, 7)$	9,606	1.065	1.309	2.121	5.529	$< 0.001^{***}$	$< 0.001^{***}$
$H(\text{CR}, 3)$	9,606	0.189	0.325	2.291	5.714	$< 0.001^{***}$	$< 0.001^{***}$
$H(\text{LS}, 3)$	9,606	0.318	0.610	2.748	8.325	$< 0.001^{***}$	$< 0.001^{***}$
$H(\text{LEV}, 3)$	9,606	0.488	0.680	2.003	4.177	$< 0.001^{***}$	$< 0.001^{***}$
$H(\text{LIQ}, 3)$	9,606	0.329	0.515	1.833	2.958	$< 0.001^{***}$	$< 0.001^{***}$
$H(\text{NEWS}, 3)$	9,606	0.313	0.562	2.546	7.116	$< 0.001^{***}$	$< 0.001^{***}$
$H(\text{ATTN}, 3)$	9,606	0.567	1.010	2.001	2.933	$< 0.001^{***}$	$< 0.001^{***}$

Notes: The table reports continuous event histories with the same diagnostics as Table A.12.1. $H(q, k)$ denotes the decayed event history for event channel q and half-life k days. CI denotes composite instability, CR crash event, LS liquidation-volume shock, LEV derivatives-positioning pressure, LIQ liquidity contraction, NEWS negative news, and ATTN search attention. The reported histories include the training, validation, and historical-evaluation periods. Interaction histories, cross-token summaries, and full ETH, BNB, SOL, and XRP event-history diagnostics are reported in Online Appendix A.3.

Figure A.12.1. Raw Input Series



Notes: Figure A.12.1 reports BTC in Panel A and the pooled token panel in Panel B. It uses within-token robust-z standardisation and 30-day rolling averages for visualization only; the forecasting models use lagged raw controls rather than the plotted smoothed series.